Quantum Theory As A Mechanism Tree

A MorphWiki mechanism synthesis

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This book reorganizes quantum theory away from historical article names and toward the recurring construction detected across the MorphWiki quantum pages: context-selected Hilbert space, state carrier, generator, spectral question, probability/readout, compatibility limits, boundary realization, many-mode extension, and protocols.

Contents

How To Use The Mechanism Tree

What The Tree Is

This book is a derivation map for quantum topics. Each named topic is treated as a contribution to one prediction-making mechanism. First specify the context: preparation, basis, boundary condition, gauge, detector arrangement, or domain. That context selects the Hilbert space and operator domain. A state is then transported by a generator, an observable defines the possible spectral answers, the Born or trace rule assigns probabilities to those answers, and compatibility constraints record which questions cannot be made jointly sharp. The page title is therefore not the root of the explanation. The root is the role the page plays in that construction.

> Mechanism-first reading: identify the operation a named topic performs in the quantum construction.
>
> **Constructor sequence:** context → Hilbert-space carrier and operator domain → generator/evolution → observable spectrum → probability readout → compatibility constraint → boundary/protocol realization.

The nonstandard result of this run is empirical rather than axiomatic. Standard quantum mechanics already contains Hilbert spaces, operators, spectra, Born probabilities, and commutators. What the rewrite adds is that a large topic system built from ordinary article names collapses into a stable constructor order. Particles, fields, protocols, interpretations, and boundary effects no longer appear as equal roots; they become later roles in one state-to-spectrum mechanism.

Why Hilbert Space Is Central

Hilbert space is central because quantum mechanics requires a space of admissible states before it requires a coordinate geometry of positions. Euclidean space labels positions, domains, detectors, or boundary conditions. Hilbert space carries phase, superposition, spin, entanglement, operator domains, and probability amplitudes. It supplies the inner product, norm, linear superposition, operator action, spectral projectors, and unitary maps required for quantum prediction. The usual position wave function illustrates the distinction. In the expression \(\psi(x)\), the coordinate \(x\in\mathbb R^3\) labels a representation. The state itself is \(\psi\in L^2(\mathbb R^3)\), and adding spin, identical particles, fields, or open-system mixtures moves the carrier to tensor products, Fock spaces, density operators, or operator algebras. Thus Euclidean space is often a basis or realization domain; Hilbert space is the admissibility selector.

$$
\begin{aligned}
|\psi\rangle \in \mathcal{H}, \quad \rho \geq 0, \quad \operatorname{Tr} \rho = 1 &\qquad \text{state carrier} \\
p_i = |\langle e_i|\psi\rangle|^2, \quad \Pr(\Delta) = \operatorname{Tr}(\rho E_A(\Delta)) &\qquad \text{probability structure} \\
A = A^\dagger, \quad A = \int_{\sigma(A)} \lambda \, dE_A(\lambda) &\qquad \text{operator-to-spectrum conversion} \\
\rho_t = U(t)\rho U(t)^\dagger, \quad U^\dagger U = I &\qquad \text{identity-preserving evolution.}
\end{aligned}
$$

In this sense Hilbert space is the selector chosen by the first branch. A context selects \((\mathcal H_C,\mathcal D_C)\): the admissible carrier and the operator domain. The later branches explain what is done with that selected carrier: states inhabit it, generators move states through it, observables resolve spectra on it, Born rules read probabilities from it, and commutators identify when two questions cannot share one sharp basis. Hilbert space selects the legal arena; it does not select the outcome.

The Compact Representation

The compact representation is the minimal formal record needed to say what a quantum mechanism is doing. It is a compressed version of standard quantum mechanics:

> **Compact tree:** selector → carrier → map → question → readout.
>
> Compatibility constrains which questions can be jointly sharp. Realization adds boundaries, fields, detectors, protocols, and scaling limits.

- **Context** means the experimental or mathematical setting in which a quantum question is posed: preparation, basis, boundary, gauge, detector, domain, or representation.
- **Hilbert-space carrier and domain** means the Hilbert space, operator domain, and constraints selected by that context. Examples include normalization, positivity of a density operator, boundary conditions, gauge constraints, and domain conditions for an unbounded operator.
- **Generator/evolution** means the Hamiltonian, unitary map, quantum channel, action, or other lawful map that transports the state before readout.
- **Observable spectrum** means the allowed answer set of a measurable question, represented by a self-adjoint operator, projection-valued measure, POVM, or channel readout.
- **Probability readout** means the Born or trace rule that turns the state and spectral channels into probabilities for observed outcomes.
- **Compatibility constraint** means a restriction on which questions can be jointly resolved, usually expressed by commutators, uncertainty relations, contextuality tests, conservation laws, or admissibility checks.
- **Boundary/protocol realization** means the physical or engineered implementation: a potential well, scattering boundary, cavity, field mode, detector, circuit, channel, or measurement protocol.

$$
\begin{aligned}
\mathfrak{M}_Q &= (C, \mathcal{H}_C, \mathcal{D}_C, \rho, H_C, A_C, E_{A_C}, \Pr, \mathcal{K}, \mathcal{R})\\
C &: \text{ context, preparation, basis, boundary, gauge, or detector arrangement}\\
\mathcal{H}_C, \mathcal{D}_C &: \text{ admissible state space and operator domain}\\
\rho_t &= U_C(t)\rho U_C(t)^\dagger, \qquad U_C(t) = \exp(-iH_C t/\hbar)\\
A_C &= \int_{\sigma(A_C)} \lambda \, dE_{A_C}(\lambda)\\
\Pr(\Delta \mid C, \rho, A) &= \mathrm{Tr}(\rho_t E_{A_C}(\Delta))\\
\mathcal{K} &: \text{compatibility tests such as commutators, constraints, and admissibility checks}\\
\mathcal{R} &: \text{realization layer: boundary, field mode, detector, circuit, or scaling limit.}
\end{aligned}
$$

This representation explains the tree. Context selects the Hilbert-space carrier and operator domain. The wave function and density matrix are states on that carrier. The Hamiltonian and unitary operator contribute the generator. Observables and projection-valued measures contribute the spectral question. The Born rule contributes the probability readout. Commutators, uncertainty, Bell-type pages, and EPR-type pages contribute compatibility tests. Tunnelling, particle-in-a-box, scattering, fields, gauge theory, circuits, and channels contribute realization or protocol layers.

Recipe For Reading A Page

1. **Locate the page in the tree.** The branch tells which part of $\mathfrak{M}_Q$ the page mainly modifies.

2. **Fill the compact tuple.** Identify C, $\mathcal{H}_C$, ρ, H_C or the relevant map, A_C, the spectral measure or readout, and any compatibility condition.

3. **Separate topic-specific pages from core-derived pages.** Topic-specific pages have native equations or explicit constructor skeletons. Core-derived pages inherit the shared constructor spine and specialize it with branch and topic evidence.

4. **Read anomalies as diagnostics.** An anomaly is a junction where the topic uses several roles at once, not a claim that the topic is wrong.

5. **Use transfer carefully.** A mechanism can be transferred only when the state carrier, operator role, readout, and compatibility test survive the move; field-specific nouns may change.

What Was Found In This Run

The current export contains 147 quantum pages. Of these, 145 have topic-specific equation skeletons or explicit constructor overrides, and 2 are expanded from the compact constructor as core-derived mechanisms. The sparse-attention profile is strongly operator/spectral: 146 pages exceed the operator/spectrum threshold, with mean signal 0.327. State evolution is also broad (146 pages), while boundary/context (54), incompatibility (59), and explicit protocol (32) are more selective. The practical result is a compact reading of quantum theory as operator-to-spectrum conversion under context and compatibility constraints. This is the main structural finding of the rewrite. Quantum theory can be introduced as a constructor in which particles, waves, fields, detectors, boundaries, circuits, and interpretations are roles attached to a smaller state-operator-readout spine. The portable unit is the legal transformation from context and state to spectral readout.

Branch Map

Role	Use in the derivation	Pages
Hilbert-space context: admissible carrier and basis	A quantum calculation first fixes the Hilbert space, operator domain, basis, preparation context, representation, gauge, or boundary condition. This is not the measured answer; it is the legal carrier on which states, transformations, observables, and probabilities can be defined.	7
State carrier inside Hilbert space	A state is the probability-bearing element of the selected Hilbert space or its density-operator state space. Wave functions, density matrices, superpositions, and coherent states are different representations of this predictive carrier.	24
Generator: lawful change before readout	Hamiltonians, unitary maps, equations of motion, and path weights describe the lawful transport of the state before a question is resolved.	22
Spectral question: what can be asked	An observable is a permitted question whose operator form determines the possible numerical answers.	10
Readout rule: how answers become probabilities	Measurement connects a state and an observable to recorded frequencies. Projection, POVMs, Born weights, and collapse language are alternative ways of presenting this state-to-spectrum readout step.	10
Compatibility limit: what cannot be jointly sharp	The non-classical part of the theory appears when two otherwise legal questions do not compose into one common sharp question. Commutators, uncertainty relations, contextuality, Bell tests, and entanglement live here.	6
Boundary realization: how effects appear	Many named quantum effects are boundary realizations of the same construction. A potential, barrier, box, cavity, detector, or medium changes the allowed spectral channels without changing the basic prediction problem.	12
Many-mode extension: fields, particles, and scaling	Quantum field theory, gauge theory, renormalization, photons, fermions, and related topics extend the same state-operator-spectrum logic to variable particle number, local fields, and scale-dependent descriptions.	22
Protocol layer: engineered transformations	Quantum computing, channels, circuits, algorithms, networks, sensors, and error correction turn the same formal machinery into controlled sequences of operations.	18
Annotations: history, interpretations, and popular frames	Some pages help readers navigate the subject but do not form steps in the mechanism. They are kept as annotations so books, historical figures, interpretations, and popular frames do not distort the constructive tree.	16

Boundary Of The Claim

The book is a mechanism-indexed synthesis generated from Wikipedia topic scaffolds, MorphWiki mechanism summaries, and ranked source-equation witness profiles. Its uses are reading, teaching, hypothesis generation, and constructor-target selection. Formal derivation, domain review, and experimental validation are the tests that promote a mechanism reading into a physical claim.

Preamble: Findings From The Rewrite

Core Insight

The rewrite reveals a compact operational reading of quantum theory. The standard formalism has a simpler explanatory order than the usual topic list. The order recovered by the sparse-attention pass is:

Constructor sequence: context → Hilbert-space carrier and operator domain → generator/evolution → observable spectrum → probability readout → compatibility constraint → boundary/protocol realization.

Here, context means the preparation, basis, boundary, gauge, domain, detector, or representation in which a question is posed. It selects the Hilbert space and operator domain allowed by that context, together with normalization, positivity, boundary, gauge, or domain constraints. This carrier is central: it defines which states are legal, how amplitudes and norms are compared, which operators may act, and how spectral projectors can be read probabilistically. The generator is the Hamiltonian, unitary map, channel, or action that carries the state before readout. The observable spectrum is the allowed answer set of the question being asked. The probability readout is the Born or trace-rule assignment to those answers. The compatibility constraint records which otherwise valid questions cannot be resolved together, for example through a non-zero commutator, uncertainty relation, contextuality test, conservation law, or admissibility condition. Boundary and protocol realization specify how the same construction is embodied as a potential well, scattering boundary, cavity, field mode, detector, circuit, or measurement sequence.

Main Finding And Why It Matters

The main finding is that the quantum corpus can be reorganized away from named topics into a smaller operational constructor: context-selected Hilbert space, state carrier, generator, observable spectrum, probability readout, compatibility limits, and realization or protocol. This changes the explanatory root. In the usual presentation, quantum theory appears as a collection of named effects and puzzles: particles, waves, measurement, tunnelling, entanglement, collapse, fields, and interpretations. In the mechanism-tree view, these become roles inside one compact state-operator-spectrum construction.

The importance is practical. The tree gives a recipe for constructing and comparing theories. A page becomes useful when one can specify what state is carried, what operator or map acts, what spectrum or readout is produced, what compatibility constraint applies, and what changes when the realization changes. This turns topic similarity into mechanism similarity: two subjects may use different nouns while preserving the same state-operator-readout role.

This organization of existing quantum theory has direct consequences for physical AI. A constructor-oriented AI can search over compact mechanisms, test whether the operator/spectral role survives changes of representation, and use anomalies as places where several roles collide. The result is a compact, falsifiable, and transferable representation of quantum theory.

Transition Sparse-Attention Result

A second sparse-attention pass treated the rewrite itself as the object of study: Wikipedia/topic view to mechanism-tree view. This comparison asks what becomes visible only after the same quantum topics are reorganized by constructor role rather than by article title.

The transition run covered 147 pages. Of these, 145 currently have topic-specific equation skeletons or explicit constructor overrides, while 2 are expanded by the shared compact constructor. The distinction is evidential, not structural: both page types are read through the same context-state-generator-spectrum-readout sequence.

The transition signal is asymmetric: the rewrite adds operational roles more strongly than it adds object names. The mechanism view exposes the state, operator, readout, compatibility, boundary, and protocol roles that are implicit in the topic view.

Route role	Mean signal	Interpretation
operator-to-spectrum readout	0.3272	constructor role preserved in the rewrite
state evolution	0.2321	constructor role preserved in the rewrite
normalization or admissibility	0.1646	constructor role preserved in the rewrite
preparation, basis, or boundary context	0.0943	constructor role preserved in the rewrite
non-commuting compatibility limits	0.0861	constructor role preserved in the rewrite
controlled update protocol	0.0753	constructor role preserved in the rewrite

New Structural Information

1. **The rewrite converts a noun-indexed encyclopedia into a derivation graph.** The new object is not a better summary of each topic; it is an ordering relation: which topic plays context, state, generator, observable, readout, compatibility, boundary, field, protocol, or annotation.

2. **The dominant stable role is operator-to-spectrum readout, not object naming.** The rewrite makes explicit that many named quantum topics become different ways of asking a legal spectral question of a state.

3. **Particles become stable role-realizations inside field/mode/readout machinery.** The particle pages are not discarded; they are relocated as field/mode/statistics/readout constructions. This is a more precise statement than 'particles are not fundamental'.

4. **Interpretations mostly act on readout semantics rather than replacing the**

formal constructor. QBism, relational quantum mechanics, collapse language, and popular frames can be kept without letting them become false roots of the derivation tree.

5. **Boundary pages are realization gates: they change allowed spectra without changing the core prediction problem.** Tunnelling, particle-in-a-box, scattering, cavities, and spectral lines become one family: boundary-shaped spectra.

6. **Protocol pages are an engineering layer over the constructor, not the root of the theory.** Quantum computing is reorganized as controlled composition of states, operators, readouts, and error constraints rather than a separate ontology of qubits.

7. **Anomalies identify where several constructor roles collide.** Anomalies are not errors in the tree. They are research handles: EPR, measurement problem, quantum gravity, quantum biology, and related pages require several roles at once.

Highest-Attention Transition Pages

- **Quantum logic** (generators, constructed): dominant roles after rewriting are measurement, context, state. Mechanism reading: This page sits between generators and annotations. The ambiguity is useful: it marks a place where two constructor roles meet and should be separated before the page is used as a derivation.

- **Schrödinger's cat** (states, constructed): dominant roles after rewriting are state, measurement, context. Mechanism reading: Schrödinger's cat is a macroscopic readout protocol, not a spectrum-first topic. It couples microscopic unitary evolution to a macroscopic boundary and forces the reader to separate three steps: coherent state transport, decoherence or apparatus coupling, and the rule by which one record is selected or conditioned.

- **Measurement problem** (measurement, constructed): dominant roles after rewriting are measurement, context, state. Mechanism reading: The measurement problem is a readout junction. It sits where unitary state transport, detector context, probability assignment, and state update meet. The useful decomposition is not 'observer versus system' but: state evolution before measurement, coupling to an apparatus or environment, POVM or projection readout, and the rule used to condition the state after the record.

- **Quantum biology** (generators, constructed): dominant roles after rewriting are state, context, observable. Mechanism reading: Quantum biology is an open-system transfer problem. The anomaly is that the biological environment is not background noise only; it is part of the boundary that may preserve, destroy, or select coherence. A rigorous constructor must name the state carrier, the environmental coupling, the coherence or transport observable, and the classical control that would remove the quantum contribution.

- **Introduction to quantum mechanics** (annotations, constructed): dominant roles after rewriting are context, measurement, state. Mechanism reading: An introductory page is anomalous because pedagogy compresses the whole constructor into one narrative. It mixes states, operators, spectra, measurement, examples, and interpretations. It should be used as a map, then decomposed into mechanism branches before making technical claims.

- **Scattering** (boundaries, constructed): dominant roles after rewriting are context, boundary, observable. Mechanism reading: Scattering is a boundary-to-spectrum mechanism. The important object is not the incoming particle name but the map from asymptotic in-states to out-states. The constructor should specify the Hamiltonian or interaction region, the boundary/asymptotic channels, the S-matrix or cross-section readout, and the conservation constraints.

- **Delayed-choice quantum eraser** (measurement, constructed): dominant roles after rewriting are measurement, state, context. Mechanism reading: The delayed-choice eraser is a protocol-order stress test. Its mechanism is the arrangement of which-path information, later measurement choice, and conditional correlation readout. The anomaly is not retrocausality by default; it is that the relevant statistics are defined only after the full measurement protocol is specified.

- **Fermi–Dirac statistics** (fields, constructed): dominant roles after rewriting are context, state, measurement. Mechanism reading: Fermi-Dirac statistics is an admissibility rule for many-particle states. The central mechanism is antisymmetry and occupation restriction, not an ordinary eigenvalue list. The constructor should expose anticommutation, exclusion, occupation numbers, and the thermodynamic readout derived from that constrained state space.

Operational Use

- **Teach quantum theory as a derivation tree.** The reader sees the required assembly order: context, state, generator, observable spectrum, probability readout, compatibility limit, and realization. This avoids presenting interpretations, particles, and protocols as equal primitives.

- **Build constructor targets for the decoder.** The 145 topic-specific pages are seed targets. The 2 core-derived pages become a supervised specialization set: each needs a topic-native state, operator/map, spectrum/readout, compatibility condition, and realization.

- **Find transfer candidates across fields.** A page can be transferred only if the role survives. This lets FieldBridge search for analogues of a mechanism rather than semantic analogues of words.

- **Use anomalies as research prompts.** Multi-role pages such as EPR, measurement problem, quantum gravity, and quantum biology are not clean branches. They mark places where compatibility, boundary, protocol, and transport signals collide.

- **Separate interpretation from machinery.** Interpretive pages can be retained as readout/probability annotations without allowing them to rewrite the Hamiltonian/operator/spectral core.

The practical consequence is a derivation tree rather than a topic list: topic-specific pages show local equations, while core-derived pages show how the same compact constructor specializes to the page role.

Main Findings

1. **Quantum theory collapses to a smaller constructor.** The constructor order is context, state space, generator/evolution, observable spectrum, Born readout, and compatibility constraint. This is not a new postulate of quantum mechanics; it is a cleaner order in which to present the existing formalism.

2. **Operator/spectral structure is the spine.** 146 of 147 pages exceed the operator/spectrum threshold, with mean signal 0.327. This is the strongest and most universal signal in the rewrite.

3. **Evolution is secondary but broad.** 146 pages carry state-evolution or transport signal. Evolution is not absent; it is the motion of the predictive carrier before a readout.
4. **Context and boundary are realization gates.** 54 pages emphasize preparation, boundary, or domain context. Tunnelling, cavities, scattering, and particle-in-a-box become boundary-shaped spectra rather than separate conceptual primitives.
5. **Incompatibility is the non-classical junction.** 59 pages carry explicit incompatibility signal. Entanglement, Bell phenomena, commutators, and uncertainty are grouped by failure of joint sharpness.
6. **Protocol is rare and therefore informative.** Only 32 pages exceed the protocol threshold. Quantum computing and information are therefore best treated as an engineering layer over the state-operator-spectrum machinery, not as the foundation of the subject.
7. **The tree compresses named topics into mechanism roles.** Hilbert-space context: admissible carrier and basis: 7, State carrier inside Hilbert space: 24, Generator: lawful change before readout: 22, Spectral question: what can be asked: 10, Readout rule: how answers become probabilities: 10, Compatibility limit: what cannot be jointly sharp: 6, Boundary realization: how effects appear: 12, Many-mode extension: fields, particles, and scaling: 22, Protocol layer: engineered transformations: 18, Annotations: history, interpretations, and popular frames: 16. This is the mechanism-level table of contents produced by the rewrite.
8. **Particles are stable role-realizations.** Particle pages resolve into field modes, spectra, statistics, and readout-stable excitations. The claim is not that particles are fake. The claim is that particle identity is a stable role inside a state-operator-readout construction.
9. **String theory and AdS/CFT sit on the field/geometry translation frontier.** String theory appears as a spectral many-mode constructor. AdS/CFT is the sharper geometry-translation case because its boundary and geometry signals are stronger in the current export.
10. **Black holes are not yet a root in this export.** The sparse-attention script flags black-hole physics as a boundary/information-closure target rather than as a finished claim. A serious black-hole reading requires an explicit rerun on black holes, horizons, Hawking radiation, entropy, holography, and information-loss topics.
11. **Interpretations sit mostly on the readout layer.** Interpretations change the meaning assigned to state, probability, collapse, update, or observer language. They do not, in this tree, replace the Hamiltonian, operator, spectrum, or Born-rule machinery.
12. **Geometry appears as realization and reconstruction.** Geometry is necessary for embodiment, boundary conditions, and physical interpretation, but source-equation stability checks preserve structure and spectral/operator roles more strongly than geometry.

What Changed Relative To Wikipedia

A conventional encyclopedia organizes quantum theory by names and historical articles. The MorphWiki rewrite organizes the same material by the operations needed to construct a prediction. The page named measurement no longer becomes the origin of the theory; it becomes the readout junction. The page named particle becomes a realization. The page named tunnelling becomes a boundary spectral channel. The page named quantum field theory becomes a scaling and many-mode extension. The tree therefore gives a new order of explanation rather than a new interpretation of consciousness, observation, or ontology.

Unusual Findings And Anomalies

The most useful anomalies are not errors. They are pages that sit between branches or violate the dominant operator/spectrum pattern. These pages mark conceptual junctions where a theory-builder should look for missing assumptions, hidden boundary conditions, or possible transfers to other fields.

The anomaly labels describe a page's role in the mechanism tree, not the literal physical object named by the page. For example, a page can be structurally anomalous because context, protocol, or compatibility carries the explanation before spectra are read out.

- **not eigenvalue-first**: the page should not be explained by spectra first; another construction step fixes the admissible question before eigenvalues become meaningful.
- **context participates in the law**: preparation, apparatus, context, representation, or boundary conditions are part of the mechanism rather than surrounding detail.
- **admissibility meets incompatibility**: the page joins the rules that make a quantum state legal with the rules that restrict jointly resolvable questions.
- **operation order matters**: the order of operations matters; the topic cannot be reduced to a static state, operator, and spectrum.
- **several mechanisms meet**: several construction steps meet in one topic, so the page is a junction rather than a clean branch leaf.
- **branch interface**: the topic belongs at an interface between two explanatory roles and should be read as a bridge before being assigned to one branch.

- **Schrödinger's cat**. Primary reading: State carrier inside Hilbert space; neighboring reading: Annotations: history, interpretations, and popular frames. Schrödinger's cat is a macroscopic readout protocol, not a spectrum-first topic. It couples microscopic unitary evolution to a macroscopic boundary and forces the reader to separate three steps: coherent state transport, decoherence or apparatus coupling, and the rule by which one record is selected or conditioned.
- **Einstein–Podolsky–Rosen paradox**. Primary reading: Compatibility limit: what cannot be jointly sharp; neighboring reading: Annotations: history, interpretations, and popular frames. EPR is a compatibility test, not a page about a peculiar object. Its mechanism is a bipartite state, separated measurement contexts, and a correlation readout that cannot be reduced to pre-existing local values. The construction should therefore start from the joint state and the allowed local observables, then ask which correlation constraints fail.
- **Quantum biology**. Primary reading: Generator: lawful change before readout; neighboring reading: Hilbert-space context: admissible carrier and basis. Quantum biology is an open-system transfer problem. The anomaly is that the biological environment is not background noise only; it is part of the boundary that may preserve, destroy, or select coherence. A rigorous constructor must name the state carrier, the environmental coupling, the coherence or transport observable, and the classical control that would remove the quantum contribution.

- **Introduction to quantum mechanics**. Primary reading: Annotations: history, interpretations, and popular frames; neighboring reading: State carrier inside Hilbert space. An introductory page is anomalous because pedagogy compresses the whole constructor into one narrative. It mixes states, operators, spectra, measurement, examples, and interpretations. It should be used as a map, then decomposed into mechanism branches before making technical claims.

- **Measurement problem**. Primary reading: Readout rule: how answers become probabilities; neighboring reading: Generator: lawful change before readout. The measurement problem is a readout junction. It sits where unitary state transport, detector context, probability assignment, and state update meet. The useful decomposition is not 'observer versus system' but: state evolution before measurement, coupling to an apparatus or environment, POVM or projection readout, and the rule used to condition the state after the record.

- **Quantum gravity**. Primary reading: Many-mode extension: fields, particles, and scaling; neighboring reading: Generator: lawful change before readout. Quantum gravity is a field/boundary junction. It asks whether geometry itself becomes part of the quantum state carrier or remains a realization layer for an operator theory. The missing constructor is explicit: a state of geometry, a constraint or evolution operator, a boundary or semiclassical readout, and a test showing which geometric quantities survive quantization.

- **Scattering**. Primary reading: Boundary realization: how effects appear; neighboring reading: Generator: lawful change before readout. Scattering is a boundary-to-spectrum mechanism. The important object is not the incoming particle name but the map from asymptotic in-states to out-states. The constructor should specify the Hamiltonian or interaction region, the boundary/asymptotic channels, the S-matrix or cross-section readout, and the conservation constraints.

- **Quantum state**. Primary reading: State carrier inside Hilbert space; neighboring reading: Generator: lawful change before readout. Quantum state is the carrier, not the final prediction. It is anomalous because it precedes several downstream roles: admissibility, evolution, observable choice, and probability readout. A clean derivation should state whether the carrier is a vector, density operator, field state, or register, and which transformations preserve its legality.

Interpretation Of The Anomalies

The anomaly interpretation is aggregated from the labels assigned by the sparse-attention pass. It is therefore a map of recurring failure modes in the mechanism tree, not a list of hand-picked examples.

- **several mechanisms meet**. several construction steps meet in one topic, so the page is a junction rather than a clean branch leaf. Diagnostic use: split the page into smaller constructor steps and test whether each step has its own state, map, readout, and constraint.

- **not eigenvalue-first**. the page should not be explained by spectra first; another construction step fixes the admissible question before eigenvalues become meaningful. Diagnostic use: check whether the page needs a non-spectral root role before forcing an eigenvalue-first derivation.

- **operation order matters**. the order of operations matters; the topic cannot be reduced to a static state, operator, and spectrum. Diagnostic use: write the ordered experimental or computational sequence and test whether changing the order changes the mechanism.
- **context participates in the law**. preparation, apparatus, context, representation, or boundary conditions are part of the mechanism rather than surrounding detail. Diagnostic use: name the preparation, apparatus, domain, detector, or context variable and test whether changing it changes the readout.
- **admissibility meets incompatibility**. the page joins the rules that make a quantum state legal with the rules that restrict jointly resolvable questions. Diagnostic use: separate the admissibility condition from the non-commuting or jointly unresolvable question.
- **branch interface**. the topic belongs at an interface between two explanatory roles and should be read as a bridge before being assigned to one branch. Diagnostic use: treat the page as a junction and compare both branch assignments against source equations.

Relation To Lagrangian, Gromov-Wasserstein, And Noether Layers

The later validation chapter explains how three global evidence layers check this tree. Noether-style tests ask which part of a formalism remains the same after a rewrite. Gromov-Wasserstein neighborhoods ask which formulations are structurally near enough to translate. The learned Lagrangian asks whether a proposed move through the atlas is a low-resistance continuation, a strained bridge, or a void-boundary target. In ordinary quantum language: the invariant is mostly operator/structural role, while geometry, boundary, and representation are where the role becomes embodied.

Practical Consequence

A reader, teacher, or AI constructor should not start from named quantum objects. The constructive order is: define the admissible context, define the predictive carrier, define lawful change, define the legal question, resolve its spectrum, assign probabilities, test incompatibility, then dress the construction in the relevant physical realization. That is the main structural result of the rewrite.

Part I

The Mechanism Tree

Chapter 1

Root Construction

Given a context-selected Hilbert space, an admissible state, and a permitted observable, quantum theory determines which numerical outcomes can occur, assigns probabilities to those outcomes, and marks which questions cannot be made simultaneously sharp.

1.1 Re-Derivation Path

1. **Context and Hilbert space.** Fix the Hilbert space, operator domain, basis, preparation condition, boundary, gauge, or representation.
2. **State carrier.** Write a normalized vector, wave function, density operator, or field/register state on that carrier.
3. **Generator.** Let a Hamiltonian, unitary map, constraint, or action move that carrier before readout.
4. **Spectral question.** Represent a measurable question by an operator with allowed answers.
5. **Readout.** Resolve the operator into outcome channels and assign probabilities.
6. **Compatibility.** Mark when two legal questions cannot be made jointly sharp.
7. **Realization and extension.** Add boundaries, fields, scaling limits, detectors, or protocols as later embodiments of the same construction.

$$
\begin{aligned}
B &\longmapsto \rho_B \\
\rho_t &= U_t \rho_B U_t^\dagger \\
O &= \sum_i \lambda_i P_i \\
p_i &= \mathrm{Tr}(P_i \rho_t)
\end{aligned}
$$

$$[O_1, O_2] \neq 0.$$

Chapter 2

Sparse Attention Summary

Operation	Mean signal	Pages above 0.10
State evolution	0.232	146
Normalization and admissibility	0.165	136
Observables and spectra	0.327	146
Preparation and boundary context	0.094	54
Incompatible questions	0.086	59
Controlled update protocol	0.075	32

The dominant recurrent signal is observables and spectra. Evolution, context, incompatibility, closure, and protocol are branch-forming operations. This is why the book begins with admissible context, state carrier, generator, and legal question rather than with particles or measurement folklore.

Chapter 3

Compact Operator Formulation

The compact form of nonrelativistic quantum theory is a map from a context and a state to a probability measure over an operator spectrum.

This chapter is the formal spine of the book. It does not introduce new physics. It rewrites familiar quantum mechanics as a sequence of necessary construction steps. Each later page is placed by asking which step it changes: admissible context, state carrier, generator, observable, readout, compatibility, realization, many-mode extension, or protocol.

3.1 The Minimal Constructor

$$
\begin{aligned}
C &\longmapsto (\mathcal{H}_C, \mathcal{D}_C) \\
\rho \in \mathcal{S}(\mathcal{H}_C), &\qquad \rho \geq 0, \quad \operatorname{Tr}\rho = 1 \\
\rho_t = U_C(t)\rho U_C(t)^\dagger, &\qquad U_C(t) = \exp(-iH_C t/\hbar) \\
A_C &= \int_{\sigma(A_C)} \lambda \, dE_{A_C}(\lambda) \\
\Pr(\Delta \mid C, \rho, A) &= \operatorname{Tr}(\rho_t \, E_{A_C}(\Delta)) \\
[A_C, B_C] \neq 0 &\quad \Rightarrow \quad \text{no generic common sharp spectral refinement.}
\end{aligned}
$$

3.2 Interpretation Of The Symbols

- C is the context: preparation, basis, boundary condition, gauge choice, detector arrangement, domain, or representation.
- $\mathcal{H}_C$ is the admissible state space selected by that context.
- $\mathcal{D}_C$ is the domain or admissibility condition for the generator and observables.
- ρ is the predictive state carrier. It may be a state vector, density operator, field state, or register state.
- H_C is the generator of lawful change before readout.
- A_C is the question being asked. Its spectral measure E_{A_C} supplies the outcome channels.
- $\Pr(\Delta \mid C, \rho, A)$ is the Born probability of an outcome set Δ.
- A non-zero commutator marks a compatibility limit: two legal questions may not share one sharp readout basis.

3.3 Why This Is More Compact

The usual presentation introduces particles, waves, measurement, operators, interpretations, and fields as separate conceptual blocks. The compact constructor shows that many of these are changes of role rather than separate roots. A particle is a stable state/readout role. A barrier is a context that changes the operator domain. A field is a many-mode extension of the state space. A quantum circuit is a protocolized composition of maps. An interpretation changes the meaning of state, probability, or update, but usually leaves the constructor intact.

Chapter 4

Mechanism Validation Layers

The mechanism tree is not validated by topic similarity. It is checked by three independent evidence layers: Noether-style currents ask what survives a rewrite, Gromov-Wasserstein (GW) neighborhoods ask which formulations are structurally near, and the learned Lagrangian asks whether a proposed move is an easy continuation, a strained bridge, or a void-boundary design target.

4.1 Noether Layer: What Stays The Same

Here a Noether-like current is not a physical conservation law. It is a stability test over equation fingerprints. When a theory is rewritten, translated, or moved through the corpus, the test asks which part of the formal apparatus keeps its role. That is exactly the question needed for a mechanism tree: what makes two formulations the same mechanism?

- The run promoted ? strict currents and ? near currents. These are role-preserving directions in the learned representation, not literal laws of nature.

4.2 Gromov-Wasserstein Layer (GW): What Can Be Translated

The Gromov-Wasserstein layer is an analogy layer, but not semantic analogy. It asks whether two equation neighborhoods have compatible relational geometry. For the mechanism tree, this means that a concept transfers when it preserves a role: context, state carrier, generator, spectral question, readout, compatibility limit, boundary realization, field extension, or protocol.

- The run scanned ? Gromov-Wasserstein candidates, hardened ? bridge records, and retained ? artifact-validated method bridges.
- The practical reading is conservative: Gromov-Wasserstein can nominate a bridge, but source equations, directed transitions, and residual checks decide whether it becomes useful.

4.3 Lagrangian Layer: Which Moves Are Easy Or Strained

The learned Lagrangian is the navigation layer of the atlas. It is a representation-space action surrogate over source-equation fingerprints, separate from the physical action of a quantum system. Given a construction-role state, it asks which next states are cheap continuations, which require a strained bridge, and which absences sit on a meaningful void front. In this sense it finds roads through the operational grammar: low-action corridors where a formal role can be continued, translated, or tested. In the current quantum tree this layer supplies the global road-map constraint. A full page-coordinate export will allow direct page-level action scoring.

- For the quantum tree, the Lagrangian is therefore a construction constraint rather than a leaf selector. It tells how open construction steps should be interpreted in the atlas, but the present export cannot yet say that a particular Wikipedia page lies on a particular low-action road. That requires full fingerprints or witness transition vectors for the page.

4.4 How These Layers Change The Quantum Tree

A quantum concept is not placed by name alone. It is placed by the role it plays in the mechanism, checked against source equation witnesses, and interpreted through stability and transfer layers. If a page preserves structure and spectral role, it belongs near the spine. If it changes the admissible domain, it belongs in boundary realization. If it introduces many modes or scale flow, it belongs in the field/scaling extension. If it describes a sequence of controlled operations, it belongs in the protocol layer. This is the human-readable reason for the tree: it unfolds quantum theory as a construction, then asks which parts survive when notation, representation, geometry, or field changes.

Chapter 5

Sparse-Attention Result

Quantum theory, in this sparse-attention reading, is a constructor that turns admissible state descriptions into spectra and probabilities under compatibility and boundary constraints.

Particles, strings, fields, measurements, and interpretations collapse into roles in the same constructor rather than separate root categories.

5.1 Minimal Constructor

The sparse-attention run returns the same constructor as a role sequence. Rendered in standard quantum notation, the sequence is:

$$
\begin{aligned}
B &\longmapsto (\mathcal{H}_B, \mathcal{D}_B) \\
\rho_B(t) &= U_B(t)\rho_B(0)U_B(t)^\dagger \\
O_B &= \sum_i \lambda_i P_i \qquad \text{or, more generally,} \qquad O_B = \int_{\sigma(O_B)} \lambda \, dE_{O_B}(\lambda) \\
p_i &= \mathrm{Tr}(P_i \rho_B(t)) \,, \qquad \Pr(\Delta) = \mathrm{Tr}(\rho_B(t) E_{O_B}(\Delta)) \\
[O_1, O_2] \neq 0 \quad &\Rightarrow \quad \text{no generic common sharp readout} \\
\mathcal{R} &: \ \text{boundary, field, detector, circuit, or scaling realization.}
\end{aligned}
$$

- B denotes the context: preparation, basis, boundary, domain, gauge, or detector arrangement.
- $(\mathcal{H}_B, \mathcal{D}_B)$ is the admissible state space and operator domain selected by that context.
- $\rho_B(t)$ is the propagated state carrier.
- O_B, P_i, and E_{O_B} express the observable question and its spectral readout channels.
- The probabilities p_i or $\Pr(\Delta)$ are Born-rule readouts of the state against those channels.
- Non-commuting observables define compatibility limits; realization layers specify how the same role is embodied.

5.2 Route Statistics

Role	Mean signal	Pages > 0.10
state evolution	0.232	146
admissibility and closure	0.165	137
observable spectra	0.327	146
preparation and boundary context	0.094	54
incompatible questions	0.086	59
controlled update protocol	0.075	32

The strongest recurrent signal is observable spectra. In the current export it is active above threshold on nearly every page, while protocol/update is much sparser. This is the quantitative reason the tree begins with state, generator, observable, spectrum, and readout before it introduces algorithms, experiments, or interpretations.

5.3 Hidden Rules

The rules below are induced from the current sparse-attention export. They are not a fixed taxonomy: a rule is included only when a route, branch, title-token family, or page subset crosses a measurable threshold. Each rule therefore remains an artifact-level claim until the corresponding equations and controls are attached.

5.3.1 R01. Operator-to-spectrum as the readout spine

Reading. The stable readout of a quantum construction is usually spectral. A state becomes experimentally or mathematically predictive when an allowed operator question exposes eigenvalues, projectors, or a spectral measure.

Use. Introduce observables, self-adjointness, spectral projectors, and the Born rule as the common readout interface before presenting particles, fields, or protocols as separate subjects.

Boundary. This does not make every quantum topic an eigenvalue problem; it says that spectra are the most reusable readout layer in this export.

5.3.2 R02. State transport as the carrier of change

Reading. Quantum topics repeatedly require a lawful carrier of change: a state vector, density operator, field mode, or channel is propagated before it is read.

Use. Derive state evolution as the map between preparation and later readout, using unitary, semigroup, or channel language depending on whether the system is closed, open, or operational.

Boundary. Transport is not the whole theory. It becomes meaningful only after the admissible state space and the observable question have been specified.

5.3.3 R03. Admissibility as the legal spine

Reading. Closure and admissibility form the legal spine of the theory: states must be normalizable or positive, maps must preserve the allowed state set, and operators must have a domain on which the question is well-defined.

Use. Every derivation should state what makes the state, operator, map, or probability assignment legal before it discusses physical interpretation.

Boundary. Admissibility is the formal condition that prevents the same symbols from representing an illegal quantum construction.

5.3.4 R04. Incompatibility as a selective limit

Reading. Incompatibility is selective. It becomes central when two questions, bases, or transformations cannot be made sharp at the same time.

Use. Use commutators, non-common eigenbases, Bell-type constraints, or contextuality tests only where the page is actually about jointly unavailable readouts.

Boundary. Use non-commutation where jointly unavailable readouts are the active mechanism; many topics are dominated by state evolution, spectra, or admissibility instead.

5.3.5 R05. Context as the active boundary of the question

Reading. Context is sometimes part of the law. Preparation, boundary conditions, detector arrangement, basis choice, or representation can change which state and operator are admissible.

Use. When context is active, write the context before the state: specify the Hilbert space, domain, boundary, apparatus, or channel that makes the later spectral question meaningful.

Boundary. The mathematical question is defined only after its carrier and boundary conditions are fixed.

5.3.6 R06. Protocol as ordered implementation

Reading. Protocols are a sparse but real layer. They become central when an ordered sequence of operations, measurements, controls, or updates determines what can be inferred.

Use. Place quantum algorithms, circuits, finite automata, Bell tests, and delayed-choice experiments in this layer: their content depends on operation order, not only on a static Hamiltonian.

Boundary. A protocol is not automatically a new physical principle. It is a realization of the state, operator, readout, and compatibility machinery in an ordered experiment or computation.

5.3.7 R07. Observables are spectral questions

Reading. The observables branch is correctly centered on spectral questions. Its pages ask which physical quantities can be represented by operators and which outcome channels those operators permit.

Use. Read this branch through self-adjoint operators, spectral decompositions, projectors, expectation values, and the distinction between compatible and incompatible observables.

Boundary. Observable names are not enough. A quantity becomes a quantum observable only when its operator domain and spectral readout are specified.

5.3.8 R08. Incompatibility becomes visible through protocols

Reading. The incompatibility branch is unexpectedly protocol-like. Bell tests, erasers, nonlocality arguments, and related pages depend on which questions are asked in which order and under which measurement arrangement.

Use. Present these topics as experimental or logical sequences that expose incompatible readouts, rather than as isolated paradoxes.

Boundary. The protocol does not replace the formal incompatibility; it is the way the incompatibility becomes observable.

5.3.9 R09. Interpretations attach to readout conflicts

Reading. Interpretation and history pages cluster around compatibility and readout problems. They usually debate what the state, probability, or update means after the formal machinery has already produced a constrained readout.

Use. Attach interpretations to the mechanism they reinterpret: state assignment, measurement update, probability, nonlocal correlations, or incompatible observables.

Boundary. Interpretations should not be roots of the mechanism tree unless they supply a different formal constructor.

5.3.10 R10. Context makes spectra well-defined

Reading. The context branch is where a spectral question becomes legally posed. Hilbert space, basis, representation, operator domain, and preparation context are not background labels; they define the admissible state space on which operators can have spectra.

Use. For pages such as Hilbert space or mathematical formulation, write the carrier first: choose the space, domain, inner product, and allowed states before asking for eigenvalues or probabilities.

Boundary. This does not say context causes the physics. It says the quantum question is not well-formed until the admissible carrier of the question has been supplied.

5.3.11 R11. Engineered protocols package spectral questions

Reading. Engineered quantum protocols still depend on spectral structure. Circuits, automata, sensors, and algorithms arrange allowed transformations so that a final readout exposes the desired spectral or probability information.

Use. Describe protocols as controlled sequences of unitary maps, channels, measurements, and conditional updates, then identify what spectral or probability readout the sequence is designed to reveal.

Boundary. The engineering layer is not separate from quantum theory; it is the operational packaging of the same constructor.

5.3.12 R12. Junction pages require decomposition

Reading. Some pages are junctions rather than leaves. EPR, wave function, delayed-choice eraser, entanglement, quantum biology, and similar topics join state, evolution, readout, boundary, protocol, and compatibility in one place.

Use. Do not force such pages into one branch. Split them into constructor steps: what is the state carrier, what transforms it, what is read out, what context is required, and what compatibility limit is being tested.

Boundary. A junction is not an error or a discovery by itself. It is a signal that the page should be decomposed before it is used as an explanatory root.

5.3.13 R13. Placement precedes derivation

Reading. Role assignment is easier than full local derivation. Sparse attention can often locate the constructor role of a page before the export contains a topic-specific equation skeleton.

Use. Treat core-derived pages as specialization targets: specify their state/operator/readout roles, then decide whether they need a topic-specific equation or can remain a specialization of the compact constructor.

Boundary. A constructor role assignment is evidence of formal similarity; a topic-specific derivation still requires local equations and controls.

5.3.14 R14. Annotation pages preserve formal signal

Reading. Annotation pages still contain formal signal. Historical and interpretive texts often preserve the same state, operator, probability, and compatibility vocabulary as technical pages.

Use. Use these pages to explain why a mechanism mattered or how it was interpreted, but attach them downstream of the formal constructor they discuss.

Boundary. A strong annotation signal should not be confused with an independent equation source.

5.3.15 R15. Operator pages split by role

Reading. The word operator is not one mechanism. Unitary operators transport states, self-adjoint operators define observable spectra, and angular-momentum operators carry symmetry and incompatibility structure.

Use. Split operator pages by role rather than by title: generator of evolution, observable question, symmetry generator, algebraic constraint, or protocol component.

Boundary. A shared word is a useful index term, not a physical equivalence class.

5.3.16 R16. Schrödinger pages split by role

Reading. The Schrödinger family is not one branch. The equation is a generator of state evolution, the picture is a representation choice, the cat is a protocol/readout junction, and the biography is an annotation.

Use. Separate the Schrödinger equation, Schrödinger picture, Schrödinger's cat, and historical material by the role each performs in the constructor.

Boundary. The shared name does not imply a shared mechanism.

5.3.17 R17. Physics labels hide different roles

Reading. Physics-labeled pages are not automatically foundations. They can be field realizations, scaling regimes, annotation pages, or constructor junctions depending on how state, operator, boundary, and readout appear.

Use. Ask what each physics-labeled page does: does it define a carrier, a generator, a spectrum, a compatibility limit, or a realization domain?

Boundary. Discipline labels should not determine the mechanism tree.

5.3.18 R18. Equation pages split by mechanism

Reading. Equation-named pages are not interchangeable. Schrödinger, Dirac, Klein-Gordon, and related equations differ by state carrier, symmetry constraints, admissible domains, and relativistic or field-theoretic realization.

Use. Derive each equation by stating its state space, generator, conserved or constrained quantities, and readout role instead of grouping all equations as a single topic type.

Boundary. The word equation identifies a presentation form, not the mechanism performed by the page.

5.3.19 R19. Heisenberg pages split by role

Reading. The Heisenberg family separates representation, uncertainty, historical annotation, and operator dynamics. Its unity is not a person-name label but a recurring role of non-commuting observables and time-dependent operators.

Use. Place Heisenberg-picture pages with generators and representations; place uncertainty and matrix mechanics pages with compatibility and observable algebra.

Boundary. A named tradition should not collapse distinct constructor roles.

5.3.20 R20. Introductions are compressed maps

Reading. Introductory pages are compressed junctions. They mix state, generator, observable, probability, interpretation, and examples because they are written pedagogically rather than mechanistically.

Use. Use introductions as maps, then redistribute their content into the mechanism tree before treating any paragraph as a derivation.

Boundary. Pedagogical order is not the same as construction order.

5.3.21 R21. Network pages split by role

Reading. Network-labeled pages split into physical networks, computational networks, and graph-like state structures. The title word hides different roles of connectivity, protocol, and state space.

Use. Ask whether the network is a physical realization, a computational protocol, an entanglement structure, or an abstract graph used to define admissible transformations.

Boundary. Connectivity language is not enough to infer a common mechanism.

5.3.22 R22. Geometry enters as realization

Reading. Geometry enters mainly through realization-heavy branches. In this export, geometric language most often specifies boundary, field, context, or reconstruction layers rather than replacing the operator-spectral core.

Use. Introduce geometry when the theory needs a domain, boundary, metric, representation, field realization, or reconstruction map. Keep the operator and admissibility structure visible underneath.

Boundary. This does not deny physical geometry. It restricts the claim to transferability in the current artifacts: geometry is essential for embodiment, but less portable than the operator/readout machinery.

5.4 Topic-Specific Readings

- **Particle**: In this reading, a particle is not the root unit. It is a stable field/mode/readout realization of the state-operator-spectrum constructor. Photon, electron, fermion, and boson do not share one identical route profile; 'particle' decomposes into spectral, boundary, closure, and statistics roles. Relevant pages: Boson, Fermion, Photon, Fock space, Creation and annihilation operators, Electron.
- **Black-hole direction**: Black holes do not appear as a primary constructor node in this export. They enter as a boundary/geometry stress test through quantum gravity, string, AdS/CFT, and horizon-like witnesses. This suggests a useful rerun: black-hole topics should be processed explicitly to test whether horizon physics is a boundary readout, an information-closure problem, or an geometry-free operator-kernel gap. Relevant pages: Quantum gravity, AdS/CFT correspondence, Quantum geometry, Loop quantum gravity, Electron, Fermion.
- **String-theory direction**: String theory appears as a many-mode field/scale extension with strong spectral content and geometry/boundary pressure, not as a separate noun ontology. The string is best read as a carrier of spectra under constraints; AdS/CFT is the sharper geometry-translation page because its boundary signal is higher. Relevant pages: String theory, AdS/CFT correspondence, Renormalization, Quantum mechanics, Quantum geometry, Loop quantum gravity.
- **Measurement and interpretations**: Measurement is a readout/probability layer, and interpretations modify the meaning of that layer rather than the constructor equations. QBism and relational pages can have strong operator/spectrum evidence while still being annotations, because they narrate the readout layer instead of adding new dynamics. Relevant pages: QBism, Wave function collapse, POVM, Interpretations of quantum mechanics, Born rule, Quantum mechanics.
- **Geometry**: Geometry behaves as realization and reconstruction: it makes the operator construction legal on domains, boundaries, gauges, or holographic duals, but it is not the strongest preserved current. Quantum gravity, AdS/CFT, and spin foam cluster near

field/boundary roles; this matches the current source-equation stability checks. Relevant pages: Spin foam, Loop quantum gravity, Quantum geometry, AdS/CFT correspondence, Electron, Fermion.

- **Void leads**: The best discovery leads are missing gates: contexts where a state, operator, boundary, or compatibility relation is implied but not closed by evidence. The black-hole/holography area and geometry-free operator kernels are natural tests for this, but they need explicit topic reruns and typed equations. Relevant pages: Quantum mechanics, Quantum geometry, Fock space, Photon, Electron, Fermion.

These statements are role hypotheses produced by deterministic sparse attention over the MorphWiki quantum pages and existing source-equation findings. They are not final physics claims. The next test is to rerun the same analysis on explicit black-hole, horizon, Hawking-radiation, entropy, holography, and information-loss topics with typed equation witnesses.

Chapter 6

Anomalies And Leads

6.1 Structural Anomalies

The labels below describe why a page is structurally interesting in the mechanism tree. They do not describe the literal object named by the page.

- **not eigenvalue-first**: the page should not be explained by spectra first; another construction step fixes the admissible question before eigenvalues become meaningful.
- **context participates in the law**: preparation, apparatus, context, representation, or boundary conditions are part of the mechanism rather than surrounding detail.
- **admissibility meets incompatibility**: the page joins the rules that make a quantum state legal with the rules that restrict jointly resolvable questions.
- **operation order matters**: the order of operations matters; the topic cannot be reduced to a static state, operator, and spectrum.
- **several mechanisms meet**: several construction steps meet in one topic, so the page is a junction rather than a clean branch leaf.
- **branch interface**: the topic belongs at an interface between two explanatory roles and should be read as a bridge before being assigned to one branch.

- **Schrödinger's cat**. Schrödinger's cat is a macroscopic readout protocol, not a spectrum-first topic. It couples microscopic unitary evolution to a macroscopic boundary and forces the reader to separate three steps: coherent state transport, decoherence or apparatus coupling, and the rule by which one record is selected or conditioned.
- **Einstein–Podolsky–Rosen paradox**. EPR is a compatibility test, not a page about a peculiar object. Its mechanism is a bipartite state, separated measurement contexts, and a correlation readout that cannot be reduced to pre-existing local values. The construction should therefore start from the joint state and the allowed local observables, then ask which correlation constraints fail.
- **Quantum biology**. Quantum biology is an open-system transfer problem. The anomaly is that the biological environment is not background noise only; it is part of the boundary that may preserve, destroy, or select coherence. A rigorous constructor must name the state carrier, the environmental coupling, the coherence or transport observable, and the classical control that would remove the quantum contribution.

- **Introduction to quantum mechanics**. An introductory page is anomalous because pedagogy compresses the whole constructor into one narrative. It mixes states, operators, spectra, measurement, examples, and interpretations. It should be used as a map, then decomposed into mechanism branches before making technical claims.

- **Measurement problem**. The measurement problem is a readout junction. It sits where unitary state transport, detector context, probability assignment, and state update meet. The useful decomposition is not 'observer versus system' but: state evolution before measurement, coupling to an apparatus or environment, POVM or projection readout, and the rule used to condition the state after the record.

- **Quantum gravity**. Quantum gravity is a field/boundary junction. It asks whether geometry itself becomes part of the quantum state carrier or remains a realization layer for an operator theory. The missing constructor is explicit: a state of geometry, a constraint or evolution operator, a boundary or semiclassical readout, and a test showing which geometric quantities survive quantization.

- **Scattering**. Scattering is a boundary-to-spectrum mechanism. The important object is not the incoming particle name but the map from asymptotic in-states to out-states. The constructor should specify the Hamiltonian or interaction region, the boundary/asymptotic channels, the S-matrix or cross-section readout, and the conservation constraints.

- **Quantum state**. Quantum state is the carrier, not the final prediction. It is anomalous because it precedes several downstream roles: admissibility, evolution, observable choice, and probability readout. A clean derivation should state whether the carrier is a vector, density operator, field state, or register, and which transformations preserve its legality.

- **Wave–particle duality**. Wave-particle duality is a representation/readout switch. The same carrier is interrogated through incompatible experimental contexts, so the observed pattern changes from interference-like to count-like. The constructor should be written as context selection plus readout channel, not as two substances alternating.

- **Quantum entanglement**. Entanglement is a tensor-factorization and correlation constraint. The anomaly is that the state is not reducible to independently readable subsystem states, while the readout is still local and spectral. A useful page must distinguish the joint state, the subsystem observables, and the correlation test.

- **Fermi–Dirac statistics**. Fermi-Dirac statistics is an admissibility rule for many-particle states. The central mechanism is antisymmetry and occupation restriction, not an ordinary eigenvalue list. The constructor should expose anticommutation, exclusion, occupation numbers, and the thermodynamic readout derived from that constrained state space.

- **Hamiltonian (quantum mechanics)**. The Hamiltonian has two roles at once: it generates time evolution and, when treated as an observable, supplies an energy spectrum. That double role explains the anomaly. A clean page must separate domain/self-adjointness, unitary transport, conserved energy, and spectral readout.

- **Wave function**. The wave function is a representation of the state carrier, not a material wave by itself. Its anomaly is that it stores amplitude, phase, normalization, basis choice, and probability potential in one object. The constructor should separate representation, admissibility, evolution, and Born readout.

- **Delayed-choice quantum eraser**. The delayed-choice eraser is a protocol-order stress test. Its mechanism is the arrangement of which-path information, later measurement choice, and conditional correlation readout. The anomaly is not retrocausality by default; it is that the relevant statistics are defined only after the full measurement protocol is specified.

- **Relativistic quantum mechanics**. Relativistic quantum mechanics is a compatibility junction between quantum state evolution and spacetime symmetry. Its construction preserves relativistic covariance, defines the correct state carrier, and explains how spin, energy, and causality constraints enter the operator algebra.
- **Quantum electrodynamics**. Quantum electrodynamics is a field-interaction constructor. Its anomaly comes from combining gauge admissibility, charged matter states, photon modes, perturbative transport, and scattering/readout. The page should be derived through field operators, gauge constraints, interaction terms, and observable amplitudes.
- **Quantum field theory**. This field-level page mixes operator-to-spectrum readout, state evolution, normalization or admissibility. It should be treated as a many-mode or geometric realization problem: first identify the state sector or field algebra, then the constraints and readout that make the field content observable.
- **Bell's theorem**. This incompatibility page mixes operator-to-spectrum readout, state evolution, normalization or admissibility. Its mechanism should state which otherwise legal questions fail to share a single sharp representation, and what experiment or inequality exposes that failure.

6.2 Research Leads

- Search for systems where a state-like carrier and a legal-question operator exist, but the incompatibility relation has not been tested. Those systems are candidates for quantum-like contextual behavior without importing quantum ontology.
- Treat tunnelling, particle-in-a-box, cavity optics, and spectral lines as one family of boundary-shaped spectra. This suggests looking for overlooked boundary controls in systems currently described only by bulk evolution.
- Quantum computing should be read as an engineering layer over the state-operator-readout constructor, not as a separate foundation. New protocols should be searched by composing lawful quantum questions and controlled maps, not by naming new qubit objects.
- Pages that are branch-ambiguous are useful: they often mark junctions where two constructions meet, such as field theory joining transport, incompatibility, and boundary context.
- Historical, interpretive, and object-name pages should be demoted to annotations. The conceptual spine is context, state, generator, spectral question, probability, compatibility, realization.

Part II

Branch Chapters

Chapter 7

Hilbert-space context: admissible carrier and basis

A quantum calculation first fixes the Hilbert space, operator domain, basis, preparation context, representation, gauge, or boundary condition. This is not the measured answer; it is the legal carrier on which states, transformations, observables, and probabilities can be defined.

7.1 Why This Branch Exists

This branch is the first step because quantum mechanics is not defined on raw objects. It begins by specifying the space, basis, representation, or admissibility condition in which states and questions make sense.

7.2 Page Map

Page	Dominant evidence signal	Score
Mathematical formulation of quantum mechanics	observables and spectra	1.41
Hilbert space	observables and spectra	1.40
Transformation theory (quantum mechanics)	observables and spectra	1.14
Quantum differential calculus	observables and spectra	0.88
Quantum complexity theory	preparation and boundary context	0.75
Quantum cellular automaton	preparation and boundary context	0.71
Relativistic quantum mechanics	state evolution	0.71

7.3 Mechanism Pages

Each page below is read as a specialization of the compact quantum constructor. Topic-specific pages use explicit equations or overrides; core-derived pages use the branch-level equation form associated with their mechanism role.

7.4 Mathematical formulation of quantum mechanics

Mechanism-tree role. Hilbert-space context: admissible carrier and basis; assignment score 1.41. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Mathematical formulation of quantum mechanics is a broad quantum constructor role in the compact quantum constructor. In this tree, mathematical formulation of quantum mechanics belongs at the first step of the constructor: it fixes the Hilbert space, operator domain, basis, representation, or preparation context before any probability statement is meaningful.

Mechanism Reading

Operationally, Mathematical formulation of quantum mechanics contributes a broad quantum constructor role. The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer. In the context step, the constructor starts by declaring the legal state carrier and the conditions under which states are admissible. In this role, Mathematical formulation of quantum mechanics specifies the mathematical setting in which states, operators, spectra, and readout probabilities can be written without ambiguity. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Mathematical formulation of quantum mechanics contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as a context-setting move: it fixes the arena in which states, domains, and questions are legal.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$

$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$

$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$

$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.40, transport flow=0.25, constraint closure=0.18

- arXiv:1604.06537, score 0.558
- arXiv:1604.05385, score 0.537
- arXiv:0912.2823, score 0.535
- arXiv:1612.00682, score 0.529
- arXiv:quant-ph0205159, score 0.528

7.5 Hilbert space

Mechanism-tree role. Hilbert-space context: admissible carrier and basis; assignment score 1.40. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Hilbert space is the admissible state carrier of quantum theory: it supplies the space in which states, operators, bases, spectra, and probabilities become legally defined.

Mechanism Reading

Hilbert space is not physical space and not a geometric background in this book. It is the legal carrier of quantum identity. A state is a vector or density operator on it; the inner product gives amplitudes and norms; observables are self-adjoint operators on it; spectral projectors define possible answers; and unitary evolution preserves norm and probability. Hilbert space is therefore central because it binds state, probability, operator spectrum, and identity preservation into one formal carrier. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Hilbert space contributes a state-carrier role to the quantum construction.
- **Placement**. This page is read first as a context-setting move: it fixes the arena in which states, domains, and questions are legal.
- **Carrier or domain**. The mathematical state object: vector, wavefunction, density operator, coherent state, field state, or register.
- **Operator or map**. Operators, maps, and observables become meaningful only after this carrier and its domain have been fixed.
- **Admissibility**. Normalization, positivity, inner product, representation, tensor factorization, or superselection conditions define legal states.
- **Readout**. Probability distributions obtained by applying the appropriate observables or measurement maps to the carrier.
- **Check**. Equivalent representations must preserve probabilities and expectation values when the change is only representational.

Role-Level Equations

Standard constructor skeleton: normalized states, density states, spectral resolution, Born readout, and unitary identity preservation.

$$
\begin{aligned}
|\psi\rangle \in \mathcal{H}, \qquad \langle\psi|\psi\rangle = 1 \\
\rho \in \mathcal{S}(\mathcal{H}), \qquad \rho \geq 0, \quad \operatorname{Tr}\rho = 1 \\
A = A^\dagger, \qquad A = \int_{\sigma(A)} \lambda \, dE_A(\lambda) \\
\Pr(\Delta \mid \rho, A) = \operatorname{Tr}(\rho E_A(\Delta)) \\
\rho_t = U(t)\rho U(t)^\dagger, \qquad U^\dagger U = I
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.42, transport flow=0.13, constraint closure=0.09

- arXiv:1604.05385, score 0.416
- arXiv:1612.00682, score 0.384
- arXiv:2308.15676, score 0.382
- arXiv:2108.07838, score 0.375
- arXiv:0809.5271, score 0.344

7.6 Transformation theory (quantum mechanics)

Mechanism-tree role. Hilbert-space context: admissible carrier and basis; assignment score 1.14. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Transformation theory (quantum mechanics) is a broad quantum constructor role in the compact quantum constructor. In this tree, transformation theory (quantum mechanics) belongs at the first step of the constructor: it fixes the Hilbert space, operator domain, basis, representation, or preparation context before any probability statement is meaningful.

Mechanism Reading

Operationally, Transformation theory (quantum mechanics) contributes a broad quantum constructor role. The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer. In the context step, the constructor starts by declaring the legal state carrier and the conditions under which states are admissible. In this role, Transformation theory (quantum mechanics) specifies the mathematical setting in which states, operators, spectra, and readout probabilities can be written without ambiguity. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Transformation theory (quantum mechanics) contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as a context-setting move: it fixes the arena in which states, domains, and questions are legal.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.43, transport flow=0.24, constraint closure=0.18

- arXiv:1801.03283, score 0.537
- arXiv:1708.03640, score 0.536
- arXiv:1612.00682, score 0.534
- arXiv:quant-ph0205159, score 0.534
- arXiv:1506.05598, score 0.534

7.7 Quantum differential calculus

Mechanism-tree role. Hilbert-space context: admissible carrier and basis; assignment score 0.88. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum differential calculus is a broad quantum constructor role in the compact quantum constructor. In this tree, quantum differential calculus belongs at the first step of the constructor: it fixes the Hilbert space, operator domain, basis, representation, or preparation context before any probability statement is meaningful.

Mechanism Reading

Operationally, Quantum differential calculus contributes a broad quantum constructor role. The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer. In the context step, the constructor starts by declaring the legal state carrier and the conditions under which states are admissible. In this role, Quantum differential calculus specifies the mathematical setting in which states, operators, spectra, and readout probabilities can be written without ambiguity. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum differential calculus contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as a context-setting move: it fixes the arena in which states, domains, and questions are legal.

- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.38, transport flow=0.19, constraint closure=0.15

- arXiv:0912.2823, score 0.509
- arXiv:1604.06537, score 0.499
- arXiv:1612.00682, score 0.478
- arXiv:1801.03283, score 0.473
- arXiv:quant-ph0205159, score 0.469

7.8 Quantum complexity theory

Mechanism-tree role. Hilbert-space context: admissible carrier and basis; assignment score 0.75. **Dominant evidence signal.** preparation and boundary context. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum complexity theory is a broad quantum constructor role in the compact quantum constructor. In this tree, quantum complexity theory belongs at the first step of the constructor: it fixes the Hilbert space, operator domain, basis, representation, or preparation context before any probability statement is meaningful.

Mechanism Reading

Operationally, Quantum complexity theory contributes a broad quantum constructor role. The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer. In the context step, the constructor starts by declaring the legal state carrier and the conditions under which states are admissible. In this role, Quantum complexity theory specifies the mathematical setting in which states, operators, spectra, and readout probabilities can be written without ambiguity. In the source-evidence profile for this page, the strongest construction signal is preparation, basis, or boundary context, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum complexity theory contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as a context-setting move: it fixes the arena in which states, domains, and questions are legal.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.

- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

boundary weak form=0.25, transport flow=0.17, constraint closure=0.16

- arXiv:0912.2823, score 0.520
- arXiv:1706.03846, score 0.515
- arXiv:1708.03640, score 0.499

- arXiv:1605.07654, score 0.494
- arXiv:0908.0752, score 0.492

7.9 Quantum cellular automaton

Mechanism-tree role. Hilbert-space context: admissible carrier and basis; assignment score 0.71. **Dominant evidence signal.** preparation and boundary context. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum cellular automaton is a broad quantum constructor role in the compact quantum constructor. In this tree, quantum cellular automaton belongs at the first step of the constructor: it fixes the Hilbert space, operator domain, basis, representation, or preparation context before any probability statement is meaningful.

Mechanism Reading

Operationally, Quantum cellular automaton contributes a broad quantum constructor role. The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer. In the context step, the constructor starts by declaring the legal state carrier and the conditions under which states are admissible. In this role, Quantum cellular automaton specifies the mathematical setting in which states, operators, spectra, and readout probabilities can be written without ambiguity. In the source-evidence profile for this page, the strongest construction signal is preparation, basis, or boundary context, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum cellular automaton contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as a context-setting move: it fixes the arena in which states, domains, and questions are legal.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

boundary weak form=0.28, transport flow=0.26, constraint closure=0.15

- arXiv:0912.2823, score 0.512
- arXiv:1109.3239, score 0.490
- arXiv:gr-qc0411110, score 0.487
- arXiv:1706.03846, score 0.486
- arXiv:astro-ph0604157, score 0.480

7.10 Relativistic quantum mechanics

Mechanism-tree role. Hilbert-space context: admissible carrier and basis; assignment score 0.71. **Dominant evidence signal.** state evolution. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Relativistic quantum mechanics is a broad quantum constructor role in the compact quantum constructor. In this tree, relativistic quantum mechanics belongs at the first step of the constructor: it fixes the Hilbert space, operator domain, basis, representation, or preparation context before any probability statement is meaningful.

Mechanism Reading

Operationally, Relativistic quantum mechanics contributes a broad quantum constructor role. The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer. In the context step, the constructor starts by declaring the legal state carrier and the conditions under which states are admissible. In this role, Relativistic quantum mechanics specifies the mathematical setting in which states, operators, spectra, and readout probabilities can be written without ambiguity. In the source-evidence profile for this page, the strongest construction signal is state evolution, operator-to-spectrum readout, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Relativistic quantum mechanics contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as a context-setting move: it fixes the arena in which states, domains, and questions are legal.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$

$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$

$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$

$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

transport flow=0.27, spectral operator=0.25, constraint closure=0.21

- arXiv:0912.2823, score 0.550
- arXiv:1604.06537, score 0.526
- arXiv:1604.05385, score 0.523
- arXiv:1801.03283, score 0.517
- arXiv:0908.0752, score 0.517

Chapter 8

State carrier inside Hilbert space

A state is the probability-bearing element of the selected Hilbert space or its density-operator state space. Wave functions, density matrices, superpositions, and coherent states are different representations of this predictive carrier.

8.1 Why This Branch Exists

The state branch should be introduced before particles. It is the predictive carrier; particles, waves, fields, and qubits are later realizations of that carrier.

8.2 Page Map

Page	Dominant evidence signal	Score
Density matrix	observables and spectra	1.54
Quantum superposition	observables and spectra	1.53
Quantum decoherence	observables and spectra	1.52
Superposition principle	observables and spectra	1.51
Coherence (physics)	observables and spectra	1.50
Wave function	state evolution	1.48
Quantum state	state evolution	1.46
Two-state quantum system	observables and spectra	1.44
Qubit	observables and spectra	1.43
Quantum fluctuation	state evolution	1.14
Bose–Einstein statistics	observables and spectra	0.86
Quantum number	observables and spectra	0.86
Fourier transform	state evolution	0.86
Wave–particle duality	state evolution	0.85
Quantum statistical mechanics	observables and spectra	0.83
Quantum machine	observables and spectra	0.82
Quantum chemistry	observables and spectra	0.82
Applications of quantum mechanics	observables and spectra	0.73
Quantum machine learning	observables and spectra	0.73
Standard Model	observables and spectra	0.73
Spin (physics)	observables and spectra	0.73
Quantum simulator	observables and spectra	0.72

Page	Dominant evidence signal	Score
Werner Heisenberg	state evolution	0.69
Schrödinger's cat	state evolution	0.64

8.3 Mechanism Pages

Each page below is read as a specialization of the compact quantum constructor. Topic-specific pages use explicit equations or overrides; core-derived pages use the branch-level equation form associated with their mechanism role.

8.4 Density matrix

Mechanism-tree role. State carrier inside Hilbert space; assignment score 1.54. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Density matrix is the mixed-state constructor: it keeps probabilistic preparation, entanglement with unobserved degrees of freedom, and partial information in the same state formalism.

Mechanism Reading

Density matrices generalize pure states without changing the state-to-spectrum readout rule. They are the correct carrier when the preparation is statistical, when a subsystem is traced out, or when decoherence is being described. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Density matrix contributes a state-carrier role to the quantum construction.
- **Placement**. This page is read first as a state-carrier move: it specifies what mathematical object carries prediction.
- **Carrier or domain**. The mathematical state object: vector, wavefunction, density operator, coherent state, field state, or register.
- **Operator or map**. Operators, maps, and observables become meaningful only after this carrier and its domain have been fixed.
- **Admissibility**. Normalization, positivity, inner product, representation, tensor factorization, or superselection conditions define legal states.
- **Readout**. Probability distributions obtained by applying the appropriate observables or measurement maps to the carrier.
- **Check**. Equivalent representations must preserve probabilities and expectation values when the change is only representational.

Topic Equations

$$
\begin{aligned}
B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)} \\
\rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)} \\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)} \\
[O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.43, transport flow=0.24, constraint closure=0.17

- arXiv:1604.05385, score 0.512
- arXiv:0912.2823, score 0.493
- arXiv:1612.00682, score 0.486
- arXiv:1801.03283, score 0.481
- arXiv:1506.05598, score 0.473

8.5 Quantum superposition

Mechanism-tree role. State carrier inside Hilbert space; assignment score 1.53. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum superposition is a state-carrier role in the compact quantum constructor. In this tree, quantum superposition supplies the carrier of prediction: the object that is propagated, transformed, restricted, or read out.

Mechanism Reading

Operationally, Quantum superposition contributes a state-carrier role. The topic contributes the mathematical carrier of prediction: vector, wavefunction, density operator, register, coherent state, or field state. In the states step, at this step the constructor names the predictive carrier, not the final physical story. The carrier may be a state vector, wave function, density operator, field state, or register state. What matters is that later operations can act on it and that probabilities can be computed from it. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum superposition contributes a state-carrier role to the quantum construction.
- **Placement**. This page is read first as a state-carrier move: it specifies what mathematical object carries prediction.
- **Carrier or domain**. The mathematical state object: vector, wavefunction, density operator, coherent state, field state, or register.
- **Operator or map**. Operators, maps, and observables become meaningful only after this carrier and its domain have been fixed.
- **Admissibility**. Normalization, positivity, inner product, representation, tensor factorization, or superselection conditions define legal states.
- **Readout**. Probability distributions obtained by applying the appropriate observables or measurement maps to the carrier.
- **Check**. Equivalent representations must preserve probabilities and expectation values when the change is only representational.

Topic Equations

$$
\begin{aligned}
B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)} \\
\rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)} \\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)} \\
[O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.41, transport flow=0.22, constraint closure=0.15

- arXiv:1604.05385, score 0.602
- arXiv:1604.06537, score 0.568
- arXiv:1612.00682, score 0.551
- arXiv:1801.03283, score 0.550
- arXiv:quant-ph0205159, score 0.544

8.6 Quantum decoherence

Mechanism-tree role. State carrier inside Hilbert space; assignment score 1.52. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum decoherence is an open-system transport and coherence role in the compact quantum constructor. In this tree, quantum decoherence supplies the carrier of prediction: the object that is propagated, transformed, restricted, or read out.

Mechanism Reading

Operationally, Quantum decoherence contributes an open-system transport and coherence role. The topic concerns quantum state transport under environmental coupling, coherence loss, biological or macroscopic boundary conditions, or effective dynamics outside an ideal closed system. In the states step, at this step the constructor names the predictive carrier, not the final physical story. The carrier may be a state vector, wave function, density operator, field state, or register state. What matters is that later operations can act on it and that probabilities can be computed from it. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum decoherence contributes an open-system transport and coherence role to the quantum construction.
- **Placement**. This page is read first as a state-carrier move: it specifies what mathematical object carries prediction.
- **Carrier or domain**. A density operator, reduced state, coherence variable, bath-coupled state, or effective mesoscopic carrier.
- **Operator or map**. Hamiltonian plus environmental coupling, Lindbladian, memory kernel, stochastic map, or effective transport operator.
- **Admissibility**. Positivity, trace preservation, timescale separation, bath assumptions, and control over classical noise determine whether the model is legal.
- **Readout**. Coherence, population transfer, relaxation rate, transport efficiency, noise spectrum, or macroscopic response.
- **Check**. The quantum contribution must survive controls against classical noise, preparation artifacts, and coarse-graining choices.

Topic Equations

$$\begin{aligned} B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)} \\ \rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)} \\ O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)} \\ [O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)} \end{aligned}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.37, transport flow=0.27, constraint closure=0.18

- arXiv:1604.05385, score 0.520
- arXiv:0912.2823, score 0.519
- arXiv:1604.06537, score 0.493
- arXiv:1612.00682, score 0.482
- arXiv:quant-ph0205159, score 0.478

8.7 Superposition principle

Mechanism-tree role. State carrier inside Hilbert space; assignment score 1.51. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Superposition principle is a state-carrier role in the compact quantum constructor. In this tree, superposition principle supplies the carrier of prediction: the object that is propagated, transformed, restricted, or read out.

Mechanism Reading

Operationally, Superposition principle contributes a state-carrier role. The topic contributes the mathematical carrier of prediction: vector, wavefunction, density operator, register, coherent state, or field state. In the states step, at this step the constructor names the predictive carrier, not the final physical story. The carrier may be a state vector, wave function, density operator, field state, or register state. What matters is that later operations can act on it and that probabilities can be computed from it. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Superposition principle contributes a state-carrier role to the quantum construction.
- **Placement**. This page is read first as a state-carrier move: it specifies what mathematical object carries prediction.
- **Carrier or domain**. The mathematical state object: vector, wavefunction, density operator, coherent state, field state, or register.
- **Operator or map**. Operators, maps, and observables become meaningful only after this carrier and its domain have been fixed.
- **Admissibility**. Normalization, positivity, inner product, representation, tensor factorization, or superselection conditions define legal states.
- **Readout**. Probability distributions obtained by applying the appropriate observables or measurement maps to the carrier.
- **Check**. Equivalent representations must preserve probabilities and expectation values when the change is only representational.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.31, transport flow=0.28, constraint closure=0.19

- arXiv:0912.2823, score 0.547
- arXiv:1801.03283, score 0.528
- arXiv:quant-ph0205159, score 0.525
- arXiv:1612.00682, score 0.521
- arXiv:1708.03640, score 0.520

8.8 Coherence (physics)

Mechanism-tree role. State carrier inside Hilbert space; assignment score 1.50. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Coherence (physics) is an open-system transport and coherence role in the compact quantum constructor. In this tree, coherence (physics) supplies the carrier of prediction: the object that is propagated, transformed, restricted, or read out.

Mechanism Reading

Operationally, Coherence (physics) contributes an open-system transport and coherence role. The topic concerns quantum state transport under environmental coupling, coherence loss, biological or macroscopic boundary conditions, or effective dynamics outside an ideal closed system. In the states step, at this step the constructor names the predictive carrier, not the final physical story. The carrier may be a state vector, wave function, density operator, field state, or register state. What matters is that later operations can act on it and that probabilities can be computed from it. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, preparation, basis, or boundary context; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Coherence (physics) contributes an open-system transport and coherence role to the quantum construction.
- **Placement**. This page is read first as a state-carrier move: it specifies what mathematical object carries prediction.

- **Carrier or domain**. A density operator, reduced state, coherence variable, bath-coupled state, or effective mesoscopic carrier.
- **Operator or map**. Hamiltonian plus environmental coupling, Lindbladian, memory kernel, stochastic map, or effective transport operator.
- **Admissibility**. Positivity, trace preservation, timescale separation, bath assumptions, and control over classical noise determine whether the model is legal.
- **Readout**. Coherence, population transfer, relaxation rate, transport efficiency, noise spectrum, or macroscopic response.
- **Check**. The quantum contribution must survive controls against classical noise, preparation artifacts, and coarse-graining choices.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$

$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$

$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$

$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.37, transport flow=0.18, boundary weak form=0.12

- arXiv:1604.05385, score 0.514
- arXiv:0912.2823, score 0.512
- arXiv:1612.00682, score 0.471
- arXiv:quant-ph0205159, score 0.460
- arXiv:2108.07838, score 0.459

8.9 Wave function

Mechanism-tree role. State carrier inside Hilbert space; assignment score 1.48. **Dominant evidence signal.** state evolution. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Wave function is a representation of the state carrier in a chosen basis, not a separate physical substance.

Mechanism Reading

The wave function is the coordinate expression of a quantum state after a representation has been chosen. Its modulus squared gives a probability density in the position representation, while operators act as transformations on that representation. In the source-evidence profile for this page, the strongest construction signal is state evolution, operator-to-spectrum readout, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Wave function contributes a state-carrier role to the quantum construction.
- **Placement**. This page is read first as a state-carrier move: it specifies what mathematical object carries prediction.
- **Carrier or domain**. The mathematical state object: vector, wavefunction, density operator, coherent state, field state, or register.
- **Operator or map**. Operators, maps, and observables become meaningful only after this carrier and its domain have been fixed.
- **Admissibility**. Normalization, positivity, inner product, representation, tensor factorization, or superselection conditions define legal states.
- **Readout**. Probability distributions obtained by applying the appropriate observables or measurement maps to the carrier.
- **Check**. Equivalent representations must preserve probabilities and expectation values when the change is only representational.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

transport flow=0.26, spectral operator=0.25, constraint closure=0.21

- arXiv:1604.05385, score 0.513
- arXiv:0912.2823, score 0.512
- arXiv:1604.06537, score 0.505
- arXiv:1801.03283, score 0.479
- arXiv:1708.03640, score 0.476

8.10 Quantum state

Mechanism-tree role. State carrier inside Hilbert space; assignment score 1.46. **Dominant evidence signal.** state evolution. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum state is a state-carrier role in the compact quantum constructor. In this tree, quantum state supplies the carrier of prediction: the object that is propagated, transformed, restricted, or read out.

Mechanism Reading

Operationally, Quantum state contributes a state-carrier role. The topic contributes the mathematical carrier of prediction: vector, wavefunction, density operator, register, coherent state, or field state. In the states step, at this step the constructor names the predictive carrier, not the final physical story. The carrier may be a state vector, wave function, density operator, field state, or register state. What matters is that later operations can act on it and that probabilities can be computed from it. In the source-evidence profile for this page, the strongest construction signal is state evolution, operator-to-spectrum readout, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum state contributes a state-carrier role to the quantum construction.
- **Placement**. This page is read first as a state-carrier move: it specifies what mathematical object carries prediction.
- **Carrier or domain**. The mathematical state object: vector, wavefunction, density operator, coherent state, field state, or register.
- **Operator or map**. Operators, maps, and observables become meaningful only after this carrier and its domain have been fixed.
- **Admissibility**. Normalization, positivity, inner product, representation, tensor factorization, or superselection conditions define legal states.
- **Readout**. Probability distributions obtained by applying the appropriate observables or measurement maps to the carrier.
- **Check**. Equivalent representations must preserve probabilities and expectation values when the change is only representational.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

transport flow=0.27, spectral operator=0.22, constraint closure=0.19

- arXiv:1604.05385, score 0.539
- arXiv:0912.2823, score 0.534
- arXiv:1604.06537, score 0.511
- arXiv:quant-ph0205159, score 0.495
- arXiv:1708.03640, score 0.495

8.11 Two-state quantum system

Mechanism-tree role. State carrier inside Hilbert space; assignment score 1.44. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Two-state quantum system is a state-carrier role in the compact quantum constructor. In this tree, two-state quantum system supplies the carrier of prediction: the object that is propagated, transformed, restricted, or read out.

Mechanism Reading

Operationally, Two-state quantum system contributes a state-carrier role. The topic contributes the mathematical carrier of prediction: vector, wavefunction, density operator, register, coherent state, or field state. In the states step, at this step the constructor names the predictive carrier, not the final physical story. The carrier may be a state vector, wave function, density operator, field state, or register state. What matters is that later operations can act on it and that probabilities can be computed from it. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Two-state quantum system contributes a state-carrier role to the quantum construction.
- **Placement**. This page is read first as a state-carrier move: it specifies what mathematical object carries prediction.
- **Carrier or domain**. The mathematical state object: vector, wavefunction, density operator, coherent state, field state, or register.
- **Operator or map**. Operators, maps, and observables become meaningful only after this carrier and its domain have been fixed.
- **Admissibility**. Normalization, positivity, inner product, representation, tensor factorization, or superselection conditions define legal states.
- **Readout**. Probability distributions obtained by applying the appropriate observables or measurement maps to the carrier.
- **Check**. Equivalent representations must preserve probabilities and expectation values when the change is only representational.

Topic Equations

$$\begin{aligned} B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)} \\ \rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)} \\ O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)} \\ [O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)} \end{aligned}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.43, transport flow=0.23, constraint closure=0.17

- arXiv:1801.03283, score 0.611
- arXiv:1604.06537, score 0.598
- arXiv:quant-ph0205159, score 0.597
- arXiv:1612.00682, score 0.596
- arXiv:1506.05598, score 0.586

8.12 Qubit

Mechanism-tree role. State carrier inside Hilbert space; assignment score 1.43. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Qubit is the two-dimensional state-carrier constructor used when the admissible state space is \(\mathbb C^2\).

Mechanism Reading

A qubit is the minimal quantum state space with a basis, amplitudes, unitary control, and measurement readout. Bloch-vector language is a representation of the same two-dimensional carrier. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Qubit contributes a state-carrier role to the quantum construction.
- **Placement**. This page is read first as a state-carrier move: it specifies what mathematical object carries prediction.
- **Carrier or domain**. The mathematical state object: vector, wavefunction, density operator, coherent state, field state, or register.
- **Operator or map**. Operators, maps, and observables become meaningful only after this carrier and its domain have been fixed.
- **Admissibility**. Normalization, positivity, inner product, representation, tensor factorization, or superselection conditions define legal states.
- **Readout**. Probability distributions obtained by applying the appropriate observables or measurement maps to the carrier.
- **Check**. Equivalent representations must preserve probabilities and expectation values when the change is only representational.

Topic Equations

$$
\begin{aligned}
B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)}\\
\rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)}\\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)}\\
[O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.40, transport flow=0.23, constraint closure=0.18

- arXiv:1604.05385, score 0.577
- arXiv:1801.03283, score 0.522
- arXiv:1612.00682, score 0.519
- arXiv:quant-ph0205159, score 0.511
- arXiv:1506.05598, score 0.511

8.13 Quantum fluctuation

Mechanism-tree role. State carrier inside Hilbert space; assignment score 1.14. **Dominant evidence signal.** state evolution. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum fluctuation is an open-system transport and coherence role in the compact quantum constructor. In this tree, quantum fluctuation supplies the carrier of prediction: the object that is propagated, transformed, restricted, or read out.

Mechanism Reading

Operationally, Quantum fluctuation contributes an open-system transport and coherence role. The topic concerns quantum state transport under environmental coupling, coherence loss, biological or macroscopic boundary conditions, or effective dynamics outside an ideal closed system. In the states step, at this step the constructor names the predictive carrier, not the final physical story. The carrier may be a state vector, wave function, density operator, field state, or register state. What matters is that later operations can act on it and that probabilities can be computed from it. In the source-evidence profile for this page, the strongest construction signal is state evolution, non-commuting compatibility limits, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum fluctuation contributes an open-system transport and coherence role to the quantum construction.
- **Placement**. This page is read first as a state-carrier move: it specifies what mathematical object carries prediction.

- **Carrier or domain**. A density operator, reduced state, coherence variable, bath-coupled state, or effective mesoscopic carrier.
- **Operator or map**. Hamiltonian plus environmental coupling, Lindbladian, memory kernel, stochastic map, or effective transport operator.
- **Admissibility**. Positivity, trace preservation, timescale separation, bath assumptions, and control over classical noise determine whether the model is legal.
- **Readout**. Coherence, population transfer, relaxation rate, transport efficiency, noise spectrum, or macroscopic response.
- **Check**. The quantum contribution must survive controls against classical noise, preparation artifacts, and coarse-graining choices.

Topic Equations

$$\begin{aligned} B &\longmapsto \rho_B \quad \text{(context specifies an admissible state)} \\ \rho_t &= U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)} \\ O &= \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)} \\ [O_1, O_2] &\neq 0 \quad \text{(incompatible observables: no common sharp basis)} \end{aligned}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

transport flow=0.26, commutator incompatibility=0.24, constraint closure=0.21

- arXiv:1604.06537, score 0.476
- arXiv:1706.03846, score 0.474
- arXiv:0908.0752, score 0.462
- arXiv:1604.05385, score 0.455
- arXiv:2106.04271, score 0.455

8.14 Bose–Einstein statistics

Mechanism-tree role. State carrier inside Hilbert space; assignment score 0.86. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Bose–Einstein statistics is a broad quantum constructor role in the compact quantum constructor. In this tree, bose–Einstein statistics supplies the carrier of prediction: the object that is propagated, transformed, restricted, or read out.

Mechanism Reading

Operationally, Bose–Einstein statistics contributes a broad quantum constructor role. The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer. In the states step, at this step the constructor names the predictive carrier, not the final physical story. The carrier may be a state vector, wave function, density operator, field state, or register state. What matters is that later operations can act on it and that probabilities can be computed from it. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, preparation, basis, or boundary context; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Bose–Einstein statistics contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as a state-carrier move: it specifies what mathematical object carries prediction.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.

- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.38, transport flow=0.12, boundary weak form=0.11

- arXiv:0912.2823, score 0.523
- arXiv:2308.15676, score 0.495
- arXiv:2105.11733, score 0.484
- arXiv:2108.07838, score 0.482
- arXiv:0805.4565, score 0.480

8.15 Quantum number

Mechanism-tree role. State carrier inside Hilbert space; assignment score 0.86. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum number is a broad quantum constructor role in the compact quantum constructor. In this tree, quantum number supplies the carrier of prediction: the object that is propagated, transformed, restricted, or read out.

Mechanism Reading

Operationally, Quantum number contributes a broad quantum constructor role. The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer. In the states step, at this step the constructor names the predictive carrier, not the final physical story. The carrier may be a state vector, wave function, density operator, field state, or register state. What matters is that later operations can act on it and that probabilities can be computed from it. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum number contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as a state-carrier move: it specifies what mathematical object carries prediction.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.36, transport flow=0.26, constraint closure=0.20

- arXiv:1604.05385, score 0.538
- arXiv:1604.06537, score 0.535
- arXiv:0912.2823, score 0.512
- arXiv:1612.00682, score 0.507
- arXiv:1801.03283, score 0.506

8.16 Fourier transform

Mechanism-tree role. State carrier inside Hilbert space; assignment score 0.86. **Dominant evidence signal.** state evolution. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Fourier transform is a broad quantum constructor role in the compact quantum constructor. In this tree, fourier transform supplies the carrier of prediction: the object that is propagated, transformed, restricted, or read out.

Mechanism Reading

Operationally, Fourier transform contributes a broad quantum constructor role. The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer. In the states step, at this step the constructor names the predictive carrier, not the final physical story. The carrier may be a state vector, wave function, density operator, field state, or register state. What matters is that later operations can act on it and that probabilities can be computed from it. In the source-evidence profile for this page, the strongest construction signal is state evolution, normalization or admissibility, non-commuting compatibility limits; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Fourier transform contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as a state-carrier move: it specifies what mathematical object carries prediction.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$

$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$

$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$

$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

transport flow=0.27, constraint closure=0.22, commutator incompatibility=0.20

- arXiv:1706.03846, score 0.563
- arXiv:1708.03640, score 0.539
- arXiv:0912.2823, score 0.538
- arXiv:1604.06537, score 0.535
- arXiv:0908.0752, score 0.533

8.17 Wave–particle duality

Mechanism-tree role. State carrier inside Hilbert space; assignment score 0.85. **Dominant evidence signal.** state evolution. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Wave–particle duality is a many-mode field or particle-realization role in the compact quantum constructor. In this tree, wave–particle duality supplies the carrier of prediction: the object that is propagated, transformed, restricted, or read out.

Mechanism Reading

Operationally, Wave–particle duality contributes a many-mode field or particle-realization role. The mechanism is the field/mode version of the constructor: a state space is decomposed into modes or sectors, operators create, annihilate, or constrain those modes, and readouts are occupation, charge, spin, momentum, energy, or scattering response. In the states step, at this step the constructor names the predictive carrier, not the final physical story. The carrier may be a state vector, wave function, density operator, field state, or register state. What matters is that later operations can act on it and that probabilities can be computed from it. In the source-evidence profile for this page, the strongest construction signal is state evolution, operator-to-spectrum readout, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Wave–particle duality contributes a many-mode field or particle-realization role to the quantum construction.
- **Placement**. This page is read first as a state-carrier move: it specifies what mathematical object carries prediction.
- **Carrier or domain**. Fock space, field configuration space, or a sector selected by charge, spin, momentum, statistics, or gauge data.
- **Operator or map**. Creation, annihilation, field, charge, spin, Hamiltonian, or scattering operators acting on the admissible sector.
- **Admissibility**. Statistics, gauge constraints, commutation or anticommutation rules, domain conditions, and sector labels decide which states are legal.
- **Readout**. Occupation number, charge, spin, momentum, energy, correlation function, cross-section, or scattering amplitude.
- **Check**. The field description must preserve the relevant observables under changes of representation and reduce to the expected particle or quasiparticle limit when that limit exists.

Topic Equations

$$
\begin{aligned}
B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)}\\
\rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)}\\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)}\\
[O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

transport flow=0.27, spectral operator=0.22, constraint closure=0.19

- arXiv:1604.05385, score 0.533
- arXiv:0912.2823, score 0.532
- arXiv:1604.06537, score 0.503
- arXiv:1708.03640, score 0.495
- arXiv:quant-ph0205159, score 0.492

8.18 Quantum statistical mechanics

Mechanism-tree role. State carrier inside Hilbert space; assignment score 0.83. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum statistical mechanics is a broad quantum constructor role in the compact quantum constructor. In this tree, quantum statistical mechanics supplies the carrier of prediction: the object that is propagated, transformed, restricted, or read out.

Mechanism Reading

Operationally, Quantum statistical mechanics contributes a broad quantum constructor role. The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer. In the states step, at this step the constructor names the predictive carrier, not the final physical story. The carrier may be a state vector, wave function, density operator, field state, or register state. What matters is that later operations can act on it and that probabilities can be computed from it. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum statistical mechanics contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as a state-carrier move: it specifies what mathematical object carries prediction.

- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$
\begin{aligned}
B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)} \\
\rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)} \\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)} \\
[O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.42, transport flow=0.22, constraint closure=0.17

- arXiv:1604.05385, score 0.592
- arXiv:0912.2823, score 0.589
- arXiv:1506.05598, score 0.582
- arXiv:1801.03283, score 0.582
- arXiv:1612.00682, score 0.581

8.19 Quantum machine

Mechanism-tree role. State carrier inside Hilbert space; assignment score 0.82. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum machine is a broad quantum constructor role in the compact quantum constructor. In this tree, quantum machine supplies the carrier of prediction: the object that is propagated, transformed, restricted, or read out.

Mechanism Reading

Operationally, Quantum machine contributes a broad quantum constructor role. The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer. In the states step, at this step the constructor names the predictive carrier, not the final physical story. The carrier may be a state vector, wave function, density operator, field state, or register state. What matters is that later operations can act on it and that probabilities can be computed from it. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, controlled update protocol; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum machine contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as a state-carrier move: it specifies what mathematical object carries prediction.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.

- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.38, transport flow=0.20, discrete protocol=0.12

- arXiv:1604.05385, score 0.531
- arXiv:0805.4565, score 0.501
- arXiv:2108.07838, score 0.500
- arXiv:1612.00682, score 0.497
- arXiv:quant-ph0205159, score 0.489

8.20 Quantum chemistry

Mechanism-tree role. State carrier inside Hilbert space; assignment score 0.82. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum chemistry is a broad quantum constructor role in the compact quantum constructor. In this tree, quantum chemistry supplies the carrier of prediction: the object that is propagated, transformed, restricted, or read out.

Mechanism Reading

Operationally, Quantum chemistry contributes a broad quantum constructor role. The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer. In the states step, at this step the constructor names the predictive carrier, not the final physical story. The carrier may be a state vector, wave function, density operator, field state, or register state. What matters is that later operations can act on it and that probabilities can be computed from it. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum chemistry contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as a state-carrier move: it specifies what mathematical object carries prediction.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.37, transport flow=0.26, constraint closure=0.19

- arXiv:0912.2823, score 0.562
- arXiv:1612.00682, score 0.523
- arXiv:quant-ph0205159, score 0.516
- arXiv:1801.03283, score 0.512
- arXiv:1506.05598, score 0.508

8.21 Applications of quantum mechanics

Mechanism-tree role. State carrier inside Hilbert space; assignment score 0.73. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Applications of quantum mechanics is a broad quantum constructor role in the compact quantum constructor. In this tree, applications of quantum mechanics supplies the carrier of prediction: the object that is propagated, transformed, restricted, or read out.

Mechanism Reading

Operationally, Applications of quantum mechanics contributes a broad quantum constructor role. The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer. In the states step, at this step the constructor names the predictive carrier, not the final physical story. The carrier may be a state vector, wave function, density operator, field state, or register state. What matters is that later operations can act on it and that probabilities can be computed from it. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Applications of quantum mechanics contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as a state-carrier move: it specifies what mathematical object carries prediction.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$\begin{aligned} B &\longmapsto \rho_B \quad \text{(context specifies an admissible state)} \\ \rho_t &= U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)} \\ O &= \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)} \\ [O_1, O_2] &\neq 0 \quad \text{(incompatible observables: no common sharp basis)} \end{aligned}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.33, transport flow=0.21, constraint closure=0.12

- arXiv:1604.05385, score 0.549
- arXiv:0912.2823, score 0.548
- arXiv:quant-ph0205159, score 0.524
- arXiv:0805.4565, score 0.523
- arXiv:1612.00682, score 0.518

8.22 Quantum machine learning

Mechanism-tree role. State carrier inside Hilbert space; assignment score 0.73. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum machine learning is a broad quantum constructor role in the compact quantum constructor. In this tree, quantum machine learning supplies the carrier of prediction: the object that is propagated, transformed, restricted, or read out.

Mechanism Reading

Operationally, Quantum machine learning contributes a broad quantum constructor role. The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer. In the states step, at this step the constructor names the predictive carrier, not the final physical story. The carrier may be a state vector, wave function, density operator, field state, or register state. What matters is that later operations can act on it and that probabilities can be computed from it. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum machine learning contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as a state-carrier move: it specifies what mathematical object carries prediction.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$

$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$

$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$

$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.33, transport flow=0.26, constraint closure=0.17

- arXiv:0912.2823, score 0.550
- arXiv:1604.05385, score 0.550
- arXiv:1612.00682, score 0.517
- arXiv:quant-ph0205159, score 0.513
- arXiv:1801.03283, score 0.509

8.23 Standard Model

Mechanism-tree role. State carrier inside Hilbert space; assignment score 0.73. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Standard Model is a many-mode field or particle-realization role in the compact quantum constructor. In this tree, standard Model supplies the carrier of prediction: the object that is propagated, transformed, restricted, or read out.

Mechanism Reading

Operationally, Standard Model contributes a many-mode field or particle-realization role. The mechanism is the field/mode version of the constructor: a state space is decomposed into modes or sectors, operators create, annihilate, or constrain those modes, and readouts are occupation, charge, spin, momentum, energy, or scattering response. In the states step, at this step the constructor names the predictive carrier, not the final physical story. The carrier may be a state vector, wave function, density operator, field state, or register state. What matters is that later operations can act on it and that probabilities can be computed from it. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Standard Model contributes a many-mode field or particle-realization role to the quantum construction.
- **Placement**. This page is read first as a state-carrier move: it specifies what mathematical object carries prediction.

- **Carrier or domain**. Fock space, field configuration space, or a sector selected by charge, spin, momentum, statistics, or gauge data.
- **Operator or map**. Creation, annihilation, field, charge, spin, Hamiltonian, or scattering operators acting on the admissible sector.
- **Admissibility**. Statistics, gauge constraints, commutation or anticommutation rules, domain conditions, and sector labels decide which states are legal.
- **Readout**. Occupation number, charge, spin, momentum, energy, correlation function, cross-section, or scattering amplitude.
- **Check**. The field description must preserve the relevant observables under changes of representation and reduce to the expected particle or quasiparticle limit when that limit exists.

Topic Equations

$$\begin{aligned} B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)} \\ \rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)} \\ O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)} \\ [O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)} \end{aligned}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.32, transport flow=0.26, constraint closure=0.21

- arXiv:0912.2823, score 0.495
- arXiv:1604.06537, score 0.491
- arXiv:1801.03283, score 0.474
- arXiv:quant-ph0205159, score 0.468
- arXiv:1612.00682, score 0.467

8.24 Spin (physics)

Mechanism-tree role. State carrier inside Hilbert space; assignment score 0.73. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Spin (physics) is a broad quantum constructor role in the compact quantum constructor. In this tree, spin (physics) supplies the carrier of prediction: the object that is propagated, transformed, restricted, or read out.

Mechanism Reading

Operationally, Spin (physics) contributes a broad quantum constructor role. The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer. In the states step, at this step the constructor names the predictive carrier, not the final physical story. The carrier may be a state vector, wave function, density operator, field state, or register state. What matters is that later operations can act on it and that probabilities can be computed from it. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Spin (physics) contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as a state-carrier move: it specifies what mathematical object carries prediction.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.

- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$
\begin{aligned}
B \longmapsto \rho_B \quad & \text{(context specifies an admissible state)} \\
\rho_t = U_t \rho_B U_t^\dagger \quad & \text{(unitary evolution from preparation to readout)} \\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad & \text{(spectral probability measure)} \\
[O_1, O_2] \neq 0 \quad & \text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.30, transport flow=0.27, constraint closure=0.20

- arXiv:0912.2823, score 0.528
- arXiv:1604.05385, score 0.527
- arXiv:quant-ph0205159, score 0.509
- arXiv:1801.03283, score 0.508
- arXiv:1708.03640, score 0.507

8.25 Quantum simulator

Mechanism-tree role. State carrier inside Hilbert space; assignment score 0.72. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum simulator is a broad quantum constructor role in the compact quantum constructor. In this tree, quantum simulator supplies the carrier of prediction: the object that is propagated, transformed, restricted, or read out.

Mechanism Reading

Operationally, Quantum simulator contributes a broad quantum constructor role. The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer. In the states step, at this step the constructor names the predictive carrier, not the final physical story. The carrier may be a state vector, wave function, density operator, field state, or register state. What matters is that later operations can act on it and that probabilities can be computed from it. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum simulator contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as a state-carrier move: it specifies what mathematical object carries prediction.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$
\begin{aligned}
B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)} \\
\rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)} \\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)} \\
[O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.30, transport flow=0.28, constraint closure=0.19

- arXiv:0912.2823, score 0.526
- arXiv:1604.05385, score 0.526
- arXiv:quant-ph0205159, score 0.506
- arXiv:1708.03640, score 0.504
- arXiv:1801.03283, score 0.503

8.26 Werner Heisenberg

Mechanism-tree role. State carrier inside Hilbert space; assignment score 0.69. **Dominant evidence signal.** state evolution. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Werner Heisenberg is a broad quantum constructor role in the compact quantum constructor. In this tree, werner Heisenberg supplies the carrier of prediction: the object that is propagated, transformed, restricted, or read out.

Mechanism Reading

Operationally, Werner Heisenberg contributes a broad quantum constructor role. The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer. In the states step, at this step the constructor names the predictive carrier, not the final physical story. The carrier may be a state vector, wave function, density operator, field state, or register state. What matters is that later operations can act on it and that probabilities can be computed from it. In the source-evidence profile for this page, the strongest construction signal is state evolution, operator-to-spectrum readout, non-commuting compatibility limits; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Werner Heisenberg contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as a state-carrier move: it specifies what mathematical object carries prediction.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$
\begin{aligned}
B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)} \\
\rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)} \\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)} \\
[O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

transport flow=0.26, spectral operator=0.25, commutator incompatibility=0.22

- arXiv:1604.06537, score 0.524
- arXiv:1604.05385, score 0.505
- arXiv:0912.2823, score 0.503
- arXiv:1708.03640, score 0.485
- arXiv:1801.03283, score 0.484

8.27 Schrödinger's cat

Mechanism-tree role. State carrier inside Hilbert space; assignment score 0.64. **Dominant evidence signal.** state evolution. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Schrödinger's cat is a lawful state-transport role in the compact quantum constructor. In this tree, schrödinger's cat supplies the carrier of prediction: the object that is propagated, transformed, restricted, or read out.

Mechanism Reading

Operationally, Schrödinger's cat contributes a lawful state-transport role. The topic supplies or modifies the generator of state evolution before readout. It should be read as a transport step, not as a measurement result. In the states step, at this step the constructor names the predictive carrier, not the final physical story. The carrier may be a state vector, wave function, density operator, field state, or register state. What matters is that later operations can act on it and that probabilities can be computed from it. In the source-evidence profile for this page, the strongest construction signal is state evolution, operator-to-spectrum readout, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Schrödinger's cat contributes a lawful state-transport role to the quantum construction.
- **Placement**. This page is read first as a state-carrier move: it specifies what mathematical object carries prediction.
- **Carrier or domain**. A state vector, density operator, wavefunction, field state, or register on a specified domain.
- **Operator or map**. Hamiltonian, unitary map, channel generator, action, constraint, or differential operator that transports the state.
- **Admissibility**. Self-adjointness, complete positivity, trace preservation, gauge constraints, and boundary/domain conditions decide whether the evolution is legal.
- **Readout**. Time-dependent probabilities, spectra, transition amplitudes, conserved quantities, or response functions.
- **Check**. The generator must predict the observed evolution while preserving the relevant normalization, positivity, symmetry, or conservation constraint.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

transport flow=0.27, spectral operator=0.22, constraint closure=0.19

- arXiv:1604.05385, score 0.514
- arXiv:0912.2823, score 0.513
- arXiv:1604.06537, score 0.481
- arXiv:quant-ph0205159, score 0.463
- arXiv:1708.03640, score 0.463

Chapter 9

Generator: lawful change before readout

Hamiltonians, unitary maps, equations of motion, and path weights describe the lawful transport of the state before a question is resolved.

9.1 Why This Branch Exists

Time evolution is a transport problem over states. The Hamiltonian and path integral are two views of the same generator role rather than unrelated formalisms.

9.2 Page Map

Page	Dominant evidence signal	Score
Unitary operator	observables and spectra	1.50
Perturbation theory	observables and spectra	1.50
Quantum dynamics	observables and spectra	1.50
Path integral formulation	observables and spectra	1.49
Hamiltonian mechanics	observables and spectra	1.49
Path integral	observables and spectra	1.48
Hamiltonian (quantum mechanics)	state evolution	1.47
Perturbation theory (quantum mechanics)	observables and spectra	1.46
Schrödinger picture	observables and spectra	1.29
Creation and annihilation operators	observables and spectra	1.28
Schrödinger equation	observables and spectra	1.22
Quantum stochastic calculus	observables and spectra	0.85
Quantum chaos	observables and spectra	0.76
Symmetry in quantum mechanics	observables and spectra	0.75
Quantum amplifier	observables and spectra	0.75
Quantum mechanics	observables and spectra	0.75
Heisenberg picture	observables and spectra	0.75
Heisenberg group	observables and spectra	0.74
Canonical commutation relation	observables and spectra	0.74
Quantum biology	state evolution	0.73
Quantum engineering	state evolution	0.72

Page	Dominant evidence signal	Score
Quantum logic	observables and spectra	0.56

9.3 Mechanism Pages

Each page below is read as a specialization of the compact quantum constructor. Topic-specific pages use explicit equations or overrides; core-derived pages use the branch-level equation form associated with their mechanism role.

9.4 Unitary operator

Mechanism-tree role. Generator: lawful change before readout; assignment score 1.50. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Unitary operator is the reversible-map constructor: it changes the state while preserving inner products and probabilities.

Mechanism Reading

Unitary maps are the admissible reversible transformations of closed-system quantum theory. They preserve normalization and carry projective geometry of the state space through time or controlled operations. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, spectral profile.

Quantum Mechanism Frame

- **Role**. Unitary operator contributes a lawful state-transport role to the quantum construction.
- **Placement**. This page is read first as a lawful-transport move: it identifies what changes the state before readout.
- **Carrier or domain**. A state vector, density operator, wavefunction, field state, or register on a specified domain.
- **Operator or map**. Hamiltonian, unitary map, channel generator, action, constraint, or differential operator that transports the state.
- **Admissibility**. Self-adjointness, complete positivity, trace preservation, gauge constraints, and boundary/domain conditions decide whether the evolution is legal.
- **Readout**. Time-dependent probabilities, spectra, transition amplitudes, conserved quantities, or response functions.
- **Check**. The generator must predict the observed evolution while preserving the relevant normalization, positivity, symmetry, or conservation constraint.

Topic Equations

$$
\begin{aligned}
B &\longmapsto \rho_B \quad \text{(context specifies an admissible state)}\\
\rho_t &= U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}\\
O &= \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}\\
[O_1, O_2] &\neq 0 \quad \text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.44, transport flow=0.24, constraint closure=0.18

- arXiv:0912.2823, score 0.500
- arXiv:1612.00682, score 0.492
- arXiv:1801.03283, score 0.477
- arXiv:quant-ph0205159, score 0.471
- arXiv:1506.05598, score 0.468

9.5 Perturbation theory

Mechanism-tree role. Generator: lawful change before readout; assignment score 1.50. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Perturbation theory is a broad quantum constructor role in the compact quantum constructor. In this tree, perturbation theory belongs to the lawful-change step: it specifies how the state changes before a question is asked.

Mechanism Reading

Operationally, Perturbation theory contributes a broad quantum constructor role. The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer. In the generators step, the generator is the part of the construction that makes the state move while preserving the admissibility conditions. In ordinary quantum mechanics this is usually a Hamiltonian or unitary map; in path-integral language it is an action weight over histories. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Perturbation theory contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as a lawful-transport move: it identifies what changes the state before readout.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.39, transport flow=0.21, constraint closure=0.12

- arXiv:0912.2823, score 0.538
- arXiv:1612.00682, score 0.501
- arXiv:0805.4565, score 0.499
- arXiv:2108.07838, score 0.498
- arXiv:quant-ph0205159, score 0.494

9.6 Quantum dynamics

Mechanism-tree role. Generator: lawful change before readout; assignment score 1.50. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum dynamics is a lawful state-transport role in the compact quantum constructor. In this tree, quantum dynamics belongs to the lawful-change step: it specifies how the state changes before a question is asked.

Mechanism Reading

Operationally, Quantum dynamics contributes a lawful state-transport role. The topic supplies or modifies the generator of state evolution before readout. It should be read as a transport step, not as a measurement result. In the generators step, the generator is the part of the construction that makes the state move while preserving the admissibility conditions. In ordinary quantum mechanics this is usually a Hamiltonian or unitary map; in path-integral language it is an action weight over histories. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum dynamics contributes a lawful state-transport role to the quantum construction.
- **Placement**. This page is read first as a lawful-transport move: it identifies what changes the state before readout.
- **Carrier or domain**. A state vector, density operator, wavefunction, field state, or register on a specified domain.
- **Operator or map**. Hamiltonian, unitary map, channel generator, action, constraint, or differential operator that transports the state.
- **Admissibility**. Self-adjointness, complete positivity, trace preservation, gauge constraints, and boundary/domain conditions decide whether the evolution is legal.
- **Readout**. Time-dependent probabilities, spectra, transition amplitudes, conserved quantities, or response functions.
- **Check**. The generator must predict the observed evolution while preserving the relevant normalization, positivity, symmetry, or conservation constraint.

Topic Equations

$$
\begin{aligned}
B \longmapsto \rho_B \quad & \text{(context specifies an admissible state)} \\
\rho_t = U_t \rho_B U_t^\dagger \quad & \text{(unitary evolution from preparation to readout)} \\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad & \text{(spectral probability measure)} \\
[O_1, O_2] \neq 0 \quad & \text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.39, transport flow=0.26, constraint closure=0.20

- arXiv:1604.06537, score 0.568
- arXiv:1801.03283, score 0.539
- arXiv:1612.00682, score 0.538
- arXiv:0912.2823, score 0.535
- arXiv:1506.05598, score 0.533

9.7 Path integral formulation

Mechanism-tree role. Generator: lawful change before readout; assignment score 1.49. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Path integral formulation is the formulation in which the generator is represented by action-weighted histories.

Mechanism Reading

This page belongs with lawful change because it changes how the transport step is calculated. The observable predictions remain probabilities after amplitudes are composed and squared or traced. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Path integral formulation contributes a lawful state-transport role to the quantum construction.
- **Placement**. This page is read first as a lawful-transport move: it identifies what changes the state before readout.
- **Carrier or domain**. A state vector, density operator, wavefunction, field state, or register on a specified domain.
- **Operator or map**. Hamiltonian, unitary map, channel generator, action, constraint, or differential operator that transports the state.
- **Admissibility**. Self-adjointness, complete positivity, trace preservation, gauge constraints, and boundary/domain conditions decide whether the evolution is legal.
- **Readout**. Time-dependent probabilities, spectra, transition amplitudes, conserved quantities, or response functions.
- **Check**. The generator must predict the observed evolution while preserving the relevant normalization, positivity, symmetry, or conservation constraint.

Topic Equations

$$
\begin{aligned}
B \longmapsto \rho_B \quad & \text{(context specifies an admissible state)} \\
\rho_t = U_t \rho_B U_t^\dagger \quad & \text{(unitary evolution from preparation to readout)} \\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad & \text{(spectral probability measure)} \\
[O_1, O_2] \neq 0 \quad & \text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.34, transport flow=0.25, constraint closure=0.19

- arXiv:0912.2823, score 0.548
- arXiv:1604.06537, score 0.545
- arXiv:1801.03283, score 0.516
- arXiv:1612.00682, score 0.515
- arXiv:quant-ph0205159, score 0.514

9.8 Hamiltonian mechanics

Mechanism-tree role. Generator: lawful change before readout; assignment score 1.49. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Hamiltonian mechanics is a lawful state-transport role in the compact quantum constructor. In this tree, hamiltonian mechanics belongs to the lawful-change step: it specifies how the state changes before a question is asked.

Mechanism Reading

Operationally, Hamiltonian mechanics contributes a lawful state-transport role. The topic supplies or modifies the generator of state evolution before readout. It should be read as a transport step, not as a measurement result. In the generators step, the generator is the part of the construction that makes the state move while preserving the admissibility conditions. In ordinary quantum mechanics this is usually a Hamiltonian or unitary map; in path-integral language it is an action weight over histories. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Hamiltonian mechanics contributes a lawful state-transport role to the quantum construction.
- **Placement**. This page is read first as a lawful-transport move: it identifies what changes the state before readout.

- **Carrier or domain**. A state vector, density operator, wavefunction, field state, or register on a specified domain.
- **Operator or map**. Hamiltonian, unitary map, channel generator, action, constraint, or differential operator that transports the state.
- **Admissibility**. Self-adjointness, complete positivity, trace preservation, gauge constraints, and boundary/domain conditions decide whether the evolution is legal.
- **Readout**. Time-dependent probabilities, spectra, transition amplitudes, conserved quantities, or response functions.
- **Check**. The generator must predict the observed evolution while preserving the relevant normalization, positivity, symmetry, or conservation constraint.

Topic Equations

$$\begin{aligned} B &\longmapsto \rho_B \quad \text{(context specifies an admissible state)} \\ \rho_t = U_t \rho_B U_t^\dagger &\quad \text{(unitary evolution from preparation to readout)} \\ O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) &\quad \text{(spectral probability measure)} \\ [O_1, O_2] \neq 0 &\quad \text{(incompatible observables: no common sharp basis)} \end{aligned}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.41, transport flow=0.21, constraint closure=0.14

- arXiv:1604.06537, score 0.481
- arXiv:2108.07838, score 0.467
- arXiv:1506.05598, score 0.464
- arXiv:1612.00682, score 0.462
- arXiv:2111.12617, score 0.454

9.9 Path integral

Mechanism-tree role. Generator: lawful change before readout; assignment score 1.48. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Path integral is an alternate generator constructor: transition amplitudes are obtained by summing phase weights over histories.

Mechanism Reading

The path integral does not replace the operator constructor. It repackages the generator step as a weighted sum over histories between boundary conditions. It is especially useful when action, symmetry, and field degrees of freedom are more natural than state-vector evolution. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Path integral contributes a lawful state-transport role to the quantum construction.
- **Placement**. This page is read first as a lawful-transport move: it identifies what changes the state before readout.
- **Carrier or domain**. A state vector, density operator, wavefunction, field state, or register on a specified domain.
- **Operator or map**. Hamiltonian, unitary map, channel generator, action, constraint, or differential operator that transports the state.
- **Admissibility**. Self-adjointness, complete positivity, trace preservation, gauge constraints, and boundary/domain conditions decide whether the evolution is legal.
- **Readout**. Time-dependent probabilities, spectra, transition amplitudes, conserved quantities, or response functions.
- **Check**. The generator must predict the observed evolution while preserving the relevant normalization, positivity, symmetry, or conservation constraint.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.24, transport flow=0.11, constraint closure=0.07

- arXiv:2107.01923, score 0.313
- arXiv:0801.3568, score 0.272
- arXiv:2105.11733, score 0.271
- arXiv:1706.07300, score 0.264
- arXiv:cond-mat0108470, score 0.263

9.10 Hamiltonian (quantum mechanics)

Mechanism-tree role. Generator: lawful change before readout; assignment score 1.47. **Dominant evidence signal.** state evolution. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Hamiltonian (quantum mechanics) is the generator observable: it both transports states and supplies the energy spectrum.

Mechanism Reading

The Hamiltonian has a dual role. Dynamically, it generates unitary time evolution. Spectrally, its eigenvalues are admissible energy readouts. This dual role is one reason the operator/spectrum branch is central. In the source-evidence profile for this page, the strongest construction signal is state evolution, operator-to-spectrum readout, preparation, basis, or boundary context; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Hamiltonian (quantum mechanics) contributes a lawful state-transport role to the quantum construction.
- **Placement**. This page is read first as a lawful-transport move: it identifies what changes the state before readout.
- **Carrier or domain**. A state vector, density operator, wavefunction, field state, or register on a specified domain.
- **Operator or map**. Hamiltonian, unitary map, channel generator, action, constraint, or differential operator that transports the state.
- **Admissibility**. Self-adjointness, complete positivity, trace preservation, gauge constraints, and boundary/domain conditions decide whether the evolution is legal.
- **Readout**. Time-dependent probabilities, spectra, transition amplitudes, conserved quantities, or response functions.
- **Check**. The generator must predict the observed evolution while preserving the relevant normalization, positivity, symmetry, or conservation constraint.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$

$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$

$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$

$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

transport flow=0.26, spectral operator=0.24, boundary weak form=0.23

- arXiv:0912.2823, score 0.502
- arXiv:1604.05385, score 0.465
- arXiv:0908.0752, score 0.447
- arXiv:2108.07838, score 0.447
- arXiv:1612.00682, score 0.446

9.11 Perturbation theory (quantum mechanics)

Mechanism-tree role. Generator: lawful change before readout; assignment score 1.46. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Perturbation theory (quantum mechanics) is a broad quantum constructor role in the compact quantum constructor. In this tree, perturbation theory (quantum mechanics) belongs to the lawful-change step: it specifies how the state changes before a question is asked.

Mechanism Reading

Operationally, Perturbation theory (quantum mechanics) contributes a broad quantum constructor role. The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer. In the generators step, the generator is the part of the construction that makes the state move while preserving the admissibility conditions. In ordinary quantum mechanics this is usually a Hamiltonian or unitary map; in path-integral language it is an action weight over histories. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, preparation, basis, or boundary context, state evolution; the strongest carrier signal is local notation, information profile, spectral profile.

Quantum Mechanism Frame

- **Role**. Perturbation theory (quantum mechanics) contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as a lawful-transport move: it identifies what changes the state before readout.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$
\begin{aligned}
B &\longmapsto \rho_B \quad \text{(context specifies an admissible state)} \\
\rho_t = U_t \rho_B U_t^\dagger &\quad \text{(unitary evolution from preparation to readout)} \\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) &\quad \text{(spectral probability measure)} \\
[O_1, O_2] \neq 0 &\quad \text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.26, boundary weak form=0.24, transport flow=0.22

- arXiv:0912.2823, score 0.509
- arXiv:1604.05385, score 0.475
- arXiv:1612.00682, score 0.449
- arXiv:1506.05598, score 0.436
- arXiv:quant-ph0205159, score 0.436

9.12 Schrödinger picture

Mechanism-tree role. Generator: lawful change before readout; assignment score 1.29. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Schrödinger picture is a lawful state-transport role in the compact quantum constructor. In this tree, schrödinger picture belongs to the lawful-change step: it specifies how the state changes before a question is asked.

Mechanism Reading

Operationally, Schrödinger picture contributes a lawful state-transport role. The topic supplies or modifies the generator of state evolution before readout. It should be read as a transport step, not as a measurement result. In the generators step, the generator is the part of the construction that makes the state move while preserving the admissibility conditions. In ordinary quantum mechanics this is usually a Hamiltonian or unitary map; in path-integral language it is an action weight over histories. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, preparation, basis, or boundary context; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Schrödinger picture contributes a lawful state-transport role to the quantum construction.
- **Placement**. This page is read first as a lawful-transport move: it identifies what changes the state before readout.
- **Carrier or domain**. A state vector, density operator, wavefunction, field state, or register on a specified domain.
- **Operator or map**. Hamiltonian, unitary map, channel generator, action, constraint, or differential operator that transports the state.
- **Admissibility**. Self-adjointness, complete positivity, trace preservation, gauge constraints, and boundary/domain conditions decide whether the evolution is legal.
- **Readout**. Time-dependent probabilities, spectra, transition amplitudes, conserved quantities, or response functions.
- **Check**. The generator must predict the observed evolution while preserving the relevant normalization, positivity, symmetry, or conservation constraint.

Topic Equations

$$
\begin{aligned}
B &\longmapsto \rho_B \quad \text{(context specifies an admissible state)}\\
\rho_t = U_t \rho_B U_t^\dagger &\quad \text{(unitary evolution from preparation to readout)}\\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) &\quad \text{(spectral probability measure)}\\
[O_1, O_2] \neq 0 &\quad \text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.27, transport flow=0.27, boundary weak form=0.20

- arXiv:0912.2823, score 0.464
- arXiv:1604.06537, score 0.442
- arXiv:1801.03283, score 0.435
- arXiv:1612.00682, score 0.418
- arXiv:1506.05598, score 0.414

9.13 Creation and annihilation operators

Mechanism-tree role. Generator: lawful change before readout; assignment score 1.28. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Creation and annihilation operators are sector-changing operators: they add or remove one quantum from a mode and make many-body or field descriptions executable.

Mechanism Reading

The page is about the algebraic move that changes occupation number. Creation raises the population of a mode, annihilation lowers it, and the commutation or anticommutation rule determines the statistics. The number operator gives the spectral readout. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, non-commuting compatibility limits; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Creation and annihilation operators contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as a lawful-transport move: it identifies what changes the state before readout.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.

- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$
\begin{aligned}
B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)} \\
\rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)} \\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)} \\
[O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.27, transport flow=0.26, commutator incompatibility=0.21

- arXiv:1604.06537, score 0.536
- arXiv:0912.2823, score 0.509
- arXiv:1801.03283, score 0.505
- arXiv:2111.12617, score 0.503
- arXiv:1506.05598, score 0.502

9.14 Schrödinger equation

Mechanism-tree role. Generator: lawful change before readout; assignment score 1.22. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Schrödinger equation is the state-transport constructor: the Hamiltonian generates lawful change of the state before readout.

Mechanism Reading

The Schrödinger equation is not a measurement rule. It is the generator step of the quantum constructor. It evolves the predictive carrier while preserving normalization when the Hamiltonian is self-adjoint. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Schrödinger equation contributes a lawful state-transport role to the quantum construction.
- **Placement**. This page is read first as a lawful-transport move: it identifies what changes the state before readout.
- **Carrier or domain**. A state vector, density operator, wavefunction, field state, or register on a specified domain.
- **Operator or map**. Hamiltonian, unitary map, channel generator, action, constraint, or differential operator that transports the state.
- **Admissibility**. Self-adjointness, complete positivity, trace preservation, gauge constraints, and boundary/domain conditions decide whether the evolution is legal.
- **Readout**. Time-dependent probabilities, spectra, transition amplitudes, conserved quantities, or response functions.
- **Check**. The generator must predict the observed evolution while preserving the relevant normalization, positivity, symmetry, or conservation constraint.

Topic Equations

$$
\begin{aligned}
B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)} \\
\rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)} \\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)} \\
[O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.40, transport flow=0.26, constraint closure=0.18

- arXiv:0912.2823, score 0.567
- arXiv:1604.06537, score 0.551
- arXiv:1604.05385, score 0.548
- arXiv:1801.03283, score 0.539
- arXiv:1612.00682, score 0.536

9.15 Quantum stochastic calculus

Mechanism-tree role. Generator: lawful change before readout; assignment score 0.85. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum stochastic calculus is a broad quantum constructor role in the compact quantum constructor. In this tree, quantum stochastic calculus belongs to the lawful-change step: it specifies how the state changes before a question is asked.

Mechanism Reading

Operationally, Quantum stochastic calculus contributes a broad quantum constructor role. The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer. In the generators step, the generator is the part of the construction that makes the state move while preserving the admissibility conditions. In ordinary quantum mechanics this is usually a Hamiltonian or unitary map; in path-integral language it is an action weight over histories. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, spectral profile.

Quantum Mechanism Frame

- **Role**. Quantum stochastic calculus contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as a lawful-transport move: it identifies what changes the state before readout.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$
\begin{aligned}
B \longmapsto \rho_B \quad & \text{(context specifies an admissible state)} \\
\rho_t = U_t \rho_B U_t^\dagger \quad & \text{(unitary evolution from preparation to readout)} \\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad & \text{(spectral probability measure)} \\
[O_1, O_2] \neq 0 \quad & \text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.42, transport flow=0.22, constraint closure=0.14

- arXiv:1604.06537, score 0.486
- arXiv:1604.05385, score 0.470
- arXiv:1612.00682, score 0.469
- arXiv:2108.07838, score 0.464
- arXiv:0912.2823, score 0.463

9.16 Quantum chaos

Mechanism-tree role. Generator: lawful change before readout; assignment score 0.76. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum chaos is a broad quantum constructor role in the compact quantum constructor. In this tree, quantum chaos belongs to the lawful-change step: it specifies how the state changes before a question is asked.

Mechanism Reading

Operationally, Quantum chaos contributes a broad quantum constructor role. The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer. In the generators step, the generator is the part of the construction that makes the state move while preserving the admissibility conditions. In ordinary quantum mechanics this is usually a Hamiltonian or unitary map; in path-integral language it is an action weight over histories. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum chaos contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as a lawful-transport move: it identifies what changes the state before readout.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.43, transport flow=0.26, constraint closure=0.19

- arXiv:0912.2823, score 0.570
- arXiv:1801.03283, score 0.563
- arXiv:1612.00682, score 0.562
- arXiv:quant-ph0205159, score 0.562
- arXiv:1506.05598, score 0.557

9.17 Symmetry in quantum mechanics

Mechanism-tree role. Generator: lawful change before readout; assignment score 0.75. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Symmetry in quantum mechanics is a broad quantum constructor role in the compact quantum constructor. In this tree, symmetry in quantum mechanics belongs to the lawful-change step: it specifies how the state changes before a question is asked.

Mechanism Reading

Operationally, Symmetry in quantum mechanics contributes a broad quantum constructor role. The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer. In the generators step, the generator is the part of the construction that makes the state move while preserving the admissibility conditions. In ordinary quantum mechanics this is usually a Hamiltonian or unitary map; in path-integral language it is an action weight over histories. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Symmetry in quantum mechanics contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as a lawful-transport move: it identifies what changes the state before readout.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.40, transport flow=0.23, constraint closure=0.18

- arXiv:1604.06537, score 0.572
- arXiv:1801.03283, score 0.544
- arXiv:0912.2823, score 0.543
- arXiv:1506.05598, score 0.537
- arXiv:1612.00682, score 0.537

9.18 Quantum amplifier

Mechanism-tree role. Generator: lawful change before readout; assignment score 0.75. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum amplifier is an instrument-mediated readout role in the compact quantum constructor. In this tree, quantum amplifier belongs to the lawful-change step: it specifies how the state changes before a question is asked.

Mechanism Reading

Operationally, Quantum amplifier contributes an instrument-mediated readout role. The mechanism is an apparatus-coupled readout: a prepared probe state interacts with a sample or field, the interaction changes phase, momentum, intensity, or counting statistics, and the instrument reconstructs an image, spectrum, trajectory, or estimate. In the generators step, the generator is the part of the construction that makes the state move while preserving the admissibility conditions. In ordinary quantum mechanics this is usually a Hamiltonian or unitary map; in path-integral language it is an action weight over histories. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum amplifier contributes an instrument-mediated readout role to the quantum construction.
- **Placement**. This page is read first as a lawful-transport move: it identifies what changes the state before readout.

- **Carrier or domain**. A probe state, sample state, field mode, detector state, or estimation register.
- **Operator or map**. An interaction Hamiltonian, transfer map, measurement channel, reconstruction map, or estimator.
- **Admissibility**. The instrument must separate sample signal from preparation, detector response, calibration, noise, and reconstruction artifacts.
- **Readout**. Counts, images, spectra, phase shifts, trajectories, intensity maps, correlation data, or parameter estimates.
- **Check**. The claimed mechanism is credible only when the same readout survives control experiments, calibration changes, and reconstruction checks.

Topic Equations

$$\begin{aligned} B &\longmapsto \rho_B \quad \text{(context specifies an admissible state)} \\ \rho_t = U_t \rho_B U_t^\dagger & \quad \text{(unitary evolution from preparation to readout)} \\ O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) & \quad \text{(spectral probability measure)} \\ [O_1, O_2] \neq 0 & \quad \text{(incompatible observables: no common sharp basis)} \end{aligned}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.41, transport flow=0.22, constraint closure=0.13

- arXiv:1604.06537, score 0.581
- arXiv:1604.05385, score 0.548
- arXiv:1612.00682, score 0.535
- arXiv:2108.07838, score 0.530
- arXiv:1506.05598, score 0.527

9.19 Quantum mechanics

Mechanism-tree role. Generator: lawful change before readout; assignment score 0.75. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum mechanics is the baseline constructor: states live in Hilbert space, physical questions are represented by operators, and probabilities are assigned to spectral projectors.

Mechanism Reading

The page supplies the general quantum assembly. A preparation gives a state vector or density operator. A self-adjoint observable or measurement operator family gives the possible readout channels. The Born or trace rule assigns probabilities, while Hamiltonian evolution transports the state between preparation and readout. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum mechanics contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as a lawful-transport move: it identifies what changes the state before readout.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.36, transport flow=0.25, constraint closure=0.19

- arXiv:1604.05385, score 0.531
- arXiv:0912.2823, score 0.526
- arXiv:1604.06537, score 0.524
- arXiv:1612.00682, score 0.499
- arXiv:1801.03283, score 0.497

9.20 Heisenberg picture

Mechanism-tree role. Generator: lawful change before readout; assignment score 0.75. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Heisenberg picture is a broad quantum constructor role in the compact quantum constructor. In this tree, heisenberg picture belongs to the lawful-change step: it specifies how the state changes before a question is asked.

Mechanism Reading

Operationally, Heisenberg picture contributes a broad quantum constructor role. The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer. In the generators step, the generator is the part of the construction that makes the state move while preserving the admissibility conditions. In ordinary quantum mechanics this is usually a Hamiltonian or unitary map; in path-integral language it is an action weight over histories. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Heisenberg picture contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as a lawful-transport move: it identifies what changes the state before readout.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$\begin{aligned}
B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)} \\
\rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)} \\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)} \\
[O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)}
\end{aligned}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.42, transport flow=0.23, constraint closure=0.15

- arXiv:1604.06537, score 0.485
- arXiv:1612.00682, score 0.443
- arXiv:2108.07838, score 0.442
- arXiv:0912.2823, score 0.436
- arXiv:1801.03283, score 0.436

9.21 Heisenberg group

Mechanism-tree role. Generator: lawful change before readout; assignment score 0.74. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Heisenberg group is a broad quantum constructor role in the compact quantum constructor. In this tree, heisenberg group belongs to the lawful-change step: it specifies how the state changes before a question is asked.

Mechanism Reading

Operationally, Heisenberg group contributes a broad quantum constructor role. The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer. In the generators step, the generator is the part of the construction that makes the state move while preserving the admissibility conditions. In ordinary quantum mechanics this is usually a Hamiltonian or unitary map; in path-integral language it is an action weight over histories. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Heisenberg group contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as a lawful-transport move: it identifies what changes the state before readout.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$\begin{aligned} B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)} \\ \rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)} \\ O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)} \\ [O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)} \end{aligned}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.39, transport flow=0.23, constraint closure=0.15

- arXiv:0912.2823, score 0.401
- arXiv:1604.06537, score 0.386
- arXiv:1612.00682, score 0.356
- arXiv:2108.07838, score 0.341
- arXiv:0805.4565, score 0.341

9.22 Canonical commutation relation

Mechanism-tree role. Generator: lawful change before readout; assignment score 0.74. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Canonical commutation relation is a broad quantum constructor role in the compact quantum constructor. In this tree, canonical commutation relation belongs to the lawful-change step: it specifies how the state changes before a question is asked.

Mechanism Reading

Operationally, Canonical commutation relation contributes a broad quantum constructor role. The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer. In the generators step, the generator is the part of the construction that makes the state move while preserving the admissibility conditions. In ordinary quantum mechanics this is usually a Hamiltonian or unitary map; in path-integral language it is an action weight over histories. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, non-commuting compatibility limits; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Canonical commutation relation contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as a lawful-transport move: it identifies what changes the state before readout.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.36, transport flow=0.20, commutator incompatibility=0.15

- arXiv:1604.06537, score 0.574
- arXiv:2111.12617, score 0.563
- arXiv:0912.2823, score 0.544
- arXiv:0908.0752, score 0.532
- arXiv:1506.05598, score 0.526

9.23 Quantum biology

Mechanism-tree role. Generator: lawful change before readout; assignment score 0.73. **Dominant evidence signal.** state evolution. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum biology is an open-system transport and coherence role in the compact quantum constructor. In this tree, quantum biology belongs to the lawful-change step: it specifies how the state changes before a question is asked.

Mechanism Reading

Operationally, Quantum biology contributes an open-system transport and coherence role. The topic concerns quantum state transport under environmental coupling, coherence loss, biological or macroscopic boundary conditions, or effective dynamics outside an ideal closed system. In the generators step, the generator is the part of the construction that makes the state move while preserving the admissibility conditions. In ordinary quantum mechanics this is usually a Hamiltonian or unitary map; in path-integral language it is an action weight over histories. In the source-evidence profile for this page, the strongest construction signal is state evolution, normalization or admissibility, operator-to-spectrum readout; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum biology contributes an open-system transport and coherence role to the quantum construction.
- **Placement**. This page is read first as a lawful-transport move: it identifies what changes the state before readout.

- **Carrier or domain**. A density operator, reduced state, coherence variable, bath-coupled state, or effective mesoscopic carrier.
- **Operator or map**. Hamiltonian plus environmental coupling, Lindbladian, memory kernel, stochastic map, or effective transport operator.
- **Admissibility**. Positivity, trace preservation, timescale separation, bath assumptions, and control over classical noise determine whether the model is legal.
- **Readout**. Coherence, population transfer, relaxation rate, transport efficiency, noise spectrum, or macroscopic response.
- **Check**. The quantum contribution must survive controls against classical noise, preparation artifacts, and coarse-graining choices.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

transport flow=0.28, constraint closure=0.19, spectral operator=0.16

- arXiv:1706.03846, score 0.447
- arXiv:0912.2823, score 0.438
- arXiv:1604.05385, score 0.437
- arXiv:1604.06537, score 0.429
- arXiv:1708.03640, score 0.422

9.24 Quantum engineering

Mechanism-tree role. Generator: lawful change before readout; assignment score 0.72. **Dominant evidence signal.** state evolution. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum engineering is a broad quantum constructor role in the compact quantum constructor. In this tree, quantum engineering belongs to the lawful-change step: it specifies how the state changes before a question is asked.

Mechanism Reading

Operationally, Quantum engineering contributes a broad quantum constructor role. The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer. In the generators step, the generator is the part of the construction that makes the state move while preserving the admissibility conditions. In ordinary quantum mechanics this is usually a Hamiltonian or unitary map; in path-integral language it is an action weight over histories. In the source-evidence profile for this page, the strongest construction signal is state evolution, operator-to-spectrum readout, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum engineering contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as a lawful-transport move: it identifies what changes the state before readout.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.

- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$
\begin{aligned}
B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)} \\
\rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)} \\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)} \\
[O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

transport flow=0.26, spectral operator=0.12, constraint closure=0.05

- arXiv:1706.03846, score 0.527
- arXiv:1708.03640, score 0.504
- arXiv:0805.4565, score 0.500
- arXiv:0908.0752, score 0.496
- arXiv:quant-ph0205159, score 0.478

9.25 Quantum logic

Mechanism-tree role. Generator: lawful change before readout; assignment score 0.56. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum logic is a broad quantum constructor role in the compact quantum constructor. In this tree, quantum logic belongs to the lawful-change step: it specifies how the state changes before a question is asked.

Mechanism Reading

Operationally, Quantum logic contributes a broad quantum constructor role. The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer. In the generators step, the generator is the part of the construction that makes the state move while preserving the admissibility conditions. In ordinary quantum mechanics this is usually a Hamiltonian or unitary map; in path-integral language it is an action weight over histories. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum logic contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as a lawful-transport move: it identifies what changes the state before readout.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.40, transport flow=0.25, constraint closure=0.18

- arXiv:1604.06537, score 0.561
- arXiv:1604.05385, score 0.534
- arXiv:0912.2823, score 0.533
- arXiv:1612.00682, score 0.529
- arXiv:quant-ph0205159, score 0.528

Chapter 10

Spectral question: what can be asked

An observable is a permitted question whose operator form determines the possible numerical answers.

10.1 Why This Branch Exists

The central unit is not a microscopic entity but a legal question. Spectra make the possible answers visible.

10.2 Page Map

Page	Dominant evidence signal	Score
Angular momentum operator	observables and spectra	1.48
Observable	observables and spectra	1.47
Self-adjoint operator	observables and spectra	1.47
Spectral theory	observables and spectra	1.41
Pauli matrices	observables and spectra	1.35
Operator theory	observables and spectra	1.34
Operator (physics)	observables and spectra	1.30
Eigenvalues and eigenvectors	observables and spectra	1.48
Von Neumann algebra	observables and spectra	1.43
Electron microscope	observables and spectra	0.67

10.3 Mechanism Pages

Each page below is read as a specialization of the compact quantum constructor. Topic-specific pages use explicit equations or overrides; core-derived pages use the branch-level equation form associated with their mechanism role.

10.4 Angular momentum operator

Mechanism-tree role. Spectral question: what can be asked; assignment score 1.48. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Angular momentum operator is a broad quantum constructor role in the compact quantum constructor. In this tree, angular momentum operator belongs to the question step: it turns a physical question into an operator with admissible answers.

Mechanism Reading

Operationally, Angular momentum operator contributes a broad quantum constructor role. The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer. In the observables step, the constructor separates the state from the question asked of it. A measurable question is represented by an operator; the allowed answers are exposed by its spectral resolution. This is why the operator/spectrum signal is the spine of the quantum tree. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Angular momentum operator contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as a question-selection move: it identifies the spectrum or answer set being read.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$
\begin{aligned}
B \longmapsto \rho_B \quad & \text{(context specifies an admissible state)} \\
\rho_t = U_t \rho_B U_t^\dagger \quad & \text{(unitary evolution from preparation to readout)} \\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad & \text{(spectral probability measure)} \\
[O_1, O_2] \neq 0 \quad & \text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation

- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.38, transport flow=0.19, constraint closure=0.16

- arXiv:1604.06537, score 0.538
- arXiv:0912.2823, score 0.516
- arXiv:1604.05385, score 0.515
- arXiv:1612.00682, score 0.511
- arXiv:1801.03283, score 0.506

10.5 Observable

Mechanism-tree role. Spectral question: what can be asked; assignment score 1.47. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Observable is the legal-question constructor: it turns a physical question into an operator with spectral outcome channels.

Mechanism Reading

An observable is the mathematical form of a question that can be asked of a state. Its spectral decomposition defines the possible answers. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Observable contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as a question-selection move: it identifies the spectrum or answer set being read.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$
\begin{aligned}
B &\longmapsto \rho_B \quad \text{(context specifies an admissible state)}\\
\rho_t = U_t \rho_B U_t^\dagger &\quad \text{(unitary evolution from preparation to readout)}\\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) &\quad \text{(spectral probability measure)}\\
[O_1, O_2] \neq 0 &\quad \text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.40, transport flow=0.23, constraint closure=0.16

- arXiv:1604.05385, score 0.567
- arXiv:1604.06537, score 0.561
- arXiv:1612.00682, score 0.522
- arXiv:1801.03283, score 0.521
- arXiv:quant-ph0205159, score 0.519

10.6 Self-adjoint operator

Mechanism-tree role. Spectral question: what can be asked; assignment score 1.47. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Self-adjoint operator is the admissible-observable condition: it gives real spectra and well-defined spectral measures.

Mechanism Reading

Self-adjointness is not a technical decoration. It is the condition that makes an operator a legitimate spectral question in ordinary quantum mechanics. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Self-adjoint operator contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as a question-selection move: it identifies the spectrum or answer set being read.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.

- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.43, transport flow=0.23, constraint closure=0.17

- arXiv:0912.2823, score 0.569
- arXiv:1612.00682, score 0.553
- arXiv:1801.03283, score 0.550
- arXiv:quant-ph0205159, score 0.546
- arXiv:1506.05598, score 0.541

10.7 Spectral theory

Mechanism-tree role. Spectral question: what can be asked; assignment score 1.41. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Spectral theory is a broad quantum constructor role in the compact quantum constructor. In this tree, spectral theory belongs to the question step: it turns a physical question into an operator with admissible answers.

Mechanism Reading

Operationally, Spectral theory contributes a broad quantum constructor role. The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer. In the observables step, the constructor separates the state from the question asked of it. A measurable question is represented by an operator; the allowed answers are exposed by its spectral resolution. This is why the operator/spectrum signal is the spine of the quantum tree. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Spectral theory contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as a question-selection move: it identifies the spectrum or answer set being read.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.44, transport flow=0.22, constraint closure=0.12

- arXiv:2108.07838, score 0.501
- arXiv:1612.00682, score 0.497
- arXiv:0805.4565, score 0.492
- arXiv:quant-ph0205159, score 0.480
- arXiv:1506.05598, score 0.472

10.8 Pauli matrices

Mechanism-tree role. Spectral question: what can be asked; assignment score 1.35. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Pauli matrices is a broad quantum constructor role in the compact quantum constructor. In this tree, pauli matrices belongs to the question step: it turns a physical question into an operator with admissible answers.

Mechanism Reading

Operationally, Pauli matrices contributes a broad quantum constructor role. The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer. In the observables step, the constructor separates the state from the question asked of it. A measurable question is represented by an operator; the allowed answers are exposed by its spectral resolution. This is why the operator/spectrum signal is the spine of the quantum tree. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Pauli matrices contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as a question-selection move: it identifies the spectrum or answer set being read.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$

$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$

$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$

$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.42, transport flow=0.22, constraint closure=0.11

- arXiv:0912.2823, score 0.564
- arXiv:2108.07838, score 0.551
- arXiv:1612.00682, score 0.550
- arXiv:0805.4565, score 0.547
- arXiv:1506.05598, score 0.537

10.9 Operator theory

Mechanism-tree role. Spectral question: what can be asked; assignment score 1.34. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Operator theory is a broad quantum constructor role in the compact quantum constructor. In this tree, operator theory belongs to the question step: it turns a physical question into an operator with admissible answers.

Mechanism Reading

Operationally, Operator theory contributes a broad quantum constructor role. The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer. In the observables step, the constructor separates the state from the question asked of it. A measurable question is represented by an operator; the allowed answers are exposed by its spectral resolution. This is why the operator/spectrum signal is the spine of the quantum tree. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Operator theory contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as a question-selection move: it identifies the spectrum or answer set being read.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$
\begin{aligned}
B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)}\\
\rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)}\\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)}\\
[O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.43, transport flow=0.24, constraint closure=0.18

- arXiv:0912.2823, score 0.550
- arXiv:1604.05385, score 0.549
- arXiv:1612.00682, score 0.541
- arXiv:1506.05598, score 0.534
- arXiv:quant-ph0205159, score 0.533

10.10 Operator (physics)

Mechanism-tree role. Spectral question: what can be asked; assignment score 1.30. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Operator (physics) is a broad quantum constructor role in the compact quantum constructor. In this tree, operator (physics) belongs to the question step: it turns a physical question into an operator with admissible answers.

Mechanism Reading

Operationally, Operator (physics) contributes a broad quantum constructor role. The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer. In the observables step, the constructor separates the state from the question asked of it. A measurable question is represented by an operator; the allowed answers are exposed by its spectral resolution. This is why the operator/spectrum signal is the spine of the quantum tree. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Operator (physics) contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as a question-selection move: it identifies the spectrum or answer set being read.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.37, transport flow=0.26, constraint closure=0.19

- arXiv:1604.05385, score 0.570
- arXiv:1801.03283, score 0.543
- arXiv:0912.2823, score 0.539
- arXiv:1612.00682, score 0.536
- arXiv:1506.05598, score 0.535

10.11 Eigenvalues and eigenvectors

Mechanism-tree role. Spectral question: what can be asked; assignment score 1.48. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Eigenvalues and eigenvectors is a broad quantum constructor role in the compact quantum constructor. In this tree, eigenvalues and eigenvectors belongs to the question step: it turns a physical question into an operator with admissible answers.

Mechanism Reading

Operationally, Eigenvalues and eigenvectors contributes a broad quantum constructor role. The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer. In the observables step, the constructor separates the state from the question asked of it. A measurable question is represented by an operator; the allowed answers are exposed by its spectral resolution. This is why the operator/spectrum signal is the spine of the quantum tree. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, preparation, basis, or boundary context; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Eigenvalues and eigenvectors contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as a question-selection move: it identifies the spectrum or answer set being read.

- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$
\begin{aligned}
B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)} \\
\rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)} \\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)} \\
[O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.38, transport flow=0.16, boundary weak form=0.14

- arXiv:0912.2823, score 0.461
- arXiv:1604.05385, score 0.422
- arXiv:1612.00682, score 0.414
- arXiv:2308.15676, score 0.400
- arXiv:2108.07838, score 0.393

10.12 Von Neumann algebra

Mechanism-tree role. Spectral question: what can be asked; assignment score 1.43. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Von Neumann algebra is a broad quantum constructor role in the compact quantum constructor. In this tree, von Neumann algebra belongs to the question step: it turns a physical question into an operator with admissible answers.

Mechanism Reading

Operationally, Von Neumann algebra contributes a broad quantum constructor role. The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer. In the observables step, the constructor separates the state from the question asked of it. A measurable question is represented by an operator; the allowed answers are exposed by its spectral resolution. This is why the operator/spectrum signal is the spine of the quantum tree. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Von Neumann algebra contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as a question-selection move: it identifies the spectrum or answer set being read.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.

- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.42, transport flow=0.25, constraint closure=0.18

- arXiv:0912.2823, score 0.437
- arXiv:1604.05385, score 0.434
- arXiv:1604.06537, score 0.426
- arXiv:1612.00682, score 0.422
- arXiv:quant-ph0205159, score 0.406

10.13 Electron microscope

Mechanism-tree role. Spectral question: what can be asked; assignment score 0.67. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Electron microscope is an instrument-mediated readout role in the compact quantum constructor. In this tree, electron microscope belongs to the question step: it turns a physical question into an operator with admissible answers.

Mechanism Reading

Operationally, Electron microscope contributes an instrument-mediated readout role. The mechanism is an apparatus-coupled readout: a prepared probe state interacts with a sample or field, the interaction changes phase, momentum, intensity, or counting statistics, and the instrument reconstructs an image, spectrum, trajectory, or estimate. In the observables step, the constructor separates the state from the question asked of it. A measurable question is represented by an operator; the allowed answers are exposed by its spectral resolution. This is why the operator/spectrum signal is the spine of the quantum tree. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Electron microscope contributes an instrument-mediated readout role to the quantum construction.
- **Placement**. This page is read first as a question-selection move: it identifies the spectrum or answer set being read.
- **Carrier or domain**. A probe state, sample state, field mode, detector state, or estimation register.
- **Operator or map**. An interaction Hamiltonian, transfer map, measurement channel, reconstruction map, or estimator.
- **Admissibility**. The instrument must separate sample signal from preparation, detector response, calibration, noise, and reconstruction artifacts.
- **Readout**. Counts, images, spectra, phase shifts, trajectories, intensity maps, correlation data, or parameter estimates.
- **Check**. The claimed mechanism is credible only when the same readout survives control experiments, calibration changes, and reconstruction checks.

Core-Derived Role Equations

This equation block is the branch-level quantum mechanism attached to the page by the compact constructor. It should be read as the role equation to check against the page's source evidence, not as a claim that every source writes the formula in this notation.

$$\rho_{\text{probe}} \mapsto \mathcal{E}_{\text{sample}}(\rho_{\text{probe}}), \quad p(y) = \mathrm{Tr}(M_y \mathcal{E}_{\text{sample}}(\rho_{\text{probe}})), \quad \hat{s} = R(\{y_i\})$$

What Remains Stable

- The mechanism is an apparatus-coupled readout: a prepared probe state interacts with a sample or field, the interaction changes phase, momentum, intensity, or counting statistics, and the instrument reconstructs an image, spectrum, trajectory, or estimate.
- Electron microscope defines the legal question being asked of the state.
- The measurable answers are encoded by the operator spectrum, projectors, or spectral measure.
- The operator role is preserved across equivalent bases even when matrix entries change.

What Changes With Realization

- The local title, representation, and physical realization may change while the constructor role is preserved.
- The same observable may be represented by matrices, differential operators, projectors, or algebraic elements.
- Degeneracy, basis choice, and domain conditions can change how the spectrum is displayed.
- Detector implementation changes the physical realization, not the operator role itself.

Validation Boundary

- Use this role by specifying the page's state carrier, operator or map, readout, and compatibility condition in its own quantum language.
- Self-adjointness, or the appropriate POVM positivity condition, is what makes the question a legal readout.
- A complete spectral resolution supplies all outcome channels for the question being asked.

Evidence Profile

spectral operator=0.42, transport flow=0.20, constraint closure=0.10

- arXiv:0912.2823, score 0.567
- arXiv:1612.00682, score 0.554
- arXiv:0805.4565, score 0.553
- arXiv:2108.07838, score 0.552
- arXiv:quant-ph0205159, score 0.543

Chapter 11

Readout rule: how answers become probabilities

Measurement connects a state and an observable to recorded frequencies. Projection, POVMs, Born weights, and collapse language are alternative ways of presenting this state-to-spectrum readout step.

11.1 Why This Branch Exists

Measurement is best placed after observables and spectra: it is the rule that turns spectral resolution into recorded probability, not the mystical starting point of the theory.

11.2 Page Map

Page	Dominant evidence signal	Score
POVM	observables and spectra	1.47
Wave function collapse	observables and spectra	1.45
Born rule	observables and spectra	1.45
Measurement in quantum mechanics	observables and spectra	1.39
Quantum jump	observables and spectra	1.32
Measurement problem	state evolution	1.28
Quantum eraser experiment	observables and spectra	1.23
Projection-valued measure	observables and spectra	1.19
Delayed-choice quantum eraser	observables and spectra	1.06
Quantum metrology	observables and spectra	0.88

11.3 Mechanism Pages

Each page below is read as a specialization of the compact quantum constructor. Topic-specific pages use explicit equations or overrides; core-derived pages use the branch-level equation form associated with their mechanism role.

11.4 POVM

Mechanism-tree role. Readout rule: how answers become probabilities; assignment score 1.47. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

POVM is the generalized-readout constructor: outcome effects need not be orthogonal projectors.

Mechanism Reading

POVMs separate the probability readout from the idealized projection assumption. They are the natural mechanism for noisy, coarse-grained, indirect, or open-system measurements. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. POVM contributes a probability/readout role to the quantum construction.
- **Placement**. This page is read first as a readout move: it connects the state and question to recorded outcomes.
- **Carrier or domain**. A state vector or density operator together with the measurement context in which outcome channels are defined.
- **Operator or map**. A projection-valued measure, POVM, update map, or instrument map connecting state to record.
- **Admissibility**. Outcome probabilities must be positive, normalized, and tied to a specified readout map rather than to informal observer language.
- **Readout**. Born probabilities, detector records, post-measurement states, ensemble frequencies, or decision probabilities.
- **Check**. The interpretation is constrained by whether it changes the probability rule, the update rule, the detector model, or only the language used for them.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.41, transport flow=0.18, constraint closure=0.16

- arXiv:1604.05385, score 0.574
- arXiv:1801.03283, score 0.552
- arXiv:1612.00682, score 0.546
- arXiv:1506.05598, score 0.533
- arXiv:quant-ph0205159, score 0.531

11.5 Wave function collapse

Mechanism-tree role. Readout rule: how answers become probabilities; assignment score 1.45. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Wave function collapse is a probability/readout role in the compact quantum constructor. In this tree, wave function collapse belongs to the readout step: it connects a prepared state and an operator spectrum to probabilities or state updates.

Mechanism Reading

Operationally, Wave function collapse contributes a probability/readout role. The topic modifies how a state is connected to recorded outcomes. The stable machinery is the spectral measure or POVM together with the probability rule. In the measurement step, the constructor reads probabilities from the pair consisting of a state and a spectral question. Interpretive pages in this branch assign meaning to probability, state, or update while preserving the formal readout rule. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Wave function collapse contributes a probability/readout role to the quantum construction.
- **Placement**. This page is read first as a readout move: it connects the state and question to recorded outcomes.
- **Carrier or domain**. A state vector or density operator together with the measurement context in which outcome channels are defined.
- **Operator or map**. A projection-valued measure, POVM, update map, or instrument map connecting state to record.
- **Admissibility**. Outcome probabilities must be positive, normalized, and tied to a specified readout map rather than to informal observer language.
- **Readout**. Born probabilities, detector records, post-measurement states, ensemble frequencies, or decision probabilities.
- **Check**. The interpretation is constrained by whether it changes the probability rule, the update rule, the detector model, or only the language used for them.

Topic Equations

$$
\begin{aligned}
B \longmapsto \rho_B \quad & \text{(context specifies an admissible state)} \\
\rho_t = U_t \rho_B U_t^\dagger \quad & \text{(unitary evolution from preparation to readout)} \\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad & \text{(spectral probability measure)} \\
[O_1, O_2] \neq 0 \quad & \text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.35, transport flow=0.25, constraint closure=0.18

- arXiv:1604.05385, score 0.535
- arXiv:0912.2823, score 0.525
- arXiv:1604.06537, score 0.524
- arXiv:1612.00682, score 0.494
- arXiv:1801.03283, score 0.493

11.6 Born rule

Mechanism-tree role. Readout rule: how answers become probabilities; assignment score 1.45. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Born rule is the probability-readout constructor: it maps a state and a spectral channel to an observed probability.

Mechanism Reading

The Born rule is the point where the constructor becomes predictive. It does not name an object; it connects state preparation and a legal question to frequencies over outcome channels. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Born rule contributes a probability/readout role to the quantum construction.
- **Placement**. This page is read first as a readout move: it connects the state and question to recorded outcomes.
- **Carrier or domain**. A state vector or density operator together with the measurement context in which outcome channels are defined.
- **Operator or map**. A projection-valued measure, POVM, update map, or instrument map connecting state to record.
- **Admissibility**. Outcome probabilities must be positive, normalized, and tied to a specified readout map rather than to informal observer language.
- **Readout**. Born probabilities, detector records, post-measurement states, ensemble frequencies, or decision probabilities.
- **Check**. The interpretation is constrained by whether it changes the probability rule, the update rule, the detector model, or only the language used for them.

Topic Equations

$$
\begin{aligned}
B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)} \\
\rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)} \\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)} \\
[O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.39, transport flow=0.27, constraint closure=0.20

- arXiv:1604.05385, score 0.568
- arXiv:1604.06537, score 0.539
- arXiv:1801.03283, score 0.537
- arXiv:1612.00682, score 0.530
- arXiv:0912.2823, score 0.527

11.7 Measurement in quantum mechanics

Mechanism-tree role. Readout rule: how answers become probabilities; assignment score 1.39. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Measurement in quantum mechanics is the complete readout junction: it combines a state, a measurement model, probabilities, and sometimes an update rule.

Mechanism Reading

Measurement is not the root of quantum theory in this book. It is the junction where a prepared state and an observable or POVM are converted into probabilities and recorded outcomes. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Measurement in quantum mechanics contributes a probability/readout role to the quantum construction.
- **Placement**. This page is read first as a readout move: it connects the state and question to recorded outcomes.
- **Carrier or domain**. A state vector or density operator together with the measurement context in which outcome channels are defined.
- **Operator or map**. A projection-valued measure, POVM, update map, or instrument map connecting state to record.

- **Admissibility**. Outcome probabilities must be positive, normalized, and tied to a specified readout map rather than to informal observer language.
- **Readout**. Born probabilities, detector records, post-measurement states, ensemble frequencies, or decision probabilities.
- **Check**. The interpretation is constrained by whether it changes the probability rule, the update rule, the detector model, or only the language used for them.

Topic Equations

$$
\begin{aligned}
B \longmapsto \rho_B \quad & \text{(context specifies an admissible state)} \\
\rho_t = U_t \rho_B U_t^\dagger \quad & \text{(unitary evolution from preparation to readout)} \\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad & \text{(spectral probability measure)} \\
[O_1, O_2] \neq 0 \quad & \text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.40, transport flow=0.23, constraint closure=0.16

- arXiv:1604.05385, score 0.579
- arXiv:1604.06537, score 0.551
- arXiv:0912.2823, score 0.544
- arXiv:1612.00682, score 0.535
- arXiv:quant-ph0205159, score 0.535

11.8 Quantum jump

Mechanism-tree role. Readout rule: how answers become probabilities; assignment score 1.32. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum jump is a broad quantum constructor role in the compact quantum constructor. In this tree, quantum jump belongs to the readout step: it connects a prepared state and an operator spectrum to probabilities or state updates.

Mechanism Reading

Operationally, Quantum jump contributes a broad quantum constructor role. The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer. In the measurement step, the constructor reads probabilities from the pair consisting of a state and a spectral question. Interpretive pages in this branch assign meaning to probability, state, or update while preserving the formal readout rule. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum jump contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as a readout move: it connects the state and question to recorded outcomes.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$
\begin{aligned}
B \longmapsto \rho_B \quad & \text{(context specifies an admissible state)} \\
\rho_t = U_t \rho_B U_t^\dagger \quad & \text{(unitary evolution from preparation to readout)} \\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad & \text{(spectral probability measure)} \\
[O_1, O_2] \neq 0 \quad & \text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.43, transport flow=0.24, constraint closure=0.13

- arXiv:1604.05385, score 0.511
- arXiv:2108.07838, score 0.508
- arXiv:1612.00682, score 0.497
- arXiv:0805.4565, score 0.494
- arXiv:quant-ph0205159, score 0.481

11.9 Measurement problem

Mechanism-tree role. Readout rule: how answers become probabilities; assignment score 1.28. **Dominant evidence signal.** state evolution. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Measurement problem is a probability/readout role in the compact quantum constructor. In this tree, measurement problem belongs to the readout step: it connects a prepared state and an operator spectrum to probabilities or state updates.

Mechanism Reading

Operationally, Measurement problem contributes a probability/readout role. The topic modifies how a state is connected to recorded outcomes. The stable machinery is the spectral measure or POVM together with the probability rule. In the measurement step, the constructor reads probabilities from the pair consisting of a state and a spectral question. Interpretive pages in this branch assign meaning to probability, state, or update while preserving the formal readout rule. In the source-evidence profile for this page, the strongest construction signal is state evolution, preparation, basis, or boundary context, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Measurement problem contributes a probability/readout role to the quantum construction.
- **Placement**. This page is read first as a readout move: it connects the state and question to recorded outcomes.
- **Carrier or domain**. A state vector or density operator together with the measurement context in which outcome channels are defined.
- **Operator or map**. A projection-valued measure, POVM, update map, or instrument map connecting state to record.
- **Admissibility**. Outcome probabilities must be positive, normalized, and tied to a specified readout map rather than to informal observer language.
- **Readout**. Born probabilities, detector records, post-measurement states, ensemble frequencies, or decision probabilities.
- **Check**. The interpretation is constrained by whether it changes the probability rule, the update rule, the detector model, or only the language used for them.

Topic Equations

$$
\begin{aligned}
B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)} \\
\rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)} \\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)} \\
[O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

transport flow=0.27, boundary weak form=0.20, constraint closure=0.20

- arXiv:1604.05385, score 0.470
- arXiv:0912.2823, score 0.467
- arXiv:1706.03846, score 0.459
- arXiv:2501.07524, score 0.443
- arXiv:1604.06537, score 0.438

11.10 Quantum eraser experiment

Mechanism-tree role. Readout rule: how answers become probabilities; assignment score 1.23. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum eraser experiment is a compatibility or joint-readout role in the compact quantum constructor. In this tree, quantum eraser experiment belongs to the readout step: it connects a prepared state and an operator spectrum to probabilities or state updates.

Mechanism Reading

Operationally, Quantum eraser experiment contributes a compatibility or joint-readout role. The topic tests whether separately legal questions can be resolved together. The mechanism is a restriction on joint spectra, correlations, or admissible hidden-variable assignments. In the measurement step, the constructor reads probabilities from the pair consisting of a state and a spectral question. Interpretive pages in this branch assign meaning to probability, state, or update while preserving the formal readout rule. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, preparation, basis, or boundary context, state evolution; the strongest carrier signal is local notation, information profile, spectral profile.

Quantum Mechanism Frame

- **Role**. Quantum eraser experiment contributes a compatibility or joint-readout role to the quantum construction.
- **Placement**. This page is read first as a readout move: it connects the state and question to recorded outcomes.
- **Carrier or domain**. One state space or a multipartite state space on which several questions can be asked.
- **Operator or map**. Two or more observables, contexts, correlation operators, or hidden-variable assignments being compared.
- **Admissibility**. Commutators, uncertainty bounds, contextuality constraints, or Bell-type inequalities decide which joint assignments are possible.
- **Readout**. Joint spectra, correlations, inequality violations, uncertainty products, or incompatible outcome statistics.
- **Check**. The non-classical content appears only if the incompatible questions cannot be replaced by one common sharp classical assignment.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.32, boundary weak form=0.16, transport flow=0.14

- arXiv:1604.05385, score 0.485
- arXiv:0912.2823, score 0.484
- arXiv:gr-qc0303063, score 0.467
- arXiv:1604.06537, score 0.445
- arXiv:2110.09771, score 0.436

11.11 Projection-valued measure

Mechanism-tree role. Readout rule: how answers become probabilities; assignment score 1.19. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Projection-valued measure is the sharp-readout constructor: mutually exclusive outcome projectors partition the identity.

Mechanism Reading

A projection-valued measure encodes an ideal sharp measurement. It defines outcome channels that are orthogonal and exhaustive. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Projection-valued measure contributes a probability/readout role to the quantum construction.
- **Placement**. This page is read first as a readout move: it connects the state and question to recorded outcomes.
- **Carrier or domain**. A state vector or density operator together with the measurement context in which outcome channels are defined.
- **Operator or map**. A projection-valued measure, POVM, update map, or instrument map connecting state to record.
- **Admissibility**. Outcome probabilities must be positive, normalized, and tied to a specified readout map rather than to informal observer language.
- **Readout**. Born probabilities, detector records, post-measurement states, ensemble frequencies, or decision probabilities.
- **Check**. The interpretation is constrained by whether it changes the probability rule, the update rule, the detector model, or only the language used for them.

Topic Equations

$$\begin{aligned} B &\longmapsto \rho_B \quad \text{(context specifies an admissible state)} \\ \rho_t &= U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)} \\ O &= \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)} \\ [O_1, O_2] &\neq 0 \quad \text{(incompatible observables: no common sharp basis)} \end{aligned}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.41, transport flow=0.18, constraint closure=0.16

- arXiv:1604.05385, score 0.533
- arXiv:1612.00682, score 0.519
- arXiv:1801.03283, score 0.509
- arXiv:1506.05598, score 0.496
- arXiv:quant-ph0205159, score 0.495

11.12 Delayed-choice quantum eraser

Mechanism-tree role. Readout rule: how answers become probabilities; assignment score 1.06. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Delayed-choice quantum eraser is a compatibility or joint-readout role in the compact quantum constructor. In this tree, delayed-choice quantum eraser belongs to the readout step: it connects a prepared state and an operator spectrum to probabilities or state updates.

Mechanism Reading

Operationally, Delayed-choice quantum eraser contributes a compatibility or joint-readout role. The topic tests whether separately legal questions can be resolved together. The mechanism is a restriction on joint spectra, correlations, or admissible hidden-variable assignments. In the measurement step, the constructor reads probabilities from the pair consisting of a state and a spectral question. Interpretive pages in this branch assign meaning to probability, state, or update while preserving the formal readout rule. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Delayed-choice quantum eraser contributes a compatibility or joint-readout role to the quantum construction.
- **Placement**. This page is read first as a readout move: it connects the state and question to recorded outcomes.

- **Carrier or domain**. One state space or a multipartite state space on which several questions can be asked.
- **Operator or map**. Two or more observables, contexts, correlation operators, or hidden-variable assignments being compared.
- **Admissibility**. Commutators, uncertainty bounds, contextuality constraints, or Bell-type inequalities decide which joint assignments are possible.
- **Readout**. Joint spectra, correlations, inequality violations, uncertainty products, or incompatible outcome statistics.
- **Check**. The non-classical content appears only if the incompatible questions cannot be replaced by one common sharp classical assignment.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.29, transport flow=0.23, constraint closure=0.18

- arXiv:1604.05385, score 0.492
- arXiv:0912.2823, score 0.492
- arXiv:1604.06537, score 0.480
- arXiv:quant-ph0205159, score 0.445
- arXiv:1708.03640, score 0.444

11.13 Quantum metrology

Mechanism-tree role. Readout rule: how answers become probabilities; assignment score 0.88. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum metrology is an instrument-mediated readout role in the compact quantum constructor. In this tree, quantum metrology belongs to the readout step: it connects a prepared state and an operator spectrum to probabilities or state updates.

Mechanism Reading

Operationally, Quantum metrology contributes an instrument-mediated readout role. The mechanism is an apparatus-coupled readout: a prepared probe state interacts with a sample or field, the interaction changes phase, momentum, intensity, or counting statistics, and the instrument reconstructs an image, spectrum, trajectory, or estimate. In the measurement step, the constructor reads probabilities from the pair consisting of a state and a spectral question. Interpretive pages in this branch assign meaning to probability, state, or update while preserving the formal readout rule. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum metrology contributes an instrument-mediated readout role to the quantum construction.
- **Placement**. This page is read first as a readout move: it connects the state and question to recorded outcomes.
- **Carrier or domain**. A probe state, sample state, field mode, detector state, or estimation register.
- **Operator or map**. An interaction Hamiltonian, transfer map, measurement channel, reconstruction map, or estimator.
- **Admissibility**. The instrument must separate sample signal from preparation, detector response, calibration, noise, and reconstruction artifacts.

- **Readout**. Counts, images, spectra, phase shifts, trajectories, intensity maps, correlation data, or parameter estimates.
- **Check**. The claimed mechanism is credible only when the same readout survives control experiments, calibration changes, and reconstruction checks.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.42, transport flow=0.22, constraint closure=0.14

- arXiv:1604.06537, score 0.556
- arXiv:1604.05385, score 0.548
- arXiv:0912.2823, score 0.547

- arXiv:1612.00682, score 0.538
- arXiv:1506.05598, score 0.536

Chapter 12

Compatibility limit: what cannot be jointly sharp

The non-classical part of the theory appears when two otherwise legal questions do not compose into one common sharp question. Commutators, uncertainty relations, contextuality, Bell tests, and entanglement live here.

12.1 Why This Branch Exists

The non-classical core appears as failure of joint sharpness. Entanglement, Bell phenomena, and uncertainty are different faces of this compatibility structure.

12.2 Page Map

Page	Dominant evidence signal	Score
Bell's theorem	observables and spectra	1.42
Quantum entanglement	state evolution	1.38
Commutator	observables and spectra	1.33
Uncertainty principle	observables and spectra	1.32
Einstein–Podolsky–Rosen paradox	state evolution	1.13
Quantum nonlocality	observables and spectra	1.10

12.3 Mechanism Pages

Each page below is read as a specialization of the compact quantum constructor. Topic-specific pages use explicit equations or overrides; core-derived pages use the branch-level equation form associated with their mechanism role.

12.4 Bell's theorem

Mechanism-tree role. Compatibility limit: what cannot be jointly sharp; assignment score 1.42. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Bell's theorem is a compatibility/locality stress test: quantum correlations violate bounds satisfied by local hidden-variable assignments.

Mechanism Reading

Bell's theorem is not a page about a mysterious object. It is a falsifier for a classical joint-assignment model of measurement outcomes. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, spectral profile.

Quantum Mechanism Frame

- **Role**. Bell's theorem contributes a compatibility or joint-readout role to the quantum construction.
- **Placement**. This page is read first as a compatibility move: it asks which otherwise legal questions cannot share one sharp answer set.
- **Carrier or domain**. One state space or a multipartite state space on which several questions can be asked.
- **Operator or map**. Two or more observables, contexts, correlation operators, or hidden-variable assignments being compared.
- **Admissibility**. Commutators, uncertainty bounds, contextuality constraints, or Bell-type inequalities decide which joint assignments are possible.
- **Readout**. Joint spectra, correlations, inequality violations, uncertainty products, or incompatible outcome statistics.
- **Check**. The non-classical content appears only if the incompatible questions cannot be replaced by one common sharp classical assignment.

Topic Equations

$$
\begin{aligned}
B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)} \\
\rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)} \\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)} \\
[O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.30, transport flow=0.24, constraint closure=0.18

- arXiv:1604.05385, score 0.511
- arXiv:1604.06537, score 0.472
- arXiv:0912.2823, score 0.466
- arXiv:1801.03283, score 0.445
- arXiv:quant-ph0205159, score 0.443

12.5 Quantum entanglement

Mechanism-tree role. Compatibility limit: what cannot be jointly sharp; assignment score 1.38. **Dominant evidence signal.** state evolution. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum entanglement is the non-factorization constructor: a composite state can carry correlations not reducible to independent subsystem states.

Mechanism Reading

Entanglement belongs to composition and compatibility. It appears when the tensor-product state cannot be written as a product or mixture of local states, producing correlations that stress classical separability. In the source-evidence profile for this page, the strongest construction signal is state evolution, operator-to-spectrum readout, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum entanglement contributes a compatibility or joint-readout role to the quantum construction.
- **Placement**. This page is read first as a compatibility move: it asks which otherwise legal questions cannot share one sharp answer set.
- **Carrier or domain**. One state space or a multipartite state space on which several questions can be asked.
- **Operator or map**. Two or more observables, contexts, correlation operators, or hidden-variable assignments being compared.
- **Admissibility**. Commutators, uncertainty bounds, contextuality constraints, or Bell-type inequalities decide which joint assignments are possible.
- **Readout**. Joint spectra, correlations, inequality violations, uncertainty products, or incompatible outcome statistics.
- **Check**. The non-classical content appears only if the incompatible questions cannot be replaced by one common sharp classical assignment.

Topic Equations

$$\begin{aligned} B &\longmapsto \rho_B \quad \text{(context specifies an admissible state)} \\ \rho_t &= U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)} \\ O &= \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)} \\ [O_1, O_2] &\neq 0 \quad \text{(incompatible observables: no common sharp basis)} \end{aligned}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

transport flow=0.27, spectral operator=0.22, constraint closure=0.21

- arXiv:1604.05385, score 0.477
- arXiv:0912.2823, score 0.476
- arXiv:1604.06537, score 0.443
- arXiv:1801.03283, score 0.432
- arXiv:quant-ph0205159, score 0.427

12.6 Commutator

Mechanism-tree role. Compatibility limit: what cannot be jointly sharp; assignment score 1.33. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Commutator is the incompatibility constructor: it measures the failure of two transformations or questions to be freely exchanged.

Mechanism Reading

The commutator is the algebraic source of many non-classical restrictions. If two observables do not commute, they generally cannot be resolved in one common sharp basis. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, non-commuting compatibility limits; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Commutator contributes a compatibility or joint-readout role to the quantum construction.
- **Placement**. This page is read first as a compatibility move: it asks which otherwise legal questions cannot share one sharp answer set.
- **Carrier or domain**. One state space or a multipartite state space on which several questions can be asked.
- **Operator or map**. Two or more observables, contexts, correlation operators, or hidden-variable assignments being compared.

- **Admissibility**. Commutators, uncertainty bounds, contextuality constraints, or Bell-type inequalities decide which joint assignments are possible.
- **Readout**. Joint spectra, correlations, inequality violations, uncertainty products, or incompatible outcome statistics.
- **Check**. The non-classical content appears only if the incompatible questions cannot be replaced by one common sharp classical assignment.

Topic Equations

$$
\begin{aligned}
B &\longmapsto \rho_B \quad \text{(context specifies an admissible state)} \\
\rho_t = U_t \rho_B U_t^\dagger &\quad \text{(unitary evolution from preparation to readout)} \\
O = \sum_i \lambda_i P_i, &\quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)} \\
[O_1, O_2] \neq 0 &\quad \text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.41, transport flow=0.19, commutator incompatibility=0.13

- arXiv:1604.06537, score 0.550
- arXiv:2111.12617, score 0.543
- arXiv:1612.00682, score 0.525
- arXiv:2108.07838, score 0.519
- arXiv:0805.4565, score 0.516

12.7 Uncertainty principle

Mechanism-tree role. Compatibility limit: what cannot be jointly sharp; assignment score 1.32. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Uncertainty principle is a compatibility-limit theorem: non-commuting observables impose lower bounds on joint sharpness.

Mechanism Reading

Uncertainty is not detector imperfection. It is a structural consequence of state variance and non-commuting observables. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Uncertainty principle contributes a compatibility or joint-readout role to the quantum construction.
- **Placement**. This page is read first as a compatibility move: it asks which otherwise legal questions cannot share one sharp answer set.
- **Carrier or domain**. One state space or a multipartite state space on which several questions can be asked.
- **Operator or map**. Two or more observables, contexts, correlation operators, or hidden-variable assignments being compared.
- **Admissibility**. Commutators, uncertainty bounds, contextuality constraints, or Bell-type inequalities decide which joint assignments are possible.
- **Readout**. Joint spectra, correlations, inequality violations, uncertainty products, or incompatible outcome statistics.
- **Check**. The non-classical content appears only if the incompatible questions cannot be replaced by one common sharp classical assignment.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.40, transport flow=0.18, constraint closure=0.13

- arXiv:1604.05385, score 0.539
- arXiv:1604.06537, score 0.525
- arXiv:2111.12617, score 0.504
- arXiv:0912.2823, score 0.502
- arXiv:1612.00682, score 0.499

12.8 Einstein–Podolsky–Rosen paradox

Mechanism-tree role. Compatibility limit: what cannot be jointly sharp; assignment score 1.13. **Dominant evidence signal.** state evolution. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Einstein–Podolsky–Rosen paradox is a broad quantum constructor role in the compact quantum constructor. In this tree, einstein–Podolsky–Rosen paradox belongs to the compatibility step: it marks when two valid questions cannot be jointly sharpened in one basis.

Mechanism Reading

Operationally, Einstein–Podolsky–Rosen paradox contributes a broad quantum constructor role. The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer. In the incompatibility step, in this role the constructor describes an algebraic obstruction. If two operators fail to commute, the same state cannot generally supply one common sharp spectral decomposition for both. In the source-evidence profile for this page, the strongest construction signal is state evolution, normalization or admissibility, non-commuting compatibility limits; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Einstein–Podolsky–Rosen paradox contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as a compatibility move: it asks which otherwise legal questions cannot share one sharp answer set.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$

$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$

$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$

$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

transport flow=0.27, constraint closure=0.21, commutator incompatibility=0.14

- arXiv:1604.05385, score 0.449
- arXiv:0912.2823, score 0.447
- arXiv:2501.07524, score 0.440
- arXiv:1706.03846, score 0.439
- arXiv:1604.06537, score 0.437

12.9 Quantum nonlocality

Mechanism-tree role. Compatibility limit: what cannot be jointly sharp; assignment score 1.10. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum nonlocality is a compatibility or joint-readout role in the compact quantum constructor. In this tree, quantum nonlocality belongs to the compatibility step: it marks when two valid questions cannot be jointly sharpened in one basis.

Mechanism Reading

Operationally, Quantum nonlocality contributes a compatibility or joint-readout role. The topic tests whether separately legal questions can be resolved together. The mechanism is a restriction on joint spectra, correlations, or admissible hidden-variable assignments. In the incompatibility step, in this role the constructor describes an algebraic obstruction. If two operators fail to commute, the same state cannot generally supply one common sharp spectral decomposition for both. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, controlled update protocol; the strongest carrier signal is local notation, information profile, spectral profile.

Quantum Mechanism Frame

- **Role**. Quantum nonlocality contributes a compatibility or joint-readout role to the quantum construction.
- **Placement**. This page is read first as a compatibility move: it asks which otherwise legal questions cannot share one sharp answer set.
- **Carrier or domain**. One state space or a multipartite state space on which several questions can be asked.
- **Operator or map**. Two or more observables, contexts, correlation operators, or hidden-variable assignments being compared.
- **Admissibility**. Commutators, uncertainty bounds, contextuality constraints, or Bell-type inequalities decide which joint assignments are possible.
- **Readout**. Joint spectra, correlations, inequality violations, uncertainty products, or incompatible outcome statistics.
- **Check**. The non-classical content appears only if the incompatible questions cannot be replaced by one common sharp classical assignment.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.30, transport flow=0.24, discrete protocol=0.20

- arXiv:1604.05385, score 0.555
- arXiv:1612.00682, score 0.485
- arXiv:1801.03283, score 0.480
- arXiv:1506.05598, score 0.477
- arXiv:quant-ph0205159, score 0.475

Chapter 13

Boundary realization: how effects appear

Many named quantum effects are boundary realizations of the same construction. A potential, barrier, box, cavity, detector, or medium changes the allowed spectral channels without changing the basic prediction problem.

13.1 Why This Branch Exists

Named effects such as tunnelling and particle-in-a-box are boundary realizations. They are not separate conceptual primitives.

13.2 Page Map

Page	Dominant evidence signal	Score
Potential well	observables and spectra	1.44
Particle in a box	observables and spectra	1.42
Scattering	state evolution	1.39
Wave interference	observables and spectra	1.37
Quantum optics	observables and spectra	1.37
Spectral line	observables and spectra	0.92
S-matrix	observables and spectra	1.44
Quantum tunnelling	observables and spectra	1.42
Quantum imaging	preparation and boundary context	1.38
Quantum harmonic oscillator	observables and spectra	1.34
Macroscopic quantum phenomena	state evolution	1.08
Quantum metamaterial	state evolution	0.78

13.3 Mechanism Pages

Each page below is read as a specialization of the compact quantum constructor. Topic-specific pages use explicit equations or overrides; core-derived pages use the branch-level equation form associated with their mechanism role.

13.4 Potential well

Mechanism-tree role. Boundary realization: how effects appear; assignment score 1.44. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Potential well is a boundary-shaped spectrum role in the compact quantum constructor. In this tree, potential well belongs to realization: it shows how the abstract state-operator construction becomes legal on a domain, interface, potential, detector geometry, or scattering boundary.

Mechanism Reading

Operationally, Potential well contributes a boundary-shaped spectrum role. The topic changes the admissible domain or boundary condition and thereby changes the allowed spectrum, transmission amplitude, resonance, or scattering channel. In the boundaries step, boundary realization is where the same operator logic receives a physical presentation. The state space and generator are restricted by a domain, potential, asymptotic condition, interface, or detector arrangement. This is where geometry enters as realization, not as the invariant core. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Potential well contributes a boundary-shaped spectrum role to the quantum construction.
- **Placement**. This page is read first as a realization move: it changes the domain, boundary, geometry, or interface in which the operator acts.
- **Carrier or domain**. A Hilbert space with a selected domain, potential, interface, asymptotic channel, cavity, well, or boundary condition.
- **Operator or map**. A Hamiltonian, wave operator, transfer operator, or scattering map whose domain is changed by the boundary.
- **Admissibility**. Boundary conditions and matching conditions determine allowed states, resonances, transmission amplitudes, and spectra.
- **Readout**. Eigenvalues, resonances, tunnelling probabilities, phase shifts, reflection/transmission amplitudes, or scattering data.
- **Check**. Changing the boundary should change the spectrum or amplitude in the predicted way, while the free or asymptotic limit is recovered when the boundary is removed.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$

$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$

$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$

$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.42, transport flow=0.22, constraint closure=0.13

- arXiv:1604.05385, score 0.579
- arXiv:0912.2823, score 0.575
- arXiv:1506.05598, score 0.570
- arXiv:1612.00682, score 0.570
- arXiv:2108.07838, score 0.570

13.5 Particle in a box

Mechanism-tree role. Boundary realization: how effects appear; assignment score 1.42. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Particle in a box is a boundary-spectrum constructor: a spatial domain and boundary condition discretize the allowed energy spectrum.

Mechanism Reading

The page shows how a boundary condition changes the domain of the Hamiltonian and therefore the allowed spectra. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Particle in a box contributes a many-mode field or particle-realization role to the quantum construction.
- **Placement**. This page is read first as a realization move: it changes the domain, boundary, geometry, or interface in which the operator acts.
- **Carrier or domain**. Fock space, field configuration space, or a sector selected by charge, spin, momentum, statistics, or gauge data.
- **Operator or map**. Creation, annihilation, field, charge, spin, Hamiltonian, or scattering operators acting on the admissible sector.
- **Admissibility**. Statistics, gauge constraints, commutation or anticommutation rules, domain conditions, and sector labels decide which states are legal.
- **Readout**. Occupation number, charge, spin, momentum, energy, correlation function, cross-section, or scattering amplitude.
- **Check**. The field description must preserve the relevant observables under changes of representation and reduce to the expected particle or quasiparticle limit when that limit exists.

Topic Equations

$$
\begin{aligned}
B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)} \\
\rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)} \\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)} \\
[O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.42, transport flow=0.22, constraint closure=0.16

- arXiv:0912.2823, score 0.599
- arXiv:1801.03283, score 0.592
- arXiv:1612.00682, score 0.591
- arXiv:1506.05598, score 0.590
- arXiv:quant-ph0205159, score 0.583

13.6 Scattering

Mechanism-tree role. Boundary realization: how effects appear; assignment score 1.39. **Dominant evidence signal.** state evolution. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Scattering is a boundary-shaped spectrum role in the compact quantum constructor. In this tree, scattering belongs to realization: it shows how the abstract state-operator construction becomes legal on a domain, interface, potential, detector geometry, or scattering boundary.

Mechanism Reading

Operationally, Scattering contributes a boundary-shaped spectrum role. The topic changes the admissible domain or boundary condition and thereby changes the allowed spectrum, transmission amplitude, resonance, or scattering channel. In the boundaries step, boundary realization is where the same operator logic receives a physical presentation. The state space and generator are restricted by a domain, potential, asymptotic condition, interface, or detector arrangement. This is where geometry enters as realization, not as the invariant core. In the source-evidence profile for this page, the strongest construction signal is state evolution, normalization or admissibility, operator-to-spectrum readout; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Scattering contributes a boundary-shaped spectrum role to the quantum construction.
- **Placement**. This page is read first as a realization move: it changes the domain, boundary, geometry, or interface in which the operator acts.
- **Carrier or domain**. A Hilbert space with a selected domain, potential, interface, asymptotic channel, cavity, well, or boundary condition.
- **Operator or map**. A Hamiltonian, wave operator, transfer operator, or scattering map whose domain is changed by the boundary.
- **Admissibility**. Boundary conditions and matching conditions determine allowed states, resonances, transmission amplitudes, and spectra.
- **Readout**. Eigenvalues, resonances, tunnelling probabilities, phase shifts, reflection/transmission amplitudes, or scattering data.
- **Check**. Changing the boundary should change the spectrum or amplitude in the predicted way, while the free or asymptotic limit is recovered when the boundary is removed.

Topic Equations

$$\begin{aligned}
B &\longmapsto \rho_B \quad \text{(context specifies an admissible state)}\\
\rho_t &= U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}\\
O &= \sum_i \lambda_i P_i, \quad p_i = \operatorname{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}\\
[O_1, O_2] &\neq 0 \quad \text{(incompatible observables: no common sharp basis)}
\end{aligned}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

transport flow=0.27, constraint closure=0.21, spectral operator=0.20

- arXiv:1604.05385, score 0.530
- arXiv:0912.2823, score 0.527
- arXiv:1604.06537, score 0.503
- arXiv:1708.03640, score 0.495
- arXiv:1801.03283, score 0.495

13.7 Wave interference

Mechanism-tree role. Boundary realization: how effects appear; assignment score 1.37. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Wave interference is a broad quantum constructor role in the compact quantum constructor. In this tree, wave interference belongs to realization: it shows how the abstract state-operator construction becomes legal on a domain, interface, potential, detector geometry, or scattering boundary.

Mechanism Reading

Operationally, Wave interference contributes a broad quantum constructor role. The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer. In the boundaries step, boundary realization is where the same operator logic receives a physical presentation. The state space and generator are restricted by a domain, potential, asymptotic condition, interface, or detector arrangement. This is where geometry enters as realization, not as the invariant core. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Wave interference contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as a realization move: it changes the domain, boundary, geometry, or interface in which the operator acts.

- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$
\begin{aligned}
B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)} \\
\rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)} \\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)} \\
[O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.36, transport flow=0.25, constraint closure=0.18

- arXiv:1604.05385, score 0.561
- arXiv:1506.05598, score 0.541
- arXiv:1801.03283, score 0.538
- arXiv:1612.00682, score 0.535
- arXiv:0908.0752, score 0.532

13.8 Quantum optics

Mechanism-tree role. Boundary realization: how effects appear; assignment score 1.37. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum optics is a boundary-shaped spectrum role in the compact quantum constructor. In this tree, quantum optics belongs to realization: it shows how the abstract state-operator construction becomes legal on a domain, interface, potential, detector geometry, or scattering boundary.

Mechanism Reading

Operationally, Quantum optics contributes a boundary-shaped spectrum role. The topic changes the admissible domain or boundary condition and thereby changes the allowed spectrum, transmission amplitude, resonance, or scattering channel. In the boundaries step, boundary realization is where the same operator logic receives a physical presentation. The state space and generator are restricted by a domain, potential, asymptotic condition, interface, or detector arrangement. This is where geometry enters as realization, not as the invariant core. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum optics contributes a boundary-shaped spectrum role to the quantum construction.
- **Placement**. This page is read first as a realization move: it changes the domain, boundary, geometry, or interface in which the operator acts.
- **Carrier or domain**. A Hilbert space with a selected domain, potential, interface, asymptotic channel, cavity, well, or boundary condition.
- **Operator or map**. A Hamiltonian, wave operator, transfer operator, or scattering map whose domain is changed by the boundary.
- **Admissibility**. Boundary conditions and matching conditions determine allowed states, resonances, transmission amplitudes, and spectra.

- **Readout**. Eigenvalues, resonances, tunnelling probabilities, phase shifts, reflection/transmission amplitudes, or scattering data.
- **Check**. Changing the boundary should change the spectrum or amplitude in the predicted way, while the free or asymptotic limit is recovered when the boundary is removed.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.34, transport flow=0.22, constraint closure=0.13

- arXiv:1604.05385, score 0.571
- arXiv:0912.2823, score 0.570
- arXiv:0805.4565, score 0.549

- arXiv:1708.03640, score 0.548
- arXiv:quant-ph0205159, score 0.548

13.9 Spectral line

Mechanism-tree role. Boundary realization: how effects appear; assignment score 0.92. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Spectral line is a boundary-shaped spectrum role in the compact quantum constructor. In this tree, spectral line belongs to realization: it shows how the abstract state-operator construction becomes legal on a domain, interface, potential, detector geometry, or scattering boundary.

Mechanism Reading

Operationally, Spectral line contributes a boundary-shaped spectrum role. The topic changes the admissible domain or boundary condition and thereby changes the allowed spectrum, transmission amplitude, resonance, or scattering channel. In the boundaries step, boundary realization is where the same operator logic receives a physical presentation. The state space and generator are restricted by a domain, potential, asymptotic condition, interface, or detector arrangement. This is where geometry enters as realization, not as the invariant core. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, preparation, basis, or boundary context, state evolution; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Spectral line contributes a boundary-shaped spectrum role to the quantum construction.
- **Placement**. This page is read first as a realization move: it changes the domain, boundary, geometry, or interface in which the operator acts.
- **Carrier or domain**. A Hilbert space with a selected domain, potential, interface, asymptotic channel, cavity, well, or boundary condition.
- **Operator or map**. A Hamiltonian, wave operator, transfer operator, or scattering map whose domain is changed by the boundary.
- **Admissibility**. Boundary conditions and matching conditions determine allowed states, resonances, transmission amplitudes, and spectra.
- **Readout**. Eigenvalues, resonances, tunnelling probabilities, phase shifts, reflection/transmission amplitudes, or scattering data.
- **Check**. Changing the boundary should change the spectrum or amplitude in the predicted way, while the free or asymptotic limit is recovered when the boundary is removed.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.27, boundary weak form=0.21, transport flow=0.18

- arXiv:0912.2823, score 0.552
- arXiv:1604.06537, score 0.516
- arXiv:1604.05385, score 0.516
- arXiv:1801.03283, score 0.497
- arXiv:quant-ph0205159, score 0.497

13.10 S-matrix

Mechanism-tree role. Boundary realization: how effects appear; assignment score 1.44. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

S-matrix is a boundary-shaped spectrum role in the compact quantum constructor. In this tree, s-matrix belongs to realization: it shows how the abstract state-operator construction becomes legal on a domain, interface, potential, detector geometry, or scattering boundary.

Mechanism Reading

Operationally, S-matrix contributes a boundary-shaped spectrum role. The topic changes the admissible domain or boundary condition and thereby changes the allowed spectrum, transmission amplitude, resonance, or scattering channel. In the boundaries step, boundary realization is where the same operator logic receives a physical presentation. The state space and generator are restricted by a domain, potential, asymptotic condition, interface, or detector arrangement. This is where geometry enters as realization, not as the invariant core. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. S-matrix contributes a boundary-shaped spectrum role to the quantum construction.
- **Placement**. This page is read first as a realization move: it changes the domain, boundary, geometry, or interface in which the operator acts.
- **Carrier or domain**. A Hilbert space with a selected domain, potential, interface, asymptotic channel, cavity, well, or boundary condition.
- **Operator or map**. A Hamiltonian, wave operator, transfer operator, or scattering map whose domain is changed by the boundary.
- **Admissibility**. Boundary conditions and matching conditions determine allowed states, resonances, transmission amplitudes, and spectra.
- **Readout**. Eigenvalues, resonances, tunnelling probabilities, phase shifts, reflection/transmission amplitudes, or scattering data.
- **Check**. Changing the boundary should change the spectrum or amplitude in the predicted way, while the free or asymptotic limit is recovered when the boundary is removed.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.39, transport flow=0.27, constraint closure=0.19

- arXiv:0912.2823, score 0.567
- arXiv:1604.06537, score 0.550
- arXiv:1801.03283, score 0.547
- arXiv:1612.00682, score 0.539
- arXiv:1506.05598, score 0.535

13.11 Quantum tunnelling

Mechanism-tree role. Boundary realization: how effects appear; assignment score 1.42. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum tunnelling is a boundary-realization constructor: a state has nonzero transmission through a classically forbidden region.

Mechanism Reading

Tunnelling shows that the realization layer matters. A potential barrier changes the admissible wave solutions and produces transmission even where classical kinetic energy would be negative. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum tunnelling contributes a boundary-shaped spectrum role to the quantum construction.
- **Placement**. This page is read first as a realization move: it changes the domain, boundary, geometry, or interface in which the operator acts.
- **Carrier or domain**. A Hilbert space with a selected domain, potential, interface, asymptotic channel, cavity, well, or boundary condition.
- **Operator or map**. A Hamiltonian, wave operator, transfer operator, or scattering map whose domain is changed by the boundary.
- **Admissibility**. Boundary conditions and matching conditions determine allowed states, resonances, transmission amplitudes, and spectra.
- **Readout**. Eigenvalues, resonances, tunnelling probabilities, phase shifts, reflection/transmission amplitudes, or scattering data.
- **Check**. Changing the boundary should change the spectrum or amplitude in the predicted way, while the free or asymptotic limit is recovered when the boundary is removed.

Topic Equations

$$\begin{aligned} B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)} \\ \rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)} \\ O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)} \\ [O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)} \end{aligned}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.42, transport flow=0.24, constraint closure=0.18

- arXiv:1604.05385, score 0.569
- arXiv:0912.2823, score 0.568
- arXiv:1612.00682, score 0.562
- arXiv:quant-ph0205159, score 0.558
- arXiv:1801.03283, score 0.555

13.12 Quantum imaging

Mechanism-tree role. Boundary realization: how effects appear; assignment score 1.38. **Dominant evidence signal.** preparation and boundary context. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum imaging is an instrument-mediated readout role in the compact quantum constructor. In this tree, quantum imaging belongs to realization: it shows how the abstract state-operator construction becomes legal on a domain, interface, potential, detector geometry, or scattering boundary.

Mechanism Reading

Operationally, Quantum imaging contributes an instrument-mediated readout role. The mechanism is an apparatus-coupled readout: a prepared probe state interacts with a sample or field, the interaction changes phase, momentum, intensity, or counting statistics, and the instrument reconstructs an image, spectrum, trajectory, or estimate. In the boundaries step, boundary realization is where the same operator logic receives a physical presentation. The state space and generator are restricted by a domain, potential, asymptotic condition, interface, or detector arrangement. This is where geometry enters as realization, not as the invariant core. In the source-evidence profile for this page, the strongest construction signal is preparation, basis, or boundary context, operator-to-spectrum readout, state evolution; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum imaging contributes an instrument-mediated readout role to the quantum construction.
- **Placement**. This page is read first as a realization move: it changes the domain, boundary, geometry, or interface in which the operator acts.
- **Carrier or domain**. A probe state, sample state, field mode, detector state, or estimation register.
- **Operator or map**. An interaction Hamiltonian, transfer map, measurement channel, reconstruction map, or estimator.
- **Admissibility**. The instrument must separate sample signal from preparation, detector response, calibration, noise, and reconstruction artifacts.
- **Readout**. Counts, images, spectra, phase shifts, trajectories, intensity maps, correlation data, or parameter estimates.
- **Check**. The claimed mechanism is credible only when the same readout survives control experiments, calibration changes, and reconstruction checks.

Topic Equations

$$\begin{aligned}
B &\longmapsto \rho_B \quad \text{(context specifies an admissible state)}\\
\rho_t = U_t \rho_B U_t^\dagger &\quad \text{(unitary evolution from preparation to readout)}\\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) &\quad \text{(spectral probability measure)}\\
[O_1, O_2] \neq 0 &\quad \text{(incompatible observables: no common sharp basis)}
\end{aligned}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

boundary weak form=0.25, spectral operator=0.25, transport flow=0.22

- arXiv:0912.2823, score 0.523
- arXiv:1604.05385, score 0.484
- arXiv:1612.00682, score 0.452
- arXiv:quant-ph0205159, score 0.449
- arXiv:gr-qc0411110, score 0.447

13.13 Quantum harmonic oscillator

Mechanism-tree role. Boundary realization: how effects appear; assignment score 1.34. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum harmonic oscillator is a broad quantum constructor role in the compact quantum constructor. In this tree, quantum harmonic oscillator belongs to realization: it shows how the abstract state-operator construction becomes legal on a domain, interface, potential, detector geometry, or scattering boundary.

Mechanism Reading

Operationally, Quantum harmonic oscillator contributes a broad quantum constructor role. The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer. In the boundaries step, boundary realization is where the same operator logic receives a physical presentation. The state space and generator are restricted by a domain, potential, asymptotic condition, interface, or detector arrangement. This is where geometry enters as realization, not as the invariant core. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum harmonic oscillator contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as a realization move: it changes the domain, boundary, geometry, or interface in which the operator acts.

- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$
\begin{aligned}
B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)} \\
\rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)} \\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)} \\
[O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.42, transport flow=0.23, constraint closure=0.16

- arXiv:1604.06537, score 0.603
- arXiv:1801.03283, score 0.568
- arXiv:1612.00682, score 0.560
- arXiv:0912.2823, score 0.556
- arXiv:1506.05598, score 0.555

13.14 Macroscopic quantum phenomena

Mechanism-tree role. Boundary realization: how effects appear; assignment score 1.08. **Dominant evidence signal.** state evolution. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Macroscopic quantum phenomena is an open-system transport and coherence role in the compact quantum constructor. In this tree, macroscopic quantum phenomena belongs to realization: it shows how the abstract state-operator construction becomes legal on a domain, interface, potential, detector geometry, or scattering boundary.

Mechanism Reading

Operationally, Macroscopic quantum phenomena contributes an open-system transport and coherence role. The topic concerns quantum state transport under environmental coupling, coherence loss, biological or macroscopic boundary conditions, or effective dynamics outside an ideal closed system. In the boundaries step, boundary realization is where the same operator logic receives a physical presentation. The state space and generator are restricted by a domain, potential, asymptotic condition, interface, or detector arrangement. This is where geometry enters as realization, not as the invariant core. In the source-evidence profile for this page, the strongest construction signal is state evolution, preparation, basis, or boundary context, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Macroscopic quantum phenomena contributes an open-system transport and coherence role to the quantum construction.
- **Placement**. This page is read first as a realization move: it changes the domain, boundary, geometry, or interface in which the operator acts.
- **Carrier or domain**. A density operator, reduced state, coherence variable, bath-coupled state, or effective mesoscopic carrier.
- **Operator or map**. Hamiltonian plus environmental coupling, Lindbladian, memory kernel, stochastic map, or effective transport operator.
- **Admissibility**. Positivity, trace preservation, timescale separation, bath assumptions, and control over classical noise determine whether the model is legal.

- **Readout**. Coherence, population transfer, relaxation rate, transport efficiency, noise spectrum, or macroscopic response.
- **Check**. The quantum contribution must survive controls against classical noise, preparation artifacts, and coarse-graining choices.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

transport flow=0.28, boundary weak form=0.23, constraint closure=0.18

- arXiv:0912.2823, score 0.544
- arXiv:1706.03846, score 0.539
- arXiv:hep-lat9608080, score 0.518

- arXiv:1708.03640, score 0.511
- arXiv:astro-ph0604157, score 0.510

13.15 Quantum metamaterial

Mechanism-tree role. Boundary realization: how effects appear; assignment score 0.78. **Dominant evidence signal.** state evolution. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum metamaterial is a boundary-shaped spectrum role in the compact quantum constructor. In this tree, quantum metamaterial belongs to realization: it shows how the abstract state-operator construction becomes legal on a domain, interface, potential, detector geometry, or scattering boundary.

Mechanism Reading

Operationally, Quantum metamaterial contributes a boundary-shaped spectrum role. The topic changes the admissible domain or boundary condition and thereby changes the allowed spectrum, transmission amplitude, resonance, or scattering channel. In the boundaries step, boundary realization is where the same operator logic receives a physical presentation. The state space and generator are restricted by a domain, potential, asymptotic condition, interface, or detector arrangement. This is where geometry enters as realization, not as the invariant core. In the source-evidence profile for this page, the strongest construction signal is state evolution, operator-to-spectrum readout, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum metamaterial contributes a boundary-shaped spectrum role to the quantum construction.
- **Placement**. This page is read first as a realization move: it changes the domain, boundary, geometry, or interface in which the operator acts.
- **Carrier or domain**. A Hilbert space with a selected domain, potential, interface, asymptotic channel, cavity, well, or boundary condition.
- **Operator or map**. A Hamiltonian, wave operator, transfer operator, or scattering map whose domain is changed by the boundary.
- **Admissibility**. Boundary conditions and matching conditions determine allowed states, resonances, transmission amplitudes, and spectra.
- **Readout**. Eigenvalues, resonances, tunnelling probabilities, phase shifts, reflection/transmission amplitudes, or scattering data.
- **Check**. Changing the boundary should change the spectrum or amplitude in the predicted way, while the free or asymptotic limit is recovered when the boundary is removed.

Topic Equations

$$
\begin{aligned}
B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)} \\
\rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)} \\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)} \\
[O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

transport flow=0.30, spectral operator=0.13, constraint closure=0.07

- arXiv:1706.03846, score 0.540
- arXiv:1708.03640, score 0.516
- arXiv:quant-ph0312187, score 0.513
- arXiv:0908.0752, score 0.506
- arXiv:0805.4565, score 0.505

Chapter 14

Many-mode extension: fields, particles, and scaling

Quantum field theory, gauge theory, renormalization, photons, fermions, and related topics extend the same state-operator-spectrum logic to variable particle number, local fields, and scale-dependent descriptions.

14.1 Why This Branch Exists

Field theory and gauge theory extend the same construction to many modes, local generators, and scale-dependent descriptions.

14.2 Page Map

Page	Dominant evidence signal	Score
Fermi–Dirac statistics	state evolution	1.50
Dirac equation	observables and spectra	1.50
Quantum electrodynamics	observables and spectra	1.50
Renormalization	observables and spectra	1.50
Gauge theory	observables and spectra	1.49
Photon	observables and spectra	1.48
Quantum field theory	observables and spectra	1.47
Fermion	state evolution	1.42
Boson	normalization and admissibility	1.40
Quantum chromodynamics	observables and spectra	1.30
AdS/CFT correspondence	state evolution	1.43
Klein–Gordon equation	observables and spectra	1.22
Quantum gravity	state evolution	1.20
Spin network	observables and spectra	1.11
Electron	observables and spectra	0.94
Loop quantum gravity	observables and spectra	0.92
Spin foam	observables and spectra	0.92
Quantum cosmology	state evolution	0.91
String theory	observables and spectra	0.90
Quantum geometry	observables and spectra	0.88

Page	Dominant evidence signal	Score
Fock space	observables and spectra	0.86
Quantum spacetime	incompatible questions	0.85

14.3 Mechanism Pages

Each page below is read as a specialization of the compact quantum constructor. Topic-specific pages use explicit equations or overrides; core-derived pages use the branch-level equation form associated with their mechanism role.

14.4 Fermi–Dirac statistics

Mechanism-tree role. Many-mode extension: fields, particles, and scaling; assignment score 1.50. **Dominant evidence signal.** state evolution. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Fermi–Dirac statistics is a lawful state-transport role in the compact quantum constructor. In this tree, fermi–Dirac statistics belongs to the many-mode extension: the same state, generator, observable, and compatibility logic is lifted from one system to fields, particles, scaling limits, or gauge constraints.

Mechanism Reading

Operationally, Fermi–Dirac statistics contributes a lawful state-transport role. The topic supplies or modifies the generator of state evolution before readout. It should be read as a transport step, not as a measurement result. In the fields step, the field layer extends the same constructor to variable numbers of modes and symmetry constraints. Creation and annihilation operators, correlation functions, gauge conditions, and renormalization flows are higher-capacity versions of the same assembly. In the source-evidence profile for this page, the strongest construction signal is state evolution, preparation, basis, or boundary context, operator-to-spectrum readout; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Fermi–Dirac statistics contributes a lawful state-transport role to the quantum construction.
- **Placement**. This page is read first as a many-mode or field-realization move: it extends the state and operator construction beyond a single-particle carrier.
- **Carrier or domain**. A state vector, density operator, wavefunction, field state, or register on a specified domain.
- **Operator or map**. Hamiltonian, unitary map, channel generator, action, constraint, or differential operator that transports the state.
- **Admissibility**. Self-adjointness, complete positivity, trace preservation, gauge constraints, and boundary/domain conditions decide whether the evolution is legal.
- **Readout**. Time-dependent probabilities, spectra, transition amplitudes, conserved quantities, or response functions.

- **Check**. The generator must predict the observed evolution while preserving the relevant normalization, positivity, symmetry, or conservation constraint.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

transport flow=0.26, boundary weak form=0.24, spectral operator=0.23

- arXiv:0912.2823, score 0.491
- arXiv:1604.05385, score 0.452
- arXiv:1801.03283, score 0.421
- arXiv:1612.00682, score 0.421
- arXiv:quant-ph0205159, score 0.417

14.5 Dirac equation

Mechanism-tree role. Many-mode extension: fields, particles, and scaling; assignment score 1.50. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Dirac equation is a lawful state-transport role in the compact quantum constructor. In this tree, dirac equation belongs to the many-mode extension: the same state, generator, observable, and compatibility logic is lifted from one system to fields, particles, scaling limits, or gauge constraints.

Mechanism Reading

Operationally, Dirac equation contributes a lawful state-transport role. The topic supplies or modifies the generator of state evolution before readout. It should be read as a transport step, not as a measurement result. In the fields step, the field layer extends the same constructor to variable numbers of modes and symmetry constraints. Creation and annihilation operators, correlation functions, gauge conditions, and renormalization flows are higher-capacity versions of the same assembly. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Dirac equation contributes a lawful state-transport role to the quantum construction.
- **Placement**. This page is read first as a many-mode or field-realization move: it extends the state and operator construction beyond a single-particle carrier.
- **Carrier or domain**. A state vector, density operator, wavefunction, field state, or register on a specified domain.
- **Operator or map**. Hamiltonian, unitary map, channel generator, action, constraint, or differential operator that transports the state.
- **Admissibility**. Self-adjointness, complete positivity, trace preservation, gauge constraints, and boundary/domain conditions decide whether the evolution is legal.
- **Readout**. Time-dependent probabilities, spectra, transition amplitudes, conserved quantities, or response functions.
- **Check**. The generator must predict the observed evolution while preserving the relevant normalization, positivity, symmetry, or conservation constraint.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$

$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$

$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$

$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.30, transport flow=0.26, constraint closure=0.21

- arXiv:0912.2823, score 0.525
- arXiv:1604.06537, score 0.523
- arXiv:1801.03283, score 0.515
- arXiv:quant-ph0205159, score 0.504
- arXiv:1612.00682, score 0.503

14.6 Quantum electrodynamics

Mechanism-tree role. Many-mode extension: fields, particles, and scaling; assignment score 1.50. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum electrodynamics is a lawful state-transport role in the compact quantum constructor. In this tree, quantum electrodynamics belongs to the many-mode extension: the same state, generator, observable, and compatibility logic is lifted from one system to fields, particles, scaling limits, or gauge constraints.

Mechanism Reading

Operationally, Quantum electrodynamics contributes a lawful state-transport role. The topic supplies or modifies the generator of state evolution before readout. It should be read as a transport step, not as a measurement result. In the fields step, the field layer extends the same constructor to variable numbers of modes and symmetry constraints. Creation and annihilation operators, correlation functions, gauge conditions, and renormalization flows are higher-capacity versions of the same assembly. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum electrodynamics contributes a lawful state-transport role to the quantum construction.
- **Placement**. This page is read first as a many-mode or field-realization move: it extends the state and operator construction beyond a single-particle carrier.
- **Carrier or domain**. A state vector, density operator, wavefunction, field state, or register on a specified domain.
- **Operator or map**. Hamiltonian, unitary map, channel generator, action, constraint, or differential operator that transports the state.
- **Admissibility**. Self-adjointness, complete positivity, trace preservation, gauge constraints, and boundary/domain conditions decide whether the evolution is legal.
- **Readout**. Time-dependent probabilities, spectra, transition amplitudes, conserved quantities, or response functions.
- **Check**. The generator must predict the observed evolution while preserving the relevant normalization, positivity, symmetry, or conservation constraint.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.26, transport flow=0.22, constraint closure=0.21

- arXiv:0912.2823, score 0.470
- arXiv:1604.06537, score 0.452
- arXiv:2111.12617, score 0.422
- arXiv:1801.03283, score 0.416
- arXiv:2501.07524, score 0.413

14.7 Renormalization

Mechanism-tree role. Many-mode extension: fields, particles, and scaling; assignment score 1.50. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Renormalization is the scale-flow constructor: the effective parameters of the theory change with resolution while predictions remain controlled.

Mechanism Reading

Renormalization explains why a mechanism can preserve its operator role while changing its apparent parameters across scales. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Renormalization contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as a many-mode or field-realization move: it extends the state and operator construction beyond a single-particle carrier.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$
\begin{aligned}
B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)} \\
\rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)} \\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)} \\
[O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.38, transport flow=0.19, constraint closure=0.15

- arXiv:0912.2823, score 0.498
- arXiv:1604.06537, score 0.494
- arXiv:1612.00682, score 0.465
- arXiv:1801.03283, score 0.464
- arXiv:2111.12617, score 0.461

14.8 Gauge theory

Mechanism-tree role. Many-mode extension: fields, particles, and scaling; assignment score 1.49. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Gauge theory is a redundancy-and-constraint constructor: different local presentations can represent the same physical state.

Mechanism Reading

Gauge theory belongs at the field/geometry frontier. It separates physical degrees of freedom from representational choices and imposes covariant transport through a connection. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Gauge theory contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as a many-mode or field-realization move: it extends the state and operator construction beyond a single-particle carrier.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.

- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$\begin{aligned} B &\longmapsto \rho_B \quad \text{(context specifies an admissible state)} \\ \rho_t &= U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)} \\ O &= \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)} \\ [O_1, O_2] &\neq 0 \quad \text{(incompatible observables: no common sharp basis)} \end{aligned}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.39, transport flow=0.25, constraint closure=0.19

- arXiv:0912.2823, score 0.511
- arXiv:1604.05385, score 0.478
- arXiv:1612.00682, score 0.468
- arXiv:1801.03283, score 0.463
- arXiv:quant-ph0205159, score 0.459

14.9 Photon

Mechanism-tree role. Many-mode extension: fields, particles, and scaling; assignment score 1.48. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Photon is a field-mode constructor: a one-quantum excitation of the electromagnetic field, constrained by massless dispersion and transverse polarization.

Mechanism Reading

The native photon mechanism is field quantization. The electromagnetic field is decomposed into modes, creation and annihilation operators act on those modes, and a one-photon state is created from the vacuum. The relevant readouts are occupation number, energy, momentum, polarization, and detector clicks; the constraints are dispersion and gauge-compatible transversality. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Photon contributes a many-mode field or particle-realization role to the quantum construction.
- **Placement**. This page is read first as a many-mode or field-realization move: it extends the state and operator construction beyond a single-particle carrier.
- **Carrier or domain**. Fock space, field configuration space, or a sector selected by charge, spin, momentum, statistics, or gauge data.
- **Operator or map**. Creation, annihilation, field, charge, spin, Hamiltonian, or scattering operators acting on the admissible sector.
- **Admissibility**. Statistics, gauge constraints, commutation or anticommutation rules, domain conditions, and sector labels decide which states are legal.
- **Readout**. Occupation number, charge, spin, momentum, energy, correlation function, cross-section, or scattering amplitude.
- **Check**. The field description must preserve the relevant observables under changes of representation and reduce to the expected particle or quasiparticle limit when that limit exists.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^{\dagger} \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.40, transport flow=0.25, constraint closure=0.19

- arXiv:1604.05385, score 0.536
- arXiv:0912.2823, score 0.504
- arXiv:1612.00682, score 0.495
- arXiv:1801.03283, score 0.494
- arXiv:quant-ph0205159, score 0.488

14.10 Quantum field theory

Mechanism-tree role. Many-mode extension: fields, particles, and scaling; assignment score 1.47. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum field theory is the many-mode local-field extension of the quantum constructor.

Mechanism Reading

Quantum field theory lifts the state-operator-spectrum construction to fields, local operators, creation and annihilation modes, and scattering amplitudes. Particles become stable excitation/readout roles of fields. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum field theory contributes a many-mode field or particle-realization role to the quantum construction.
- **Placement**. This page is read first as a many-mode or field-realization move: it extends the state and operator construction beyond a single-particle carrier.
- **Carrier or domain**. Fock space, field configuration space, or a sector selected by charge, spin, momentum, statistics, or gauge data.
- **Operator or map**. Creation, annihilation, field, charge, spin, Hamiltonian, or scattering operators acting on the admissible sector.
- **Admissibility**. Statistics, gauge constraints, commutation or anticommutation rules, domain conditions, and sector labels decide which states are legal.
- **Readout**. Occupation number, charge, spin, momentum, energy, correlation function, cross-section, or scattering amplitude.
- **Check**. The field description must preserve the relevant observables under changes of representation and reduce to the expected particle or quasiparticle limit when that limit exists.

Topic Equations

$$\begin{aligned} B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)} \\ \rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)} \\ O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)} \\ [O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)} \end{aligned}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.27, transport flow=0.26, constraint closure=0.19

- arXiv:0912.2823, score 0.518
- arXiv:1604.06537, score 0.508
- arXiv:1708.03640, score 0.473
- arXiv:quant-ph0205159, score 0.471
- arXiv:1801.03283, score 0.468

14.11 Fermion

Mechanism-tree role. Many-mode extension: fields, particles, and scaling; assignment score 1.42. **Dominant evidence signal.** state evolution. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Fermion is an exchange-symmetry constructor: identical fermions live in antisymmetric sectors and obey Pauli exclusion.

Mechanism Reading

The mechanism is an admissibility rule on many-body state space. Antisymmetry, anticommutation, and zero-or-one mode occupation define the portable role. In the source-evidence profile for this page, the strongest construction signal is state evolution, normalization or admissibility, preparation, basis, or boundary context; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Fermion contributes a many-mode field or particle-realization role to the quantum construction.
- **Placement**. This page is read first as a many-mode or field-realization move: it extends the state and operator construction beyond a single-particle carrier.
- **Carrier or domain**. Fock space, field configuration space, or a sector selected by charge, spin, momentum, statistics, or gauge data.
- **Operator or map**. Creation, annihilation, field, charge, spin, Hamiltonian, or scattering operators acting on the admissible sector.
- **Admissibility**. Statistics, gauge constraints, commutation or anticommutation rules, domain conditions, and sector labels decide which states are legal.
- **Readout**. Occupation number, charge, spin, momentum, energy, correlation function, cross-section, or scattering amplitude.
- **Check**. The field description must preserve the relevant observables under changes of representation and reduce to the expected particle or quasiparticle limit when that limit exists.

Topic Equations

$$
\begin{aligned}
B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)} \\
\rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)} \\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)} \\
[O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

transport flow=0.28, constraint closure=0.26, boundary weak form=0.18

- arXiv:1706.03846, score 0.487
- arXiv:0912.2823, score 0.479
- arXiv:1708.03640, score 0.465
- arXiv:0908.0752, score 0.453
- arXiv:astro-ph0604157, score 0.452

14.12 Boson

Mechanism-tree role. Many-mode extension: fields, particles, and scaling; assignment score 1.40. **Dominant evidence signal.** normalization and admissibility. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Boson is an exchange-symmetry constructor: identical bosons live in symmetric sectors and can share the same mode.

Mechanism Reading

The mechanism is symmetric exchange. Commuting creation and annihilation operators allow arbitrary nonnegative occupation of a mode, which supports field modes, coherent states, condensates, and photon-like readouts. In the source-evidence profile for this page, the strongest construction signal is normalization or admissibility, state evolution, operator-to-spectrum readout; the strongest carrier signal is local notation, information profile, spectral profile.

Quantum Mechanism Frame

- **Role**. Boson contributes a many-mode field or particle-realization role to the quantum construction.
- **Placement**. This page is read first as a many-mode or field-realization move: it extends the state and operator construction beyond a single-particle carrier.
- **Carrier or domain**. Fock space, field configuration space, or a sector selected by charge, spin, momentum, statistics, or gauge data.
- **Operator or map**. Creation, annihilation, field, charge, spin, Hamiltonian, or scattering operators acting on the admissible sector.
- **Admissibility**. Statistics, gauge constraints, commutation or anticommutation rules, domain conditions, and sector labels decide which states are legal.
- **Readout**. Occupation number, charge, spin, momentum, energy, correlation function, cross-section, or scattering amplitude.
- **Check**. The field description must preserve the relevant observables under changes of representation and reduce to the expected particle or quasiparticle limit when that limit exists.

Role-Level Equations

Topic-specific constructor: the equations express symmetric sectors, commutation, and unrestricted mode occupation.

$$\mathcal{F}_+(\mathcal{H}) = \bigoplus_{n=0}^{\infty} \operatorname{Sym}^n \mathcal{H}$$
$$[a_i, a_j^\dagger] = \delta_{ij}, \qquad [a_i, a_j] = 0$$
$$n_i = a_i^\dagger a_i \in \{0, 1, 2, \ldots\}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

constraint closure=0.26, transport flow=0.16, spectral operator=0.16

- arXiv:1706.03846, score 0.448
- arXiv:1708.03640, score 0.430
- arXiv:1801.08949, score 0.424
- arXiv:0908.0752, score 0.421
- arXiv:1801.03283, score 0.413

14.13 Quantum chromodynamics

Mechanism-tree role. Many-mode extension: fields, particles, and scaling; assignment score 1.30. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum chromodynamics is a lawful state-transport role in the compact quantum constructor. In this tree, quantum chromodynamics belongs to the many-mode extension: the same state, generator, observable, and compatibility logic is lifted from one system to fields, particles, scaling limits, or gauge constraints.

Mechanism Reading

Operationally, Quantum chromodynamics contributes a lawful state-transport role. The topic supplies or modifies the generator of state evolution before readout. It should be read as a transport step, not as a measurement result. In the fields step, the field layer extends the same constructor to variable numbers of modes and symmetry constraints. Creation and annihilation operators, correlation functions, gauge conditions, and renormalization flows are higher-capacity versions of the same assembly. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum chromodynamics contributes a lawful state-transport role to the quantum construction.
- **Placement**. This page is read first as a many-mode or field-realization move: it extends the state and operator construction beyond a single-particle carrier.

- **Carrier or domain**. A state vector, density operator, wavefunction, field state, or register on a specified domain.
- **Operator or map**. Hamiltonian, unitary map, channel generator, action, constraint, or differential operator that transports the state.
- **Admissibility**. Self-adjointness, complete positivity, trace preservation, gauge constraints, and boundary/domain conditions decide whether the evolution is legal.
- **Readout**. Time-dependent probabilities, spectra, transition amplitudes, conserved quantities, or response functions.
- **Check**. The generator must predict the observed evolution while preserving the relevant normalization, positivity, symmetry, or conservation constraint.

Topic Equations

$$\begin{aligned} B \longmapsto \rho_B \quad & \text{(context specifies an admissible state)} \\ \rho_t = U_t \rho_B U_t^\dagger \quad & \text{(unitary evolution from preparation to readout)} \\ O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad & \text{(spectral probability measure)} \\ [O_1, O_2] \neq 0 \quad & \text{(incompatible observables: no common sharp basis)} \end{aligned}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.33, transport flow=0.27, constraint closure=0.21

- arXiv:0912.2823, score 0.480
- arXiv:1604.05385, score 0.480
- arXiv:1604.06537, score 0.476
- arXiv:1801.03283, score 0.462
- arXiv:quant-ph0205159, score 0.455

14.14 AdS/CFT correspondence

Mechanism-tree role. Many-mode extension: fields, particles, and scaling; assignment score 1.43. **Dominant evidence signal.** state evolution. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

AdS/CFT correspondence is a geometry-translation constructor: bulk gravitational data and boundary field data are treated as dual presentations of one operator structure.

Mechanism Reading

AdS/CFT belongs at the frontier where geometry becomes a realization rather than the invariant root. The practical content is a dictionary between bulk fields and boundary operators. In the source-evidence profile for this page, the strongest construction signal is state evolution, operator-to-spectrum readout, preparation, basis, or boundary context; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. AdS/CFT correspondence contributes a geometry or holographic realization role to the quantum construction.
- **Placement**. This page is read first as a many-mode or field-realization move: it extends the state and operator construction beyond a single-particle carrier.
- **Carrier or domain**. A spacetime, boundary algebra, gauge orbit, spin network, bulk/boundary pair, or geometric representation of a quantum state space.
- **Operator or map**. Hamiltonian, action, constraint, boundary operator, correlation map, or dictionary between two representations.
- **Admissibility**. Gauge, boundary, metric, covariance, and constraint conditions decide which geometric descriptions represent the same physical content.
- **Readout**. Boundary correlators, spectra, entropies, scattering data, geometric invariants, or reconstructed bulk quantities.
- **Check**. A geometric reformulation is physical only to the extent that it preserves observables or correlation functions across the representation change.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

transport flow=0.26, spectral operator=0.26, boundary weak form=0.22

- arXiv:0912.2823, score 0.576
- arXiv:quant-ph0205159, score 0.514
- arXiv:1801.03283, score 0.511
- arXiv:1708.03640, score 0.511
- arXiv:1506.05598, score 0.509

14.15 Klein–Gordon equation

Mechanism-tree role. Many-mode extension: fields, particles, and scaling; assignment score 1.22. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Klein–Gordon equation is a lawful state-transport role in the compact quantum constructor. In this tree, klein–Gordon equation belongs to the many-mode extension: the same state, generator, observable, and compatibility logic is lifted from one system to fields, particles, scaling limits, or gauge constraints.

Mechanism Reading

Operationally, Klein–Gordon equation contributes a lawful state-transport role. The topic supplies or modifies the generator of state evolution before readout. It should be read as a transport step, not as a measurement result. In the fields step, the field layer extends the same constructor to variable numbers of modes and symmetry constraints. Creation and annihilation operators, correlation functions, gauge conditions, and renormalization flows are higher-capacity versions of the same assembly. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, non-commuting compatibility limits; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Klein–Gordon equation contributes a lawful state-transport role to the quantum construction.
- **Placement**. This page is read first as a many-mode or field-realization move: it extends the state and operator construction beyond a single-particle carrier.
- **Carrier or domain**. A state vector, density operator, wavefunction, field state, or register on a specified domain.
- **Operator or map**. Hamiltonian, unitary map, channel generator, action, constraint, or differential operator that transports the state.
- **Admissibility**. Self-adjointness, complete positivity, trace preservation, gauge constraints, and boundary/domain conditions decide whether the evolution is legal.
- **Readout**. Time-dependent probabilities, spectra, transition amplitudes, conserved quantities, or response functions.
- **Check**. The generator must predict the observed evolution while preserving the relevant normalization, positivity, symmetry, or conservation constraint.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.27, transport flow=0.26, commutator incompatibility=0.21

- arXiv:1604.06537, score 0.573
- arXiv:0908.0752, score 0.537
- arXiv:1506.05598, score 0.537
- arXiv:1708.03640, score 0.536
- arXiv:quant-ph0205159, score 0.536

14.16 Quantum gravity

Mechanism-tree role. Many-mode extension: fields, particles, and scaling; assignment score 1.20. **Dominant evidence signal.** state evolution. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum gravity is a geometry or holographic realization role in the compact quantum constructor. In this tree, quantum gravity belongs to the many-mode extension: the same state, generator, observable, and compatibility logic is lifted from one system to fields, particles, scaling limits, or gauge constraints.

Mechanism Reading

Operationally, Quantum gravity contributes a geometry or holographic realization role. Geometry supplies the realization, boundary, gauge, or dual description in which the operator construction becomes physically legible. In the fields step, the field layer extends the same constructor to variable numbers of modes and symmetry constraints. Creation and annihilation operators, correlation functions, gauge conditions, and renormalization flows are higher-capacity versions of the same assembly. In the source-evidence profile for this page, the strongest construction signal is state evolution, normalization or admissibility, operator-to-spectrum readout; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum gravity contributes a geometry or holographic realization role to the quantum construction.
- **Placement**. This page is read first as a many-mode or field-realization move: it extends the state and operator construction beyond a single-particle carrier.
- **Carrier or domain**. A spacetime, boundary algebra, gauge orbit, spin network, bulk/boundary pair, or geometric representation of a quantum state space.
- **Operator or map**. Hamiltonian, action, constraint, boundary operator, correlation map, or dictionary between two representations.
- **Admissibility**. Gauge, boundary, metric, covariance, and constraint conditions decide which geometric descriptions represent the same physical content.
- **Readout**. Boundary correlators, spectra, entropies, scattering data, geometric invariants, or reconstructed bulk quantities.
- **Check**. A geometric reformulation is physical only to the extent that it preserves observables or correlation functions across the representation change.

Topic Equations

$$\begin{aligned} B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)} \\ \rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)} \\ O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)} \\ [O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)} \end{aligned}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

transport flow=0.27, constraint closure=0.20, spectral operator=0.20

- arXiv:1604.05385, score 0.535
- arXiv:0912.2823, score 0.535
- arXiv:1708.03640, score 0.500
- arXiv:1801.03283, score 0.497
- arXiv:quant-ph0205159, score 0.495

14.17 Spin network

Mechanism-tree role. Many-mode extension: fields, particles, and scaling; assignment score 1.11. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Spin network is an engineered operation-sequence role in the compact quantum constructor. In this tree, spin network belongs to the many-mode extension: the same state, generator, observable, and compatibility logic is lifted from one system to fields, particles, scaling limits, or gauge constraints.

Mechanism Reading

Operationally, Spin network contributes an engineered operation-sequence role. The mechanism is a controlled composition of allowed maps: a sequence that prepares, transforms, protects, transmits, or reads a quantum state. In the fields step, the field layer extends the same constructor to variable numbers of modes and symmetry constraints. Creation and annihilation operators, correlation functions, gauge conditions, and renormalization flows are higher-capacity versions of the same assembly. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Spin network contributes an engineered operation-sequence role to the quantum construction.
- **Placement**. This page is read first as a many-mode or field-realization move: it extends the state and operator construction beyond a single-particle carrier.
- **Carrier or domain**. An input state, register, channel state, error syndrome, key, or controlled experimental configuration.
- **Operator or map**. An ordered sequence of gates, channels, measurements, corrections, encodings, or conditional maps.
- **Admissibility**. Each step must belong to the claimed map class: unitary, completely positive, trace-preserving, projective, conditional, or corrective.
- **Readout**. Output state, key, error rate, fidelity, channel capacity, algorithmic success probability, or sensor estimate.
- **Check**. Changing operation order, inserting classical controls, or replacing a quantum channel should identify which step carries the effect.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.35, transport flow=0.23, constraint closure=0.17

- arXiv:0912.2823, score 0.587
- arXiv:1801.03283, score 0.545
- arXiv:1612.00682, score 0.545
- arXiv:quant-ph0205159, score 0.540
- arXiv:1506.05598, score 0.539

14.18 Electron

Mechanism-tree role. Many-mode extension: fields, particles, and scaling; assignment score 0.94. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Electron is a charged spinor constructor: its identity is fixed by mass, charge, spin-1/2 representation, fermionic statistics, and electromagnetic coupling.

Mechanism Reading

The electron is not a generic object label in this tree. Its native mechanism combines a spinor state, a Schrödinger/Pauli/Dirac generator depending on regime, conserved charge, and fermionic anticommutation. The readouts are charge, spin, momentum, energy, and scattering response. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, preparation, basis, or boundary context, state evolution; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Electron contributes a many-mode field or particle-realization role to the quantum construction.
- **Placement**. This page is read first as a many-mode or field-realization move: it extends the state and operator construction beyond a single-particle carrier.
- **Carrier or domain**. Fock space, field configuration space, or a sector selected by charge, spin, momentum, statistics, or gauge data.

- **Operator or map**. Creation, annihilation, field, charge, spin, Hamiltonian, or scattering operators acting on the admissible sector.
- **Admissibility**. Statistics, gauge constraints, commutation or anticommutation rules, domain conditions, and sector labels decide which states are legal.
- **Readout**. Occupation number, charge, spin, momentum, energy, correlation function, cross-section, or scattering amplitude.
- **Check**. The field description must preserve the relevant observables under changes of representation and reduce to the expected particle or quasiparticle limit when that limit exists.

Topic Equations

$$\begin{aligned} B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)} \\ \rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)} \\ O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)} \\ [O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)} \end{aligned}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.26, boundary weak form=0.25, transport flow=0.22

- arXiv:0912.2823, score 0.507
- arXiv:1604.05385, score 0.467
- arXiv:1612.00682, score 0.436
- arXiv:quant-ph0205159, score 0.435
- arXiv:gr-qc0411110, score 0.431

14.19 Loop quantum gravity

Mechanism-tree role. Many-mode extension: fields, particles, and scaling; assignment score 0.92. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Loop quantum gravity is a geometry or holographic realization role in the compact quantum constructor. In this tree, loop quantum gravity belongs to the many-mode extension: the same state, generator, observable, and compatibility logic is lifted from one system to fields, particles, scaling limits, or gauge constraints.

Mechanism Reading

Operationally, Loop quantum gravity contributes a geometry or holographic realization role. Geometry supplies the realization, boundary, gauge, or dual description in which the operator construction becomes physically legible. In the fields step, the field layer extends the same constructor to variable numbers of modes and symmetry constraints. Creation and annihilation operators, correlation functions, gauge conditions, and renormalization flows are higher-capacity versions of the same assembly. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Loop quantum gravity contributes a geometry or holographic realization role to the quantum construction.
- **Placement**. This page is read first as a many-mode or field-realization move: it extends the state and operator construction beyond a single-particle carrier.
- **Carrier or domain**. A spacetime, boundary algebra, gauge orbit, spin network, bulk/boundary pair, or geometric representation of a quantum state space.
- **Operator or map**. Hamiltonian, action, constraint, boundary operator, correlation map, or dictionary between two representations.
- **Admissibility**. Gauge, boundary, metric, covariance, and constraint conditions decide which geometric descriptions represent the same physical content.

- **Readout**. Boundary correlators, spectra, entropies, scattering data, geometric invariants, or reconstructed bulk quantities.
- **Check**. A geometric reformulation is physical only to the extent that it preserves observables or correlation functions across the representation change.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.35, transport flow=0.26, constraint closure=0.18

- arXiv:0912.2823, score 0.577
- arXiv:quant-ph0205159, score 0.543
- arXiv:1612.00682, score 0.542

- arXiv:1801.03283, score 0.542
- arXiv:1506.05598, score 0.535

14.20 Spin foam

Mechanism-tree role. Many-mode extension: fields, particles, and scaling; assignment score 0.92. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Spin foam is a geometry or holographic realization role in the compact quantum constructor. In this tree, spin foam belongs to the many-mode extension: the same state, generator, observable, and compatibility logic is lifted from one system to fields, particles, scaling limits, or gauge constraints.

Mechanism Reading

Operationally, Spin foam contributes a geometry or holographic realization role. Geometry supplies the realization, boundary, gauge, or dual description in which the operator construction becomes physically legible. In the fields step, the field layer extends the same constructor to variable numbers of modes and symmetry constraints. Creation and annihilation operators, correlation functions, gauge conditions, and renormalization flows are higher-capacity versions of the same assembly. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Spin foam contributes a geometry or holographic realization role to the quantum construction.
- **Placement**. This page is read first as a many-mode or field-realization move: it extends the state and operator construction beyond a single-particle carrier.
- **Carrier or domain**. A spacetime, boundary algebra, gauge orbit, spin network, bulk/boundary pair, or geometric representation of a quantum state space.
- **Operator or map**. Hamiltonian, action, constraint, boundary operator, correlation map, or dictionary between two representations.
- **Admissibility**. Gauge, boundary, metric, covariance, and constraint conditions decide which geometric descriptions represent the same physical content.
- **Readout**. Boundary correlators, spectra, entropies, scattering data, geometric invariants, or reconstructed bulk quantities.
- **Check**. A geometric reformulation is physical only to the extent that it preserves observables or correlation functions across the representation change.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.44, transport flow=0.23, constraint closure=0.13

- arXiv:0912.2823, score 0.487
- arXiv:1604.06537, score 0.485
- arXiv:1612.00682, score 0.470
- arXiv:2108.07838, score 0.456
- arXiv:quant-ph0205159, score 0.449

14.21 Quantum cosmology

Mechanism-tree role. Many-mode extension: fields, particles, and scaling; assignment score 0.91. **Dominant evidence signal.** state evolution. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum cosmology is a broad quantum constructor role in the compact quantum constructor. In this tree, quantum cosmology belongs to the many-mode extension: the same state, generator, observable, and compatibility logic is lifted from one system to fields, particles, scaling limits, or gauge constraints.

Mechanism Reading

Operationally, Quantum cosmology contributes a broad quantum constructor role. The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer. In the fields step, the field layer extends the same constructor to variable numbers of modes and symmetry constraints. Creation and annihilation operators, correlation functions, gauge conditions, and renormalization flows are higher-capacity versions of the same assembly. In the source-evidence profile for this page, the strongest construction signal is state evolution, operator-to-spectrum readout, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum cosmology contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as a many-mode or field-realization move: it extends the state and operator construction beyond a single-particle carrier.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Core-Derived Role Equations

This equation block is the branch-level quantum mechanism attached to the page by the compact constructor. It should be read as the role equation to check against the page's source evidence, not as a claim that every source writes the formula in this notation.

$$C \mapsto (\mathcal{H}_C, \mathcal{D}_C), \quad \rho \mapsto U\rho U^\dagger, \quad A = \int \lambda \, dE_A(\lambda)$$

What Remains Stable

- The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer.
- Quantum cosmology extends the state-operator-spectrum constructor to many modes, fields, particles, gauge structure, or scale.
- Particle identity is treated as a stable excitation or representation role rather than as the starting object.
- Creation/annihilation, field operators, gauge constraints, and scale flow preserve operator structure across realizations.

What Changes With Realization

- The local title, representation, and physical realization may change while the constructor role is preserved.
- The carrier can be a field state, occupation-number state, gauge orbit, spin network, or effective theory.
- The same formal role may be displayed through particles, modes, amplitudes, correlation functions, or boundary dictionaries.
- Scale and geometry can change the realization while preserving operator or spectral content.

Validation Boundary

- Use this role by specifying the page's state carrier, operator or map, readout, and compatibility condition in its own quantum language.
- Commutation, anticommutation, gauge, and occupation rules define which many-mode states are admissible.
- The field or many-mode construction must reduce to the appropriate single-particle, quasi-particle, or low-energy limit when those limits exist.

Evidence Profile

transport flow=0.28, spectral operator=0.16, constraint closure=0.09

- arXiv:1706.03846, score 0.466
- arXiv:0908.0752, score 0.443
- arXiv:1708.03640, score 0.441
- arXiv:0805.4565, score 0.440
- arXiv:0806.4515, score 0.423

14.22 String theory

Mechanism-tree role. Many-mode extension: fields, particles, and scaling; assignment score 0.90. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

String theory is a geometry or holographic realization role in the compact quantum constructor. In this tree, string theory belongs to the many-mode extension: the same state, generator, observable, and compatibility logic is lifted from one system to fields, particles, scaling limits, or gauge constraints.

Mechanism Reading

Operationally, String theory contributes a geometry or holographic realization role. Geometry supplies the realization, boundary, gauge, or dual description in which the operator construction becomes physically legible. In the fields step, the field layer extends the same constructor to variable numbers of modes and symmetry constraints. Creation and annihilation operators, correlation functions, gauge conditions, and renormalization flows are higher-capacity versions of the same assembly. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. String theory contributes a geometry or holographic realization role to the quantum construction.
- **Placement**. This page is read first as a many-mode or field-realization move: it extends the state and operator construction beyond a single-particle carrier.
- **Carrier or domain**. A spacetime, boundary algebra, gauge orbit, spin network, bulk/boundary pair, or geometric representation of a quantum state space.
- **Operator or map**. Hamiltonian, action, constraint, boundary operator, correlation map, or dictionary between two representations.
- **Admissibility**. Gauge, boundary, metric, covariance, and constraint conditions decide which geometric descriptions represent the same physical content.
- **Readout**. Boundary correlators, spectra, entropies, scattering data, geometric invariants, or reconstructed bulk quantities.
- **Check**. A geometric reformulation is physical only to the extent that it preserves observables or correlation functions across the representation change.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation

- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.42, transport flow=0.22, constraint closure=0.15

- arXiv:0912.2823, score 0.599
- arXiv:1612.00682, score 0.586
- arXiv:quant-ph0205159, score 0.582
- arXiv:1506.05598, score 0.578
- arXiv:1801.03283, score 0.578

14.23 Quantum geometry

Mechanism-tree role. Many-mode extension: fields, particles, and scaling; assignment score 0.88. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum geometry is a geometry-realization page: geometric quantities are promoted to quantum observables rather than assumed as a smooth background.

Mechanism Reading

Quantum geometry uses a quantum state of geometry, often represented by graph or spin-network data. The operator-to-spectrum step asks for eigenvalues of geometric observables such as area or volume. This places the page near the geometry/boundary frontier rather than inside a generic many-mode field layer. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, normalization or admissibility, state evolution; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum geometry contributes a geometry or holographic realization role to the quantum construction.
- **Placement**. This page is read first as a many-mode or field-realization move: it extends the state and operator construction beyond a single-particle carrier.
- **Carrier or domain**. A spacetime, boundary algebra, gauge orbit, spin network, bulk/boundary pair, or geometric representation of a quantum state space.
- **Operator or map**. Hamiltonian, action, constraint, boundary operator, correlation map, or dictionary between two representations.
- **Admissibility**. Gauge, boundary, metric, covariance, and constraint conditions decide which geometric descriptions represent the same physical content.
- **Readout**. Boundary correlators, spectra, entropies, scattering data, geometric invariants, or reconstructed bulk quantities.
- **Check**. A geometric reformulation is physical only to the extent that it preserves observables or correlation functions across the representation change.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.37, constraint closure=0.16, transport flow=0.14

- arXiv:1708.03640, score 0.596
- arXiv:quant-ph0205159, score 0.595
- arXiv:1801.03283, score 0.594
- arXiv:1612.00682, score 0.593
- arXiv:1506.05598, score 0.592

14.24 Fock space

Mechanism-tree role. Many-mode extension: fields, particles, and scaling; assignment score 0.86. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Fock space is the occupation-number state space: the construction that replaces a fixed-particle Hilbert space by a direct sum over particle number.

Mechanism Reading

Fock space changes the carrier of the quantum state. Instead of describing one system in one Hilbert space, it builds sectors with zero, one, two, and more identical quanta, then imposes the bosonic or fermionic exchange rule. Creation and annihilation operators are the native coordinates of this page because they move the state between occupation sectors. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Fock space contributes a many-mode field or particle-realization role to the quantum construction.
- **Placement**. This page is read first as a many-mode or field-realization move: it extends the state and operator construction beyond a single-particle carrier.
- **Carrier or domain**. Fock space, field configuration space, or a sector selected by charge, spin, momentum, statistics, or gauge data.
- **Operator or map**. Creation, annihilation, field, charge, spin, Hamiltonian, or scattering operators acting on the admissible sector.
- **Admissibility**. Statistics, gauge constraints, commutation or anticommutation rules, domain conditions, and sector labels decide which states are legal.
- **Readout**. Occupation number, charge, spin, momentum, energy, correlation function, cross-section, or scattering amplitude.
- **Check**. The field description must preserve the relevant observables under changes of representation and reduce to the expected particle or quasiparticle limit when that limit exists.

Role-Level Equations

Topic-specific constructor: the equations express variable particle number, exchange symmetry, and occupation-number readout.

$$\mathcal{F}_{\pm}(\mathcal{H}) = \bigoplus_{n=0}^{\infty} \mathcal{S}_{\pm}\mathcal{H}^{\otimes n}$$

$$[a_i, a_j^{\dagger}]_{\mp} = \delta_{ij}, \qquad [a_i, a_j]_{\mp} = 0$$

$$N = \sum_i a_i^{\dagger} a_i, \qquad N|n_1, n_2, \ldots\rangle = \left(\sum_i n_i\right)|n_1, n_2, \ldots\rangle$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.44, transport flow=0.09, constraint closure=0.06

- arXiv:2308.15676, score 0.548
- arXiv:2108.07838, score 0.539
- arXiv:1612.00682, score 0.533
- arXiv:0809.5271, score 0.528
- arXiv:2105.11733, score 0.520

14.25 Quantum spacetime

Mechanism-tree role. Many-mode extension: fields, particles, and scaling; assignment score 0.85. **Dominant evidence signal.** incompatible questions. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum spacetime is a geometry or holographic realization role in the compact quantum constructor. In this tree, quantum spacetime belongs to the many-mode extension: the same state, generator, observable, and compatibility logic is lifted from one system to fields, particles, scaling limits, or gauge constraints.

Mechanism Reading

Operationally, Quantum spacetime contributes a geometry or holographic realization role. Geometry supplies the realization, boundary, gauge, or dual description in which the operator construction becomes physically legible. In the fields step, the field layer extends the same constructor to variable numbers of modes and symmetry constraints. Creation and annihilation operators, correlation functions, gauge conditions, and renormalization flows are higher-capacity versions of the same assembly. In the source-evidence profile for this page, the strongest construction signal is non-commuting compatibility limits, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum spacetime contributes a geometry or holographic realization role to the quantum construction.
- **Placement**. This page is read first as a many-mode or field-realization move: it extends the state and operator construction beyond a single-particle carrier.

- **Carrier or domain**. A spacetime, boundary algebra, gauge orbit, spin network, bulk/boundary pair, or geometric representation of a quantum state space.
- **Operator or map**. Hamiltonian, action, constraint, boundary operator, correlation map, or dictionary between two representations.
- **Admissibility**. Gauge, boundary, metric, covariance, and constraint conditions decide which geometric descriptions represent the same physical content.
- **Readout**. Boundary correlators, spectra, entropies, scattering data, geometric invariants, or reconstructed bulk quantities.
- **Check**. A geometric reformulation is physical only to the extent that it preserves observables or correlation functions across the representation change.

Topic Equations

$$
\begin{aligned}
B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)} \\
\rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)} \\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)} \\
[O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

commutator incompatibility=0.26, transport flow=0.26, constraint closure=0.21

- arXiv:1604.06537, score 0.500
- arXiv:1706.03846, score 0.489
- arXiv:1604.05385, score 0.478
- arXiv:2111.12617, score 0.476
- arXiv:1708.03640, score 0.470

Chapter 15

Protocol layer: engineered transformations

Quantum computing, channels, circuits, algorithms, networks, sensors, and error correction turn the same formal machinery into controlled sequences of operations.

15.1 Why This Branch Exists

Quantum information is the engineering layer: the same state-operator-readout machinery becomes a controlled sequence of transformations.

15.2 Page Map

Page	Dominant evidence signal	Score
Quantum information science	observables and spectra	1.42
Quantum network	observables and spectra	1.42
Quantum algorithm	observables and spectra	1.42
Quantum error correction	observables and spectra	1.41
Quantum channel	observables and spectra	1.40
Quantum logic gate	observables and spectra	1.39
Quantum neural network	observables and spectra	1.39
Quantum circuit	observables and spectra	1.39
Quantum information	observables and spectra	1.38
Quantum computing	observables and spectra	1.20
Quantum key distribution	observables and spectra	0.81
Quantum sensor	observables and spectra	1.39
Quantum teleportation	observables and spectra	1.39
Quantum bus	observables and spectra	1.24
Quantum image processing	observables and spectra	1.23
Quantum finite automaton	state evolution	1.15
Quantum cryptography	observables and spectra	1.13
Quantum programming	observables and spectra	0.83

15.3 Mechanism Pages

Each page below is read as a specialization of the compact quantum constructor. Topic-specific pages use explicit equations or overrides; core-derived pages use the branch-level equation form associated with their mechanism role.

15.4 Quantum information science

Mechanism-tree role. Protocol layer: engineered transformations; assignment score 1.42. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum information science is a broad quantum constructor role in the compact quantum constructor. In this tree, quantum information science belongs to the protocol layer: it packages the quantum constructor into engineered sequences of admissible transformations and readouts.

Mechanism Reading

Operationally, Quantum information science contributes a broad quantum constructor role. The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer. In the protocols step, protocols are built after the state, operation, and readout rules exist. A circuit, channel, sensor, network, or algorithm is a controlled composition of maps whose output is checked by a final measurement. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, controlled update protocol; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum information science contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as an operation-sequence move: it specifies an ordered composition of allowed maps.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.35, transport flow=0.13, discrete protocol=0.10

- arXiv:1604.05385, score 0.561
- arXiv:2308.15676, score 0.535
- arXiv:2105.11733, score 0.533
- arXiv:0809.5271, score 0.531
- arXiv:cond-mat0108470, score 0.530

15.5 Quantum network

Mechanism-tree role. Protocol layer: engineered transformations; assignment score 1.42. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum network is an engineered operation-sequence role in the compact quantum constructor. In this tree, quantum network belongs to the protocol layer: it packages the quantum constructor into engineered sequences of admissible transformations and readouts.

Mechanism Reading

Operationally, Quantum network contributes an engineered operation-sequence role. The mechanism is a controlled composition of allowed maps: a sequence that prepares, transforms, protects, transmits, or reads a quantum state. In the protocols step, protocols are built after the state, operation, and readout rules exist. A circuit, channel, sensor, network, or algorithm is a controlled composition of maps whose output is checked by a final measurement. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum network contributes an engineered operation-sequence role to the quantum construction.
- **Placement**. This page is read first as an operation-sequence move: it specifies an ordered composition of allowed maps.
- **Carrier or domain**. An input state, register, channel state, error syndrome, key, or controlled experimental configuration.
- **Operator or map**. An ordered sequence of gates, channels, measurements, corrections, encodings, or conditional maps.
- **Admissibility**. Each step must belong to the claimed map class: unitary, completely positive, trace-preserving, projective, conditional, or corrective.
- **Readout**. Output state, key, error rate, fidelity, channel capacity, algorithmic success probability, or sensor estimate.
- **Check**. Changing operation order, inserting classical controls, or replacing a quantum channel should identify which step carries the effect.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.32, transport flow=0.22, constraint closure=0.13

- arXiv:1604.05385, score 0.577
- arXiv:0912.2823, score 0.574
- arXiv:0805.4565, score 0.555
- arXiv:1708.03640, score 0.551
- arXiv:quant-ph0205159, score 0.551

15.6 Quantum algorithm

Mechanism-tree role. Protocol layer: engineered transformations; assignment score 1.42. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum algorithm is an engineered operation-sequence role in the compact quantum constructor. In this tree, quantum algorithm belongs to the protocol layer: it packages the quantum constructor into engineered sequences of admissible transformations and readouts.

Mechanism Reading

Operationally, Quantum algorithm contributes an engineered operation-sequence role. The mechanism is a controlled composition of allowed maps: a sequence that prepares, transforms, protects, transmits, or reads a quantum state. In the protocols step, protocols are built after the state, operation, and readout rules exist. A circuit, channel, sensor, network, or algorithm is a controlled composition of maps whose output is checked by a final measurement. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum algorithm contributes an engineered operation-sequence role to the quantum construction.
- **Placement**. This page is read first as an operation-sequence move: it specifies an ordered composition of allowed maps.
- **Carrier or domain**. An input state, register, channel state, error syndrome, key, or controlled experimental configuration.
- **Operator or map**. An ordered sequence of gates, channels, measurements, corrections, encodings, or conditional maps.
- **Admissibility**. Each step must belong to the claimed map class: unitary, completely positive, trace-preserving, projective, conditional, or corrective.
- **Readout**. Output state, key, error rate, fidelity, channel capacity, algorithmic success probability, or sensor estimate.
- **Check**. Changing operation order, inserting classical controls, or replacing a quantum channel should identify which step carries the effect.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.32, transport flow=0.22, constraint closure=0.16

- arXiv:0912.2823, score 0.576
- arXiv:1604.05385, score 0.576
- arXiv:1708.03640, score 0.559
- arXiv:quant-ph0205159, score 0.556
- arXiv:1801.03283, score 0.556

15.7 Quantum error correction

Mechanism-tree role. Protocol layer: engineered transformations; assignment score 1.41. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum error correction is an engineered operation-sequence role in the compact quantum constructor. In this tree, quantum error correction belongs to the protocol layer: it packages the quantum constructor into engineered sequences of admissible transformations and readouts.

Mechanism Reading

Operationally, Quantum error correction contributes an engineered operation-sequence role. The mechanism is a controlled composition of allowed maps: a sequence that prepares, transforms, protects, transmits, or reads a quantum state. In the protocols step, protocols are built after the state, operation, and readout rules exist. A circuit, channel, sensor, network, or algorithm is a controlled composition of maps whose output is checked by a final measurement. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum error correction contributes an engineered operation-sequence role to the quantum construction.
- **Placement**. This page is read first as an operation-sequence move: it specifies an ordered composition of allowed maps.
- **Carrier or domain**. An input state, register, channel state, error syndrome, key, or controlled experimental configuration.
- **Operator or map**. An ordered sequence of gates, channels, measurements, corrections, encodings, or conditional maps.
- **Admissibility**. Each step must belong to the claimed map class: unitary, completely positive, trace-preserving, projective, conditional, or corrective.
- **Readout**. Output state, key, error rate, fidelity, channel capacity, algorithmic success probability, or sensor estimate.
- **Check**. Changing operation order, inserting classical controls, or replacing a quantum channel should identify which step carries the effect.

Topic Equations

$$
\begin{aligned}
B &\longmapsto \rho_B \quad \text{(context specifies an admissible state)} \\
\rho_t &= U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)} \\
O &= \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)} \\
[O_1, O_2] &\neq 0 \quad \text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.39, transport flow=0.22, constraint closure=0.15

- arXiv:1604.05385, score 0.553
- arXiv:1604.06537, score 0.525
- arXiv:0912.2823, score 0.523
- arXiv:1612.00682, score 0.516
- arXiv:quant-ph0205159, score 0.513

15.8 Quantum channel

Mechanism-tree role. Protocol layer: engineered transformations; assignment score 1.40. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum channel is the open-system protocol constructor: it maps input states to output states while preserving complete positivity and trace.

Mechanism Reading

A channel is the mechanism for noisy transformations, measurements with forgotten outcomes, and subsystem evolution. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum channel contributes an engineered operation-sequence role to the quantum construction.
- **Placement**. This page is read first as an operation-sequence move: it specifies an ordered composition of allowed maps.
- **Carrier or domain**. An input state, register, channel state, error syndrome, key, or controlled experimental configuration.
- **Operator or map**. An ordered sequence of gates, channels, measurements, corrections, encodings, or conditional maps.

- **Admissibility**. Each step must belong to the claimed map class: unitary, completely positive, trace-preserving, projective, conditional, or corrective.
- **Readout**. Output state, key, error rate, fidelity, channel capacity, algorithmic success probability, or sensor estimate.
- **Check**. Changing operation order, inserting classical controls, or replacing a quantum channel should identify which step carries the effect.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.40, transport flow=0.21, constraint closure=0.14

- arXiv:1604.05385, score 0.560
- arXiv:1604.06537, score 0.555
- arXiv:1506.05598, score 0.542
- arXiv:0805.4565, score 0.542
- arXiv:1708.03640, score 0.539

15.9 Quantum logic gate

Mechanism-tree role. Protocol layer: engineered transformations; assignment score 1.39. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum logic gate is an engineered operation-sequence role in the compact quantum constructor. In this tree, quantum logic gate belongs to the protocol layer: it packages the quantum constructor into engineered sequences of admissible transformations and readouts.

Mechanism Reading

Operationally, Quantum logic gate contributes an engineered operation-sequence role. The mechanism is a controlled composition of allowed maps: a sequence that prepares, transforms, protects, transmits, or reads a quantum state. In the protocols step, protocols are built after the state, operation, and readout rules exist. A circuit, channel, sensor, network, or algorithm is a controlled composition of maps whose output is checked by a final measurement. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, spectral profile.

Quantum Mechanism Frame

- **Role**. Quantum logic gate contributes an engineered operation-sequence role to the quantum construction.
- **Placement**. This page is read first as an operation-sequence move: it specifies an ordered composition of allowed maps.
- **Carrier or domain**. An input state, register, channel state, error syndrome, key, or controlled experimental configuration.
- **Operator or map**. An ordered sequence of gates, channels, measurements, corrections, encodings, or conditional maps.
- **Admissibility**. Each step must belong to the claimed map class: unitary, completely positive, trace-preserving, projective, conditional, or corrective.
- **Readout**. Output state, key, error rate, fidelity, channel capacity, algorithmic success probability, or sensor estimate.

- **Check**. Changing operation order, inserting classical controls, or replacing a quantum channel should identify which step carries the effect.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.43, transport flow=0.26, constraint closure=0.19

- arXiv:1604.05385, score 0.581
- arXiv:1801.03283, score 0.552
- arXiv:1612.00682, score 0.550
- arXiv:quant-ph0205159, score 0.545
- arXiv:1506.05598, score 0.539

15.10 Quantum neural network

Mechanism-tree role. Protocol layer: engineered transformations; assignment score 1.39. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum neural network is an engineered operation-sequence role in the compact quantum constructor. In this tree, quantum neural network belongs to the protocol layer: it packages the quantum constructor into engineered sequences of admissible transformations and readouts.

Mechanism Reading

Operationally, Quantum neural network contributes an engineered operation-sequence role. The mechanism is a controlled composition of allowed maps: a sequence that prepares, transforms, protects, transmits, or reads a quantum state. In the protocols step, protocols are built after the state, operation, and readout rules exist. A circuit, channel, sensor, network, or algorithm is a controlled composition of maps whose output is checked by a final measurement. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum neural network contributes an engineered operation-sequence role to the quantum construction.
- **Placement**. This page is read first as an operation-sequence move: it specifies an ordered composition of allowed maps.
- **Carrier or domain**. An input state, register, channel state, error syndrome, key, or controlled experimental configuration.
- **Operator or map**. An ordered sequence of gates, channels, measurements, corrections, encodings, or conditional maps.
- **Admissibility**. Each step must belong to the claimed map class: unitary, completely positive, trace-preserving, projective, conditional, or corrective.
- **Readout**. Output state, key, error rate, fidelity, channel capacity, algorithmic success probability, or sensor estimate.
- **Check**. Changing operation order, inserting classical controls, or replacing a quantum channel should identify which step carries the effect.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.42, transport flow=0.22, constraint closure=0.11

- arXiv:1604.05385, score 0.550
- arXiv:2108.07838, score 0.545
- arXiv:0805.4565, score 0.544
- arXiv:1612.00682, score 0.537
- arXiv:quant-ph0205159, score 0.530

15.11 Quantum circuit

Mechanism-tree role. Protocol layer: engineered transformations; assignment score 1.39. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum circuit is the engineered-composition constructor: a finite sequence of admissible maps prepares, transforms, and measures a register.

Mechanism Reading

A circuit is the protocol layer of the same state-operator-readout machinery. Gates are controlled unitary or channel maps; measurement converts final states into output probabilities. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum circuit contributes an engineered operation-sequence role to the quantum construction.
- **Placement**. This page is read first as an operation-sequence move: it specifies an ordered composition of allowed maps.
- **Carrier or domain**. An input state, register, channel state, error syndrome, key, or controlled experimental configuration.
- **Operator or map**. An ordered sequence of gates, channels, measurements, corrections, encodings, or conditional maps.
- **Admissibility**. Each step must belong to the claimed map class: unitary, completely positive, trace-preserving, projective, conditional, or corrective.
- **Readout**. Output state, key, error rate, fidelity, channel capacity, algorithmic success probability, or sensor estimate.
- **Check**. Changing operation order, inserting classical controls, or replacing a quantum channel should identify which step carries the effect.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^{\dagger} \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.41, transport flow=0.22, constraint closure=0.12

- arXiv:1604.05385, score 0.552
- arXiv:0912.2823, score 0.551
- arXiv:1612.00682, score 0.537
- arXiv:2108.07838, score 0.532
- arXiv:quant-ph0205159, score 0.530

15.12 Quantum information

Mechanism-tree role. Protocol layer: engineered transformations; assignment score 1.38. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum information is a broad quantum constructor role in the compact quantum constructor. In this tree, quantum information belongs to the protocol layer: it packages the quantum constructor into engineered sequences of admissible transformations and readouts.

Mechanism Reading

Operationally, Quantum information contributes a broad quantum constructor role. The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer. In the protocols step, protocols are built after the state, operation, and readout rules exist. A circuit, channel, sensor, network, or algorithm is a controlled composition of maps whose output is checked by a final measurement. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, spectral profile.

Quantum Mechanism Frame

- **Role**. Quantum information contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as an operation-sequence move: it specifies an ordered composition of allowed maps.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$
\begin{aligned}
B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)} \\
\rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)} \\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)} \\
[O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.38, transport flow=0.26, constraint closure=0.21

- arXiv:1604.06537, score 0.521
- arXiv:0912.2823, score 0.501
- arXiv:1604.05385, score 0.500
- arXiv:1612.00682, score 0.493
- arXiv:1801.03283, score 0.492

15.13 Quantum computing

Mechanism-tree role. Protocol layer: engineered transformations; assignment score 1.20. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum computing is an engineered operation-sequence role in the compact quantum constructor. In this tree, quantum computing belongs to the protocol layer: it packages the quantum constructor into engineered sequences of admissible transformations and readouts.

Mechanism Reading

Operationally, Quantum computing contributes an engineered operation-sequence role. The mechanism is a controlled composition of allowed maps: a sequence that prepares, transforms, protects, transmits, or reads a quantum state. In the protocols step, protocols are built after the state, operation, and readout rules exist. A circuit, channel, sensor, network, or algorithm is a controlled composition of maps whose output is checked by a final measurement. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, controlled update protocol; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum computing contributes an engineered operation-sequence role to the quantum construction.
- **Placement**. This page is read first as an operation-sequence move: it specifies an ordered composition of allowed maps.

- **Carrier or domain**. An input state, register, channel state, error syndrome, key, or controlled experimental configuration.
- **Operator or map**. An ordered sequence of gates, channels, measurements, corrections, encodings, or conditional maps.
- **Admissibility**. Each step must belong to the claimed map class: unitary, completely positive, trace-preserving, projective, conditional, or corrective.
- **Readout**. Output state, key, error rate, fidelity, channel capacity, algorithmic success probability, or sensor estimate.
- **Check**. Changing operation order, inserting classical controls, or replacing a quantum channel should identify which step carries the effect.

Topic Equations

$$\begin{aligned} B \longmapsto \rho_B &\quad \text{(context specifies an admissible state)} \\ \rho_t = U_t \rho_B U_t^\dagger &\quad \text{(unitary evolution from preparation to readout)} \\ O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) &\quad \text{(spectral probability measure)} \\ [O_1, O_2] \neq 0 &\quad \text{(incompatible observables: no common sharp basis)} \end{aligned}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.30, transport flow=0.23, discrete protocol=0.20

- arXiv:1604.05385, score 0.579
- arXiv:1801.03283, score 0.510
- arXiv:quant-ph0205159, score 0.507
- arXiv:1708.03640, score 0.505
- arXiv:1612.00682, score 0.503

15.14 Quantum key distribution

Mechanism-tree role. Protocol layer: engineered transformations; assignment score 0.81. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum key distribution is a broad quantum constructor role in the compact quantum constructor. In this tree, quantum key distribution belongs to the protocol layer: it packages the quantum constructor into engineered sequences of admissible transformations and readouts.

Mechanism Reading

Operationally, Quantum key distribution contributes a broad quantum constructor role. The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer. In the protocols step, protocols are built after the state, operation, and readout rules exist. A circuit, channel, sensor, network, or algorithm is a controlled composition of maps whose output is checked by a final measurement. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum key distribution contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as an operation-sequence move: it specifies an ordered composition of allowed maps.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.41, transport flow=0.18, constraint closure=0.16

- arXiv:1604.05385, score 0.583
- arXiv:1612.00682, score 0.573
- arXiv:quant-ph0205159, score 0.566
- arXiv:1801.03283, score 0.563
- arXiv:1506.05598, score 0.560

15.15 Quantum sensor

Mechanism-tree role. Protocol layer: engineered transformations; assignment score 1.39. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum sensor is an instrument-mediated readout role in the compact quantum constructor. In this tree, quantum sensor belongs to the protocol layer: it packages the quantum constructor into engineered sequences of admissible transformations and readouts.

Mechanism Reading

Operationally, Quantum sensor contributes an instrument-mediated readout role. The mechanism is an apparatus-coupled readout: a prepared probe state interacts with a sample or field, the interaction changes phase, momentum, intensity, or counting statistics, and the instrument reconstructs an image, spectrum, trajectory, or estimate. In the protocols step, protocols are built after the state, operation, and readout rules exist. A circuit, channel, sensor, network, or algorithm is a controlled composition of maps whose output is checked by a final measurement. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum sensor contributes an instrument-mediated readout role to the quantum construction.
- **Placement**. This page is read first as an operation-sequence move: it specifies an ordered composition of allowed maps.
- **Carrier or domain**. A probe state, sample state, field mode, detector state, or estimation register.
- **Operator or map**. An interaction Hamiltonian, transfer map, measurement channel, reconstruction map, or estimator.
- **Admissibility**. The instrument must separate sample signal from preparation, detector response, calibration, noise, and reconstruction artifacts.
- **Readout**. Counts, images, spectra, phase shifts, trajectories, intensity maps, correlation data, or parameter estimates.
- **Check**. The claimed mechanism is credible only when the same readout survives control experiments, calibration changes, and reconstruction checks.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.38, transport flow=0.19, constraint closure=0.12

- arXiv:1604.05385, score 0.528
- arXiv:0912.2823, score 0.528
- arXiv:1604.06537, score 0.526
- arXiv:quant-ph0205159, score 0.503
- arXiv:1612.00682, score 0.501

15.16 Quantum teleportation

Mechanism-tree role. Protocol layer: engineered transformations; assignment score 1.39. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum teleportation is a broad quantum constructor role in the compact quantum constructor. In this tree, quantum teleportation belongs to the protocol layer: it packages the quantum constructor into engineered sequences of admissible transformations and readouts.

Mechanism Reading

Operationally, Quantum teleportation contributes a broad quantum constructor role. The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer. In the protocols step, protocols are built after the state, operation, and readout rules exist. A circuit, channel, sensor, network, or algorithm is a controlled composition of maps whose output is checked by a final measurement. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, controlled update protocol; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum teleportation contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as an operation-sequence move: it specifies an ordered composition of allowed maps.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$

$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$

$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$

$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.38, transport flow=0.17, discrete protocol=0.11

- arXiv:1604.05385, score 0.536
- arXiv:0912.2823, score 0.497
- arXiv:2308.15676, score 0.486
- arXiv:1612.00682, score 0.484
- arXiv:quant-ph0205159, score 0.474

15.17 Quantum bus

Mechanism-tree role. Protocol layer: engineered transformations; assignment score 1.24. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum bus is a broad quantum constructor role in the compact quantum constructor. In this tree, quantum bus belongs to the protocol layer: it packages the quantum constructor into engineered sequences of admissible transformations and readouts.

Mechanism Reading

Operationally, Quantum bus contributes a broad quantum constructor role. The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer. In the protocols step, protocols are built after the state, operation, and readout rules exist. A circuit, channel, sensor, network, or algorithm is a controlled composition of maps whose output is checked by a final measurement. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, preparation, basis, or boundary context; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum bus contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as an operation-sequence move: it specifies an ordered composition of allowed maps.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.37, transport flow=0.21, boundary weak form=0.11

- arXiv:0912.2823, score 0.574
- arXiv:2108.07838, score 0.556
- arXiv:0805.4565, score 0.546
- arXiv:0908.0752, score 0.545
- arXiv:1612.00682, score 0.545

15.18 Quantum image processing

Mechanism-tree role. Protocol layer: engineered transformations; assignment score 1.23. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum image processing is a broad quantum constructor role in the compact quantum constructor. In this tree, quantum image processing belongs to the protocol layer: it packages the quantum constructor into engineered sequences of admissible transformations and readouts.

Mechanism Reading

Operationally, Quantum image processing contributes a broad quantum constructor role. The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer. In the protocols step, protocols are built after the state, operation, and readout rules exist. A circuit, channel, sensor, network, or algorithm is a controlled composition of maps whose output is checked by a final measurement. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, non-commuting compatibility limits; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum image processing contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as an operation-sequence move: it specifies an ordered composition of allowed maps.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.

- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$
\begin{aligned}
B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)} \\
\rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)} \\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)} \\
[O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.42, transport flow=0.15, commutator incompatibility=0.14

- arXiv:2111.12617, score 0.559
- arXiv:1604.06537, score 0.541
- arXiv:2308.15676, score 0.509
- arXiv:1506.05598, score 0.508
- arXiv:1612.00682, score 0.502

15.19 Quantum finite automaton

Mechanism-tree role. Protocol layer: engineered transformations; assignment score 1.15. **Dominant evidence signal.** state evolution. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum finite automaton is a broad quantum constructor role in the compact quantum constructor. In this tree, quantum finite automaton belongs to the protocol layer: it packages the quantum constructor into engineered sequences of admissible transformations and readouts.

Mechanism Reading

Operationally, Quantum finite automaton contributes a broad quantum constructor role. The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer. In the protocols step, protocols are built after the state, operation, and readout rules exist. A circuit, channel, sensor, network, or algorithm is a controlled composition of maps whose output is checked by a final measurement. In the source-evidence profile for this page, the strongest construction signal is state evolution, normalization or admissibility, operator-to-spectrum readout; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum finite automaton contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as an operation-sequence move: it specifies an ordered composition of allowed maps.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

transport flow=0.28, constraint closure=0.24, spectral operator=0.19

- arXiv:1604.05385, score 0.446
- arXiv:1706.03846, score 0.442
- arXiv:1708.03640, score 0.420
- arXiv:0908.0752, score 0.410
- arXiv:1801.03283, score 0.408

15.20 Quantum cryptography

Mechanism-tree role. Protocol layer: engineered transformations; assignment score 1.13. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum cryptography is an engineered operation-sequence role in the compact quantum constructor. In this tree, quantum cryptography belongs to the protocol layer: it packages the quantum constructor into engineered sequences of admissible transformations and readouts.

Mechanism Reading

Operationally, Quantum cryptography contributes an engineered operation-sequence role. The mechanism is a controlled composition of allowed maps: a sequence that prepares, transforms, protects, transmits, or reads a quantum state. In the protocols step, protocols are built after the state, operation, and readout rules exist. A circuit, channel, sensor, network, or algorithm is a controlled composition of maps whose output is checked by a final measurement. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum cryptography contributes an engineered operation-sequence role to the quantum construction.
- **Placement**. This page is read first as an operation-sequence move: it specifies an ordered composition of allowed maps.
- **Carrier or domain**. An input state, register, channel state, error syndrome, key, or controlled experimental configuration.
- **Operator or map**. An ordered sequence of gates, channels, measurements, corrections, encodings, or conditional maps.
- **Admissibility**. Each step must belong to the claimed map class: unitary, completely positive, trace-preserving, projective, conditional, or corrective.
- **Readout**. Output state, key, error rate, fidelity, channel capacity, algorithmic success probability, or sensor estimate.
- **Check**. Changing operation order, inserting classical controls, or replacing a quantum channel should identify which step carries the effect.

Topic Equations

$$
\begin{aligned}
B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)}\\
\rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)}\\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)}\\
[O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.33, transport flow=0.26, constraint closure=0.20

- arXiv:0912.2823, score 0.527
- arXiv:1604.05385, score 0.526
- arXiv:1801.03283, score 0.497
- arXiv:quant-ph0205159, score 0.496
- arXiv:1612.00682, score 0.495

15.21 Quantum programming

Mechanism-tree role. Protocol layer: engineered transformations; assignment score 0.83. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Quantum programming is an engineered operation-sequence role in the compact quantum constructor. In this tree, quantum programming belongs to the protocol layer: it packages the quantum constructor into engineered sequences of admissible transformations and readouts.

Mechanism Reading

Operationally, Quantum programming contributes an engineered operation-sequence role. The mechanism is a controlled composition of allowed maps: a sequence that prepares, transforms, protects, transmits, or reads a quantum state. In the protocols step, protocols are built after the state, operation, and readout rules exist. A circuit, channel, sensor, network, or algorithm is a controlled composition of maps whose output is checked by a final measurement. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum programming contributes an engineered operation-sequence role to the quantum construction.
- **Placement**. This page is read first as an operation-sequence move: it specifies an ordered composition of allowed maps.
- **Carrier or domain**. An input state, register, channel state, error syndrome, key, or controlled experimental configuration.
- **Operator or map**. An ordered sequence of gates, channels, measurements, corrections, encodings, or conditional maps.
- **Admissibility**. Each step must belong to the claimed map class: unitary, completely positive, trace-preserving, projective, conditional, or corrective.
- **Readout**. Output state, key, error rate, fidelity, channel capacity, algorithmic success probability, or sensor estimate.
- **Check**. Changing operation order, inserting classical controls, or replacing a quantum channel should identify which step carries the effect.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.40, transport flow=0.27, constraint closure=0.15

- arXiv:1604.05385, score 0.480
- arXiv:0805.4565, score 0.447
- arXiv:1612.00682, score 0.442
- arXiv:2108.07838, score 0.441
- arXiv:quant-ph0205159, score 0.435

Chapter 16

Annotations: history, interpretations, and popular frames

Some pages help readers navigate the subject but do not form steps in the mechanism. They are kept as annotations so books, historical figures, interpretations, and popular frames do not distort the constructive tree.

16.1 Why This Branch Exists

These pages remain useful for orientation, but they are deliberately not treated as construction steps. This prevents biographies, books, and interpretations from becoming false roots of the mechanism.

16.2 Page Map

Page	Dominant evidence signal	Score
Introduction to quantum mechanics	observables and spectra	0.72
Old quantum theory	observables and spectra	0.91
History of quantum mechanics *(annotation)*	observables and spectra	1.50
Introduction to Quantum Mechanics (book) *(annotation)*	state evolution	1.45
Quantum mind *(annotation)*	observables and spectra	1.44
History of quantum field theory *(annotation)*	state evolution	1.41
Erwin Schrödinger *(annotation)*	observables and spectra	1.39
Interpretations of quantum mechanics *(annotation)*	observables and spectra	1.38
Quantum mysticism *(annotation)*	incompatible questions	1.37
David Hilbert *(annotation)*	observables and spectra	1.36
Quantum Computing: A Gentle Introduction *(annotation)*	state evolution	1.30
QBism *(annotation)*	observables and spectra	1.43
Relational quantum mechanics *(annotation)*	observables and spectra	1.43

Page	Dominant evidence signal	Score
Modern Quantum Mechanics *(annotation)*	observables and spectra	1.23
The Quantum Universe *(annotation)*	state evolution	1.20
Quantum Theory: Concepts and Methods *(annotation)*	observables and spectra	1.10

16.3 Mechanism Pages

Each page below is read as a specialization of the compact quantum constructor. Topic-specific pages use explicit equations or overrides; core-derived pages use the branch-level equation form associated with their mechanism role.

16.4 Introduction to quantum mechanics

Mechanism-tree role. Annotations: history, interpretations, and popular frames; assignment score 0.72. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Introduction to quantum mechanics is a broad quantum constructor role in the compact quantum constructor. In this tree, introduction to quantum mechanics is an annotation layer attached to the formal constructor.

Mechanism Reading

Operationally, Introduction to quantum mechanics contributes a broad quantum constructor role. The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer. In the annotations step, this page changes how the formalism is narrated, interpreted, taught, or historically situated. The underlying assembly remains the same: a context admits states, operators expose spectra, and probability rules connect states to outcomes. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure. Its constructive use is to identify which formal layer is being interpreted: state assignment, probability, update, readout, or ontology.

Quantum Mechanism Frame

- **Role**. Introduction to quantum mechanics contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as an interpretive or historical move: it clarifies which formal layer is being discussed.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.

- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.29, transport flow=0.26, constraint closure=0.19

- arXiv:1604.05385, score 0.508
- arXiv:1604.06537, score 0.496
- arXiv:0912.2823, score 0.476

- arXiv:1708.03640, score 0.462
- arXiv:quant-ph0205159, score 0.462

16.5 Old quantum theory

Mechanism-tree role. Annotations: history, interpretations, and popular frames; assignment score 0.91. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

Central Claim

Old quantum theory is a broad quantum constructor role in the compact quantum constructor. In this tree, old quantum theory is an annotation layer attached to the formal constructor.

Mechanism Reading

Operationally, Old quantum theory contributes a broad quantum constructor role. The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer. In the annotations step, this page changes how the formalism is narrated, interpreted, taught, or historically situated. The underlying assembly remains the same: a context admits states, operators expose spectra, and probability rules connect states to outcomes. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure. Its constructive use is to identify which formal layer is being interpreted: state assignment, probability, update, readout, or ontology.

Quantum Mechanism Frame

- **Role**. Old quantum theory contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as an interpretive or historical move: it clarifies which formal layer is being discussed.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.34, transport flow=0.24, constraint closure=0.18

- arXiv:1604.06537, score 0.564
- arXiv:0912.2823, score 0.564
- arXiv:1604.05385, score 0.563
- arXiv:1708.03640, score 0.547
- arXiv:1801.03283, score 0.546

16.6 History of quantum mechanics

Mechanism-tree role. Annotations: history, interpretations, and popular frames; assignment score 1.50. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

This page is treated as historical, interpretive, or popular context rather than as a conceptual root.

Central Claim

History of quantum mechanics is a broad quantum constructor role in the compact quantum constructor. In this tree, history of quantum mechanics is an annotation layer attached to the formal constructor.

Mechanism Reading

Operationally, History of quantum mechanics contributes a broad quantum constructor role. The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer. In the annotations step, this page changes how the formalism is narrated, interpreted, taught, or historically situated. The underlying assembly remains the same: a context admits states, operators expose spectra, and probability rules connect states to outcomes. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure. Its constructive use is to identify which formal layer is being interpreted: state assignment, probability, update, readout, or ontology.

Quantum Mechanism Frame

- **Role**. History of quantum mechanics contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as an interpretive or historical move: it clarifies which formal layer is being discussed.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.42, transport flow=0.25, constraint closure=0.19

- arXiv:1604.05385, score 0.532
- arXiv:1604.06537, score 0.531
- arXiv:1612.00682, score 0.522
- arXiv:quant-ph0205159, score 0.516
- arXiv:1801.03283, score 0.514

16.7 Introduction to Quantum Mechanics (book)

Mechanism-tree role. Annotations: history, interpretations, and popular frames; assignment score 1.45. **Dominant evidence signal.** state evolution. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

This page is treated as historical, interpretive, or popular context rather than as a conceptual root.

Central Claim

Introduction to Quantum Mechanics (book) is a broad quantum constructor role in the compact quantum constructor. In this tree, introduction to Quantum Mechanics (book) is an annotation layer attached to the formal constructor.

Mechanism Reading

Operationally, Introduction to Quantum Mechanics (book) contributes a broad quantum constructor role. The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer. In the annotations step, this page changes how the formalism is narrated, interpreted, taught, or historically situated. The underlying assembly remains the same: a context admits states, operators expose spectra, and probability rules connect states to outcomes. In the source-evidence profile for this page, the strongest construction signal is state evolution, normalization or admissibility, operator-to-spectrum readout; the strongest carrier signal is local notation, information profile, spectral profile. Its constructive use is to identify which formal layer is being interpreted: state assignment, probability, update, readout, or ontology.

Quantum Mechanism Frame

- **Role**. Introduction to Quantum Mechanics (book) contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as an interpretive or historical move: it clarifies which formal layer is being discussed.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

transport flow=0.28, constraint closure=0.24, spectral operator=0.18

- arXiv:1706.03846, score 0.454
- arXiv:1708.03640, score 0.427
- arXiv:0806.4515, score 0.425
- arXiv:0908.0752, score 0.418
- arXiv:1801.03283, score 0.416

16.8 Quantum mind

Mechanism-tree role. Annotations: history, interpretations, and popular frames; assignment score 1.44. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

This page is treated as historical, interpretive, or popular context rather than as a conceptual root.

Central Claim

Quantum mind is a broad quantum constructor role in the compact quantum constructor. In this tree, quantum mind is an annotation layer attached to the formal constructor.

Mechanism Reading

Operationally, Quantum mind contributes a broad quantum constructor role. The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer. In the annotations step, this page changes how the formalism is narrated, interpreted, taught, or historically situated. The underlying assembly remains the same: a context admits states, operators expose spectra, and probability rules connect states to outcomes. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, controlled update protocol; the strongest carrier signal is local notation, information profile, formula structure. Its constructive use is to identify which formal layer is being interpreted: state assignment, probability, update, readout, or ontology.

Quantum Mechanism Frame

- **Role**. Quantum mind contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as an interpretive or historical move: it clarifies which formal layer is being discussed.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.31, transport flow=0.20, discrete protocol=0.17

- arXiv:1604.05385, score 0.547
- arXiv:0912.2823, score 0.510
- arXiv:quant-ph0205159, score 0.483
- arXiv:1612.00682, score 0.480
- arXiv:0805.4565, score 0.480

16.9 History of quantum field theory

Mechanism-tree role. Annotations: history, interpretations, and popular frames; assignment score 1.41. **Dominant evidence signal.** state evolution. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

This page is treated as historical, interpretive, or popular context rather than as a conceptual root.

Central Claim

History of quantum field theory is a many-mode field or particle-realization role in the compact quantum constructor. In this tree, history of quantum field theory is an annotation layer attached to the formal constructor.

Mechanism Reading

Operationally, History of quantum field theory contributes a many-mode field or particle-realization role. The mechanism is the field/mode version of the constructor: a state space is decomposed into modes or sectors, operators create, annihilate, or constrain those modes, and readouts are occupation, charge, spin, momentum, energy, or scattering response. In the annotations step, this page changes how the formalism is narrated, interpreted, taught, or historically situated. The underlying assembly remains the same: a context admits states, operators expose spectra, and probability rules connect states to outcomes. In the source-evidence profile for this page, the strongest construction signal is state evolution, operator-to-spectrum readout, non-commuting compatibility limits; the strongest carrier signal is local notation, information profile, formula structure. Its constructive use is to identify which formal layer is being interpreted: state assignment, probability, update, readout, or ontology.

Quantum Mechanism Frame

- **Role**. History of quantum field theory contributes a many-mode field or particle-realization role to the quantum construction.
- **Placement**. This page is read first as an interpretive or historical move: it clarifies which formal layer is being discussed.
- **Carrier or domain**. Fock space, field configuration space, or a sector selected by charge, spin, momentum, statistics, or gauge data.
- **Operator or map**. Creation, annihilation, field, charge, spin, Hamiltonian, or scattering operators acting on the admissible sector.
- **Admissibility**. Statistics, gauge constraints, commutation or anticommutation rules, domain conditions, and sector labels decide which states are legal.
- **Readout**. Occupation number, charge, spin, momentum, energy, correlation function, cross-section, or scattering amplitude.
- **Check**. The field description must preserve the relevant observables under changes of representation and reduce to the expected particle or quasiparticle limit when that limit exists.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

transport flow=0.26, spectral operator=0.24, commutator incompatibility=0.22

- arXiv:1604.06537, score 0.536
- arXiv:1604.05385, score 0.517
- arXiv:0912.2823, score 0.515
- arXiv:1708.03640, score 0.511
- arXiv:1801.03283, score 0.509

16.10 Erwin Schrödinger

Mechanism-tree role. Annotations: history, interpretations, and popular frames; assignment score 1.39. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

This page is treated as historical, interpretive, or popular context rather than as a conceptual root.

Central Claim

Erwin Schrödinger is a lawful state-transport role in the compact quantum constructor. In this tree, erwin Schrödinger is an annotation layer attached to the formal constructor.

Mechanism Reading

Operationally, Erwin Schrödinger contributes a lawful state-transport role. The topic supplies or modifies the generator of state evolution before readout. It should be read as a transport step, not as a measurement result. In the annotations step, this page changes how the formalism is narrated, interpreted, taught, or historically situated. The underlying assembly remains the same: a context admits states, operators expose spectra, and probability rules connect states to outcomes. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure. Its constructive use is to identify which formal layer is being interpreted: state assignment, probability, update, readout, or ontology.

Quantum Mechanism Frame

- **Role**. Erwin Schrödinger contributes a lawful state-transport role to the quantum construction.
- **Placement**. This page is read first as an interpretive or historical move: it clarifies which formal layer is being discussed.
- **Carrier or domain**. A state vector, density operator, wavefunction, field state, or register on a specified domain.
- **Operator or map**. Hamiltonian, unitary map, channel generator, action, constraint, or differential operator that transports the state.
- **Admissibility**. Self-adjointness, complete positivity, trace preservation, gauge constraints, and boundary/domain conditions decide whether the evolution is legal.
- **Readout**. Time-dependent probabilities, spectra, transition amplitudes, conserved quantities, or response functions.
- **Check**. The generator must predict the observed evolution while preserving the relevant normalization, positivity, symmetry, or conservation constraint.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.32, transport flow=0.26, constraint closure=0.22

- arXiv:1604.05385, score 0.519
- arXiv:0912.2823, score 0.518
- arXiv:1604.06537, score 0.516
- arXiv:1801.03283, score 0.507
- arXiv:1708.03640, score 0.500

16.11 Interpretations of quantum mechanics

Mechanism-tree role. Annotations: history, interpretations, and popular frames; assignment score 1.38. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

This page is treated as historical, interpretive, or popular context rather than as a conceptual root.

Central Claim

Interpretations of quantum mechanics is a probability/readout role in the compact quantum constructor. In this tree, interpretations of quantum mechanics is an annotation layer attached to the formal constructor.

Mechanism Reading

Operationally, Interpretations of quantum mechanics contributes a probability/readout role. The topic modifies how a state is connected to recorded outcomes. The stable machinery is the spectral measure or POVM together with the probability rule. In the annotations step, this page changes how the formalism is narrated, interpreted, taught, or historically situated. The underlying assembly remains the same: a context admits states, operators expose spectra, and probability rules connect states to outcomes. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure. Its constructive use is to identify which formal layer is being interpreted: state assignment, probability, update, readout, or ontology.

Quantum Mechanism Frame

- **Role**. Interpretations of quantum mechanics contributes a probability/readout role to the quantum construction.
- **Placement**. This page is read first as an interpretive or historical move: it clarifies which formal layer is being discussed.
- **Carrier or domain**. A state vector or density operator together with the measurement context in which outcome channels are defined.
- **Operator or map**. A projection-valued measure, POVM, update map, or instrument map connecting state to record.
- **Admissibility**. Outcome probabilities must be positive, normalized, and tied to a specified readout map rather than to informal observer language.
- **Readout**. Born probabilities, detector records, post-measurement states, ensemble frequencies, or decision probabilities.
- **Check**. The interpretation is constrained by whether it changes the probability rule, the update rule, the detector model, or only the language used for them.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^{\dagger} \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.26, transport flow=0.26, constraint closure=0.19

- arXiv:1604.05385, score 0.515
- arXiv:0912.2823, score 0.514
- arXiv:1604.06537, score 0.507
- arXiv:1708.03640, score 0.479
- arXiv:quant-ph0205159, score 0.475

16.12 Quantum mysticism

Mechanism-tree role. Annotations: history, interpretations, and popular frames; assignment score 1.37. **Dominant evidence signal.** incompatible questions. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

This page is treated as historical, interpretive, or popular context rather than as a conceptual root.

Central Claim

Quantum mysticism is a broad quantum constructor role in the compact quantum constructor. In this tree, quantum mysticism is an annotation layer attached to the formal constructor.

Mechanism Reading

Operationally, Quantum mysticism contributes a broad quantum constructor role. The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer. In the annotations step, this page changes how the formalism is narrated, interpreted, taught, or historically situated. The underlying assembly remains the same: a context admits states, operators expose spectra, and probability rules connect states to outcomes. In the source-evidence profile for this page, the strongest construction signal is non-commuting compatibility limits, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure. Its constructive use is to identify which formal layer is being interpreted: state assignment, probability, update, readout, or ontology.

Quantum Mechanism Frame

- **Role**. Quantum mysticism contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as an interpretive or historical move: it clarifies which formal layer is being discussed.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

commutator incompatibility=0.27, transport flow=0.25, constraint closure=0.17

- arXiv:2111.12617, score 0.373
- arXiv:1604.06537, score 0.361
- arXiv:1604.05385, score 0.349
- arXiv:1706.03846, score 0.325
- arXiv:2410.23449, score 0.319

16.13 David Hilbert

Mechanism-tree role. Annotations: history, interpretations, and popular frames; assignment score 1.36. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

This page is treated as historical, interpretive, or popular context rather than as a conceptual root.

Central Claim

David Hilbert is a state-carrier role in the compact quantum constructor. In this tree, david Hilbert is an annotation layer attached to the formal constructor.

Mechanism Reading

Operationally, David Hilbert contributes a state-carrier role. The topic contributes the mathematical carrier of prediction: vector, wavefunction, density operator, register, coherent state, or field state. In the annotations step, this page changes how the formalism is narrated, interpreted, taught, or historically situated. The underlying assembly remains the same: a context admits states, operators expose spectra, and probability rules connect states to outcomes. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure. Its constructive use is to identify which formal layer is being interpreted: state assignment, probability, update, readout, or ontology.

Quantum Mechanism Frame

- **Role**. David Hilbert contributes a state-carrier role to the quantum construction.
- **Placement**. This page is read first as an interpretive or historical move: it clarifies which formal layer is being discussed.
- **Carrier or domain**. The mathematical state object: vector, wavefunction, density operator, coherent state, field state, or register.
- **Operator or map**. Operators, maps, and observables become meaningful only after this carrier and its domain have been fixed.
- **Admissibility**. Normalization, positivity, inner product, representation, tensor factorization, or superselection conditions define legal states.
- **Readout**. Probability distributions obtained by applying the appropriate observables or measurement maps to the carrier.
- **Check**. Equivalent representations must preserve probabilities and expectation values when the change is only representational.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$

$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$

$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$

$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.42, transport flow=0.17, constraint closure=0.15

- arXiv:0912.2823, score 0.528
- arXiv:1604.06537, score 0.523
- arXiv:1612.00682, score 0.518
- arXiv:quant-ph0205159, score 0.505
- arXiv:1801.03283, score 0.503

16.14 Quantum Computing: A Gentle Introduction

Mechanism-tree role. Annotations: history, interpretations, and popular frames; assignment score 1.30. **Dominant evidence signal.** state evolution. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

This page is treated as historical, interpretive, or popular context rather than as a conceptual root.

Central Claim

Quantum Computing: A Gentle Introduction is an engineered operation-sequence role in the compact quantum constructor. In this tree, quantum Computing: A Gentle Introduction is an annotation layer attached to the formal constructor.

Mechanism Reading

Operationally, Quantum Computing: A Gentle Introduction contributes an engineered operation-sequence role. The mechanism is a controlled composition of allowed maps: a sequence that prepares, transforms, protects, transmits, or reads a quantum state. In the annotations step, this page changes how the formalism is narrated, interpreted, taught, or historically situated. The underlying assembly remains the same: a context admits states, operators expose spectra, and probability rules connect states to outcomes. In the source-evidence profile for this page, the strongest construction signal is state evolution, normalization or admissibility, operator-to-spectrum readout; the strongest carrier signal is local notation, information profile, formula structure. Its constructive use is to identify which formal layer is being interpreted: state assignment, probability, update, readout, or ontology.

Quantum Mechanism Frame

- **Role**. Quantum Computing: A Gentle Introduction contributes an engineered operation-sequence role to the quantum construction.
- **Placement**. This page is read first as an interpretive or historical move: it clarifies which formal layer is being discussed.
- **Carrier or domain**. An input state, register, channel state, error syndrome, key, or controlled experimental configuration.
- **Operator or map**. An ordered sequence of gates, channels, measurements, corrections, encodings, or conditional maps.
- **Admissibility**. Each step must belong to the claimed map class: unitary, completely positive, trace-preserving, projective, conditional, or corrective.
- **Readout**. Output state, key, error rate, fidelity, channel capacity, algorithmic success probability, or sensor estimate.
- **Check**. Changing operation order, inserting classical controls, or replacing a quantum channel should identify which step carries the effect.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

transport flow=0.27, constraint closure=0.25, spectral operator=0.13

- arXiv:1706.03846, score 0.534
- arXiv:1708.03640, score 0.510
- arXiv:0908.0752, score 0.505
- arXiv:1801.03283, score 0.490
- arXiv:quant-ph0205159, score 0.487

16.15 QBism

Mechanism-tree role. Annotations: history, interpretations, and popular frames; assignment score 1.43. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

This page is treated as historical, interpretive, or popular context rather than as a conceptual root.

Central Claim

QBism is a probability/readout role in the compact quantum constructor. In this tree, qBism is an annotation layer attached to the formal constructor.

Mechanism Reading

Operationally, QBism contributes a probability/readout role. The topic modifies how a state is connected to recorded outcomes. The stable machinery is the spectral measure or POVM together with the probability rule. In the annotations step, this page changes how the formalism is narrated, interpreted, taught, or historically situated. The underlying assembly remains the same: a context admits states, operators expose spectra, and probability rules connect states to outcomes. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure. Its constructive use is to identify which formal layer is being interpreted: state assignment, probability, update, readout, or ontology.

Quantum Mechanism Frame

- **Role**. QBism contributes a probability/readout role to the quantum construction.
- **Placement**. This page is read first as an interpretive or historical move: it clarifies which formal layer is being discussed.
- **Carrier or domain**. A state vector or density operator together with the measurement context in which outcome channels are defined.
- **Operator or map**. A projection-valued measure, POVM, update map, or instrument map connecting state to record.
- **Admissibility**. Outcome probabilities must be positive, normalized, and tied to a specified readout map rather than to informal observer language.
- **Readout**. Born probabilities, detector records, post-measurement states, ensemble frequencies, or decision probabilities.
- **Check**. The interpretation is constrained by whether it changes the probability rule, the update rule, the detector model, or only the language used for them.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.30, transport flow=0.26, constraint closure=0.19

- arXiv:1604.05385, score 0.486
- arXiv:1604.06537, score 0.471
- arXiv:0912.2823, score 0.452
- arXiv:quant-ph0205159, score 0.429
- arXiv:1612.00682, score 0.428

16.16 Relational quantum mechanics

Mechanism-tree role. Annotations: history, interpretations, and popular frames; assignment score 1.43. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

This page is treated as historical, interpretive, or popular context rather than as a conceptual root.

Central Claim

Relational quantum mechanics is a probability/readout role in the compact quantum constructor. In this tree, relational quantum mechanics is an annotation layer attached to the formal constructor.

Mechanism Reading

Operationally, Relational quantum mechanics contributes a probability/readout role. The topic modifies how a state is connected to recorded outcomes. The stable machinery is the spectral measure or POVM together with the probability rule. In the annotations step, this page changes how the formalism is narrated, interpreted, taught, or historically situated. The underlying assembly remains the same: a context admits states, operators expose spectra, and probability rules connect states to outcomes. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, controlled update protocol, state evolution; the strongest carrier signal is local notation, information profile, formula structure. Its constructive use is to identify which formal layer is being interpreted: state assignment, probability, update, readout, or ontology.

Quantum Mechanism Frame

- **Role**. Relational quantum mechanics contributes a probability/readout role to the quantum construction.
- **Placement**. This page is read first as an interpretive or historical move: it clarifies which formal layer is being discussed.
- **Carrier or domain**. A state vector or density operator together with the measurement context in which outcome channels are defined.
- **Operator or map**. A projection-valued measure, POVM, update map, or instrument map connecting state to record.
- **Admissibility**. Outcome probabilities must be positive, normalized, and tied to a specified readout map rather than to informal observer language.
- **Readout**. Born probabilities, detector records, post-measurement states, ensemble frequencies, or decision probabilities.
- **Check**. The interpretation is constrained by whether it changes the probability rule, the update rule, the detector model, or only the language used for them.

Topic Equations

$$
\begin{aligned}
B &\longmapsto \rho_B \quad \text{(context specifies an admissible state)} \\
\rho_t &= U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)} \\
O &= \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)} \\
[O_1, O_2] &\neq 0 \quad \text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.30, discrete protocol=0.19, transport flow=0.17

- arXiv:1604.05385, score 0.508
- arXiv:0912.2823, score 0.453
- arXiv:quant-ph0607206, score 0.435
- arXiv:2306.13129, score 0.430
- arXiv:1612.00682, score 0.424

16.17 Modern Quantum Mechanics

Mechanism-tree role. Annotations: history, interpretations, and popular frames; assignment score 1.23. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

This page is treated as historical, interpretive, or popular context rather than as a conceptual root.

Central Claim

Modern Quantum Mechanics is a broad quantum constructor role in the compact quantum constructor. In this tree, modern Quantum Mechanics is an annotation layer attached to the formal constructor.

Mechanism Reading

Operationally, Modern Quantum Mechanics contributes a broad quantum constructor role. The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer. In the annotations step, this page changes how the formalism is narrated, interpreted, taught, or historically situated. The underlying assembly remains the same: a context admits states, operators expose spectra, and probability rules connect states to outcomes. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, spectral profile. Its constructive use is to identify which formal layer is being interpreted: state assignment, probability, update, readout, or ontology.

Quantum Mechanism Frame

- **Role**. Modern Quantum Mechanics contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as an interpretive or historical move: it clarifies which formal layer is being discussed.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.41, transport flow=0.27, constraint closure=0.21

- arXiv:1612.00682, score 0.507
- arXiv:1801.03283, score 0.505
- arXiv:quant-ph0205159, score 0.504
- arXiv:1708.03640, score 0.502
- arXiv:1506.05598, score 0.498

16.18 The Quantum Universe

Mechanism-tree role. Annotations: history, interpretations, and popular frames; assignment score 1.20. **Dominant evidence signal.** state evolution. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

This page is treated as historical, interpretive, or popular context rather than as a conceptual root.

Central Claim

The Quantum Universe is a broad quantum constructor role in the compact quantum constructor. In this tree, the Quantum Universe is an annotation layer attached to the formal constructor.

Mechanism Reading

Operationally, The Quantum Universe contributes a broad quantum constructor role. The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer. In the annotations step, this page changes how the formalism is narrated, interpreted, taught, or historically situated. The underlying assembly remains the same: a context admits states, operators expose spectra, and probability rules connect states to outcomes. In the source-evidence profile for this page, the strongest construction signal is state evolution, non-commuting compatibility limits, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure. Its constructive use is to identify which formal layer is being interpreted: state assignment, probability, update, readout, or ontology.

Quantum Mechanism Frame

- **Role**. The Quantum Universe contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as an interpretive or historical move: it clarifies which formal layer is being discussed.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

transport flow=0.25, commutator incompatibility=0.25, constraint closure=0.16

- arXiv:1706.03846, score 0.533
- arXiv:1604.06537, score 0.516
- arXiv:2111.12617, score 0.515
- arXiv:1708.03640, score 0.510
- arXiv:0908.0752, score 0.501

16.19 Quantum Theory: Concepts and Methods

Mechanism-tree role. Annotations: history, interpretations, and popular frames; assignment score 1.10. **Dominant evidence signal.** observables and spectra. **Lagrangian construction road.** no Lagrangian road map available; road score 0.00.

This page is treated as historical, interpretive, or popular context rather than as a conceptual root.

Central Claim

Quantum Theory: Concepts and Methods is a broad quantum constructor role in the compact quantum constructor. In this tree, quantum Theory: Concepts and Methods is an annotation layer attached to the formal constructor.

Mechanism Reading

Operationally, Quantum Theory: Concepts and Methods contributes a broad quantum constructor role. The mechanism is read through the shared quantum constructor: state carrier, legal transformation, readout, compatibility condition, and realization layer. In the annotations step, this page changes how the formalism is narrated, interpreted, taught, or historically situated. The underlying assembly remains the same: a context admits states, operators expose spectra, and probability rules connect states to outcomes. In the source-evidence profile for this page, the strongest construction signal is operator-to-spectrum readout, state evolution, normalization or admissibility; the strongest carrier signal is local notation, information profile, formula structure. Its constructive use is to identify which formal layer is being interpreted: state assignment, probability, update, readout, or ontology.

Quantum Mechanism Frame

- **Role**. Quantum Theory: Concepts and Methods contributes a broad quantum constructor role to the quantum construction.
- **Placement**. This page is read first as an interpretive or historical move: it clarifies which formal layer is being discussed.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

$$
\begin{aligned}
B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)} \\
\rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)} \\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)} \\
[O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Validation Boundary

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Evidence Profile

spectral operator=0.36, transport flow=0.26, constraint closure=0.20

- arXiv:1604.05385, score 0.485
- arXiv:1604.06537, score 0.477
- arXiv:1612.00682, score 0.452
- arXiv:quant-ph0205159, score 0.452
- arXiv:1801.03283, score 0.451

Source Boundary

The source pages are stored under discoveries/morphwiki_quantum/pages. The mechanism tree is stored as discoveries/morphwiki_quantum/quantum_mechanism_tree.json. ArXiv links are evidence pointers, not claims that any individual paper proves the whole branch.